

Philosophy Of Nursing And The Nurse

What is Philosophy?

A philosophy is a system of motivating beliefs, concepts, and principles that direct the action of a particular group in their pursuit of goals (Gillies, 1989). It is set of beliefs or an attitude to life that guides somebody's behaviour. Philosophy begins when someone contemplates or wonders about something. Every one has a personal philosophy of life that is unique. People develop personal philosophy as they mature which entails individual's values and belief system. A nurse therefore has personal philosophy of nursing profession, nurses and nursing practice (Gillies, 1989).

Each individual nurse needs to understand his/her personal philosophy of nursing, nursing practices and nurses in order to know what influences the individual nurse's behaviour towards the clients, the professionalism of nursing and the fellow nurses. Philosophy of nursing-g is an attitude towards life and reality that evolves from each nurses beliefs (Edward, 1997;1089)

Philosophy of Nursing

This is statement of beliefs about nursing. It is worth noting that evolving professionalism of nursing and the dynamism of the profession may affect philosophy of nursing which makes it possible to witness ever-changing philosophy of nursing year after year and at different locations where nursing services are rendered. Philosophy of nursing is a set of beliefs or attitude to life that guides the behavior of a nurse in professional practice area.

In the opinion of Renpenning and Taylor (2003), philosophy of Nursing refers to statement that reflects beliefs and values about nursing as a discipline and a profession; the beliefs and values held by a particular health care institution about the nursing participants and healthcare, and the beliefs and values of the nurses who work in the institution about nursing participants, the environment, health and methodologies for nursing. This takes cognizance of view of both the nurses and nurses' public.

The philosophy of the nursing services determines the goals of the work force, which in turn influences the patient care objectives established by each nurse. A nurse develops personal philosophy of nursing based on the understanding towards Virginian Henderson's definition of professional nursing, which sees nursing as the art of assisting an individual, sick or well, in the performance of those activities that contribute to health, or its recovery, or to a peaceful death,

which the individual would have performed unaided if he has the necessary strength, will or knowledge.

Nursing aims at assisting people to regain independent in the performance of activities as soon as possible. It is by acquisition of knowledge and skill via formal training and experience that the nurses carry out these activities that assist the clients to gain comfort and independent without negatively affecting the integrity and self esteem of the clients. In the performance of these activities, the nurse does not regard the person's creed, colour or race. Philosophy of nursing therefore reflects the vision and values of nursing profession (Baxter, 1989).

Scholars have been arguing for years, whether Nursing is an art or a science. But what stops it from being both? To be a successful nurse, one must be able to provide the variety of caring and have the drive to continue to attain and apply the knowledge "Science" throughout his/her career. Nursing was being compared to art in metaphorical sense ie they simply alluded to nursing quality what is the excellent level of practice (Austgard, 2006;16). Benner (1984), was of the opinion that the best way to separate the instrument is to regulate caring as the art of Nursing. Without caring, the nurse cannot connect and trust will not develop. Without trusting relationship, therapeutic nursing will not take place. Therefore caring is at the centre of all successful nursing encounters. This does not in any form mean to down play the value of science.

A competent nurse must be able to use the technical resources available. The nurse should have knowledge of Anatomy & Physiology of human body, pathophysiological changes and outcomes and current guideline for pharmacological therapy and the ability to forecast outcomes from present to render assistance to those that need nursing cares. (Benner, 1984). Science also encompassed the skill required to perform technical tasks in nursing. Science and art of nursing are dimensions necessary in developing individual philosophy of nursing among nurses.

In the opinion of Benner (1984), Nurses are on a continuum to achieve "expert" in each of the seven domains of competencies.

- Her philosophy of nursing addresses four "theories":

1. Society
2. Environments
3. The recipient of nursing care
4. The interaction between the society, person and environment

Society: A group of people with common interest of providing care to their patients. Society for nursing could be hospital ward where nurses are held together by the common interest of providing care to live in (Benner, 1984). Nurses function in the society where common interest of people is to gain their health and remain productive.

Environment: This is the platform for the society to live in. It can be discussed as physical environment such as the area the person lives in, the person's mental awareness. This includes the person's spiritual awareness, emotions and thought about his current situation (Benner, 1984). Nurses strive to bring the sick back to full consciousness of his environment to which he should adapt or strike a balance.

The recipient of nursing care: The recipient of nursing care is not limited to just the patient. The family, the friends, clergy and other members of the health care team are all recipients of the care being given by the nurse. Everyone that comes in contact with the nursing process is affected either positively or negatively (Benner, 1984). Illness and wellness are in a continuum and there is high level of unpredictability on when an individual should fall sick. This makes everyone a potential recipient of nursing care.

The interaction between the society, person and the environment: All interactions between the society, the environment and the recipient of care are dynamic and synergistic (Benner, 1984). This calls for holistic approach mindedness on the part of the nurse when rendering care.

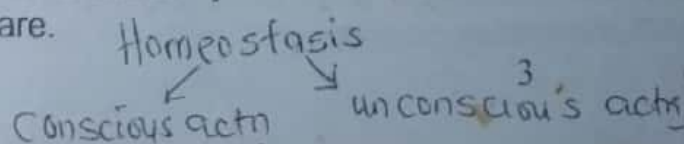
HEALTH AND ILLNESS.

WHO, 2002

Health is a process of becoming a total being with ability to be productive and enjoy life experiences. According to World Health Organization (WHO, 2002), it is a state of mental, physical and emotional well being and not only the absence of illness or infirmity. Health denotes a blending of intellectual and physical, emotional, social and spiritual resources as they assist one in mastering tasks necessary to enjoy a satisfying and protective life. Health is not static but fluctuates due to constant change and adaptation to stress resulting to homeostasis.

* Illness is any state that interrupts or hinders the process of health. The state of health or disease is expressions of one's ability to adapt to the environmental challenges. Illness therefore is said to have occurred when an individual is unable to successfully strike a balance between internal and external environments.

Once the ability to maintain homeostasis is lost an individual triggers to the negative side of health commonly referred to as ill-health which ranges from mild to severe state depending on the extent of destabilization of homeostatic mechanism. This state can be reversed through some adaptation mechanisms. The adaptation mechanisms occur through conscious actions such as covering of self with blanket, withdrawal of self from hot environment, etc; and unconscious actions such as shivering, sweating, etc. It is therefore worthy to note that Health and Illness are in a continuum for which one happens at a time. It is therefore not uncommon to witness an individual who has been "well" or "up and about" to suddenly take ill to the extent of being rushed to health institution for healthcare.



In view of the above assertion, humans engage in health-promotional activities such as exercise, adequate rest, eating of balanced diet, fruits and vegetables to remain healthy. Good still, some engage into preventive activities such as vaccination or immunization to avoid being overwhelmed by diseases and infections. Where situation arises that results to illness, curative actions come into play for which the nurse has significant role to play. An individual seeks curative services when there is trauma (physical, psychological and emotional) or infection (bacterial, fungal and viral). It is essential to note that in infections the sole aim of medical and nursing system remains to break the Chain of Infection. This will be discussed in details in Vol. 2 of this Series.

THE NURSE AND NURSING

A nurse is not just a person on a white uniform taking care of patients in a hospital, but an individual that could make difference in another person's life i.e bringing health into play where illness has disrupted it or maintaining health where it exists. In the opinion of Henderson (1960) cited in Tomey and Alligood (1998: p102), the unique function of the nurse is to assist the individual, sick or well, in the performance of those activities contributing to health or its recovery (or to a peaceful death) that he would have performed unaided if he had the necessary strength, will or knowledge. And to do this in such a way as to help him gain independence as rapidly as possible. In the light of Henderson's functional definition of Nursing, it can be deduced that nursing is combination of science and art with goal of assisting the unable to gain independence. It is invariably giving hope to the hopeless and succor to the dying. Nursing is more than assisting people to recover but making them to experience a healthy transition in every aspect of life challenges and process. It does not involve the use of Standing Order in caring but application of critical thinking in the light of practice experiences and skills which are Evidence Based oriented.

Nursing is caring and compassion delivered with technology needed to provide an environment that supports healing. This can be provided at any location, and delivered by professionally certified person(s) with the ability to show love and support in the entire process. This person (professionally) must provide competent care along the guideline set forth by the Regulatory body and Professional Associations of the state such as International Council of Nurses (ICN), National Association of Nigeria Nurses & Midwives (NANNM) and Nursing & Midwifery Council of Nigeria N&MCN).

Nursing is a way nurses give back the expectations of their "publics" to the society. The ability to truly show a person that you care involves interacting with that person in a holistic manner which transcends the physical, psychological, socio-cultural and spiritual beings of the individual. This provides therapeutic basis to promote trust, thereby aiding in the healing process vis a-vis gaining

health. Nursing is guided by the Nurses' mission statement which describes their purpose of existence and what they give to the society.

Mission statement of Nurses at Frank First Regional Medical Centre reads thus: "to compassionately deliver quality health care in an innovative and efficient manner"(Regional Medical Centre,2007). Generally, the nurse aims at being innovative and renders efficient and effective services even when her life is being jeopardized. It is common for a nurse to apply the management principle of **Subordinating individual interest to organizational interest** for the purposes of ensuring that the needs of her clients are met.

Things involved

- ☐ Team attitude of treating others as you would like to be treated.
- ☐ Competent caring in team approval.

The nurse does not work alone as a care giver but in conjunction with other health professionals with team initiative. Inter disciplinary collaborative approach to client care assists the nurse in achieving her goal of efficient, effective and timely care to the clients to whom she acts as an advocate. The nurse may be under pressure to do the best for the client with consideration given to the individuality and dignity of the client.

Nursing care requires the application of theories and models related to the professional practice. A nurse must therefore be conversant with theories and models implicit in nursing and their application in nursing care. Such theories include:

- ☐ Abraham Maslow's Hierarchy of needs theory
- ☐ Nightingale's Environmental theory
- ☐ Callister Roy's Adaptation theory
- ☐ Dorothy Orem's Self care theory
- ☐ Virginia Henderson's Activity of daily need model
- ☐ Afaf Meleis's Transitional Theory, etc.

The application of any of theory in client care demands the nurse's critical thinking ability in achieving good decision with utmost sense of judgment.

In line with the above, the nurse applies the Nursing Process in every phase of care be it for individual client, family, and community. The Nursing process gives the nurse a guide in rendering individualized care to different clients in their different dimensions of life with positive outcomes. In the Nursing Process lies the Nursing Care Plan which is an intermediate stage of the nursing process that guides the nurse in the on-going provision of nursing cares and assists in the evaluation of cares rendered. Nursing care plan outlines the nursing care to be provided to an individual, family, and community.

Nursing Code Of Ethics And Duty Of Care

NURSING CODE OF ETHICS

Code of Ethics is a statement to guide behaviour and a frame work for decision making. Code of ethics for nurses is a formal statement of the nursing profession's expected conduct at practice arena. Code of ethics remains essential tool the nurse uses when faced with ethical dilemmas at work place. A code of ethics is an implied contract through which the profession informs the society of the principles and rules by which it functions (Chally & Hough, 2005). Nursing code of ethics serves as guideline and self regulation to all professional nurses in their day to day practice to meet societal needs. Code of ethics has related term like, morals, beliefs, mores, values and attitudes.

Morals: Moral is established rules in situation in which a decision about right or wrong must be made. Morals provide standard of behavior and depicts proper behavior. It deals with principle of right conduct or decision between right and wrong, "judgment" of goodness, or badness of human action and character.

Mores: Customs and behaviour that are considered typical of a particular social group. Mores mould an individual.

Attitudes: This is behaviour an individual directs towards a person or thing. It shows ones feelings towards issues, persons or thing which is often exhibited in the person's actions.

Value: Concept of worth or broad preferences concerning appropriate course of action or outcomes. Values reflect person's sense of right and wrong or what "ought" to be "equal rights for all", People deserve admirations and to be treated with respect and dignity. Value tends to influence attitude and behaviour.

Weiss and Tappen (2010), view value inn diverse forms as below:

Personal Values: Provide internal preference for what is good, beneficial important, useful, beautiful desirables, constructive, etc. Value generates behaviour. Value an individual has over a thing determines the person's attitude towards that thing.

Intrinsic value: This is value inherent in an object for which that object is desired. Intrinsic value is inbuilt in an object, material or substance and that is the reason people want to have that object, substance or material.

Professional Value: This is a quality considered important by a professional group such as autonomy, integrity and commitment (Weiss & Tappen, 2010).

Extrinsic value: This is a value possessed by a commodity as its make-up.

Cultural Value: Cultural Value identifies what should be judged as "good" or "evil" in the society. Different cultures reflect different values. Cultural values differ from personal value though the former may have dominion over the later depending on the place and situation on ground.

Professional values: These are qualities considered important by a professional group. Autonomy, integrity and commitment are examples of professional values necessary for a polyvalent nurse to discharge publicly accepted societal role.

Nurses require good knowledge of ethics, ethical theories and principles, moral development, ethical dilemmas and ethical decision making models to achieve client oriented care. (Chally & Hough, 2005). To actualize this, the value implicit in nursing must be explored and defined by every nurse.

ETHICAL THEORIES

Several opinions abound on approaches to ethical issues. Ethical theories are ideas that tend to explain how ethical codes came into existence. Several of these theories abound but selected few will be discussed in this text.

1. The Natural (moral) Law.
2. Deontological approach/duty based theory.
3. Utilitarianism
4. Teleological approach/goal based
5. Virtue ethics.

The Natural (Moral) Law

The natural law holds that it is inherent in human beings to have good morals. The supporters of this theory were of the opinion that morals exist naturally as there exists natural phenomena. For instance, it is natural for what goes up to come down (Law of gravity)(Anarodo, 2002). In her assertion, the Natural (Moral) Law theorists are of the opinion that moral (ethics) is inherent in humans to enable one determine what is morally right or wrong at any given situation. They tend to see moral to exist in human just as caring naturally exists in human. The big question is, "does it occur in all humans?"

Anarado (2002), asserted that an act is considered right or wrong in relation to the extent to which it adheres to **natural moral law**. A little dipper into the Natural (Moral) law lays bare the fact that though it is inherent in humans to determine

what is right or wrong, there are moulding mechanisms that enable this to take proper shape. Such mechanisms are those moral imbueement achieved through teaching from Holy books (Holy Koran, the Holy Bible, which are transferred to sharpen the life style of individuals through divine understanding as the Imam, priest, pastors and teachers of the holy books teach their members. For instance, "thou shell not kill" is divine injunction from the Holy books and who so ever kills incurs the wrath of God (Anarado, 2002).

The assertion that good must be done and evil eschewed is a precept of general law and must be applied in the ethics. The natural (moral) law is expected to guide individuals on their day-to-day decision makings and when faced with ethical dilemmas but this may not be absolute truth as there exists the need to explain why some individuals are not guided by the natural (moral) law.

It is on the ground of discovered gaps in Natural (moral) law that ethical code of conduct exists to regulate the actions of members of the society for sanity (Anarado, 2002). This injunction sharpens the understanding some people have about sanctity of life and decisions on certain issues like euthanasia (mercy killing), abortion and use of certain contraceptives. Such people are justified in their actions based on this. In the opinion of Rest 1979, cited in Brent (2001), four types of psychological process are required for a person to take moral action: **moral sensitivity, moral reasoning, moral commitment and moral action.** These redirect ones moral values with one's ethical choices.

Deontological Approach

The major proponent of deontological approach was Immanuel Kant (1724-1804). In his opinion, an act is termed to be right or wrong depending on the inherent moral significant. According to Kant, an act is morally justified if it originated from good will (Brannigen & Boss, 2001). If an act originating from good will is adjudged morally right what prevents majority from going by acts from good will to make a better society?

A moral agent is expected to do what is right and refrain from what is wrong to maintain the relevance of ethics (Brannigen & Boss, 2001). Deontological approach or Duty-Based theory looks at an act to be either good or bad depending on whether it conforms to our duty or obligation (Anarado, 2002;41). Duty or obligation is used to judge the moral state of an act in this approach.

Utilitarianism

This theory is based on a fundamental belief that the moral rightness of an action is determined solely by its consequence (Chally & Hough, 2005). In the opinion of Mill (1985) cited in Chitty 2005, a "right action" conforms to the "greatest happiness principle". Chally & Hough (2005), asserted that it is right to maximize the greatest good for the happiness or pleasure of the greatest number of people.

Utilitarianism theory considers the effect actions may produce on the welfare of people at present and in time to come on the people. The supporters of the theory believe that it is possible to balance good and evil with the desire that good should outweigh the evil.

Application of "triage" in the management of the injured illustrates the implementation of utilitarianism.

Triage is the act of sorting according to quality or assignment of degree of urgency to the wounded or ill patients. It is a process for sorting injured people into groups based on their needs for or likely benefit from immediate medical treatment.

- It is a system used to allocate a scarce commodity, such as food, only to those capable of deriving the greatest benefit from it.
- It is a process, in which things are ranked in terms of importance or primary.

Triangle has gained world wide acceptance and at present the determination of who gets the highest health care financial allocation is considered based on the number that will be affected positively by such allocation.

Teleological Approach Ethics: This is an ethical theory that holds that the end of consequences of one's conduct are the ultimate basis for any judgment about the rightness of that conduct. That is a morally right act is one that will produce a good outcome or consequence. It is different from deontology in that deontology derives rightness or wrongness of one's conduct from character of the behaviour itself rather than the out come of the conduct. It is also different from virtue ethics which focuses on the character of the agent rather than on the nature or consequences of the act itself and pragmatic ethics which treats morality like sciences advancing socially over the course of many life times, such that any moral criterion is subject to revision.

Bethham (1996), noted that people are driven by their interests and their fears but their interests take precedence over their fears and their interest are carried out in accordance with how people view the consequences that might be involved with their interest. There is also the desire to make people happy by individual conducts. "Happiness" on this account is defined as the maximization of pleasure and the minimization of pain. Aggregate happiness of everyone is and not the happiness of any particular person is considered by the consequentiality.

Virtue Ethics: In the words of Aristotle "virtues are tendencies to act, feel, and judge that develop through appropriate training but some from natural tendencies". According to Burkhardt & Nathaniel (2002), individuals' actions are built from a degree of inborn moral virtues. Virtues refer to character traits like honesty, courage, kindness, respectfulness, and integrity.

Virtue ethics is one of the normative ethics that describes the character of a moral agent as a driving force for ethical behaviour, rather than rules (deontology or consequentialism) which derives rightness or wrongness from the outer of the act itself rather than character or social context (pragmatic ethics).

Moral virtues include prudence, justice, fortitude, courage, liberality, magnificence, magnanimity, temperance. According to Aristotle, each of the moral virtue was a mean between two corresponding vices. Playing the Golden rule is better ie the desirable middle between two extremes, one of excess and the other of deficiency. For instance, "courage" is a virtue, if taken to excess would manifest as recklessness and if deficient manifests as cowardice.

McClosky (2007), stated that virtue ethics can provide a basis for a balanced approach to understanding capitalism and capitalist societies. Virtue ethics can provide a rationale and foundation for peace. Virtue is an integral part of nursing profession and a defining factor between a professional nurse and any kind of nurse found anywhere.

Ethical Principles (Principles isolated in duty of care)

These are guidelines that can apply to situations to decide whether they are moral or immoral. Ethical Principles provide a foundation for nursing practice and as part of being a great nurse understanding ethics is the most critical aspect of nursing. If nurses must be successful in their professional arena, the need to be guided by the Ethics of Nursing is not debatable. Ethics in a simple form is the moral obligations of right and wrong played in the onus of "what you would like people to do for you, do the same to them".

Nursing profession, as a noble and caring profession is guided by general rules which over the years have assisted to sharpen the image of the profession before its publics. Adherence to these ethically related rules assists the nurse in rendering cares void of public criticisms. Sometimes, the nurse is caught in a dilemma involving "what she ought to do" on the basis of one ethical principle and "what she out not be do" on the basis of another principle. When situation like this occurs, ethical decision-making model assists the nurse to take the best decision for the benefit of the client.

The principles to consider include;

1. Autonomy
2. Non-maleficence
3. Beneficence
4. Justice
5. Veracity
6. Confidentiality
7. Fidelity

Autonomy

Autonomy implies the ability to act and make decisions without being controlled by anyone. It denotes self independence, self reliance, freedom of choice, etc. It is personal liberty of action or self determination. Autonomy spells ones right to decide on what will or will not be done to his or her body. Generally an individual has the freedom to think, accept or reject others' decisions and decide on what he wants to have. Essentially there are autonomies of will and action.

In nursing profession, the nurse acts as the client's advocate and so defends the client when his/her autonomy is tempered with by the healthcare providers. The nurse ensures that the client's autonomy is maintained through respect of his/her values and choices, and obtaining of consent before procedures which also promotes respect.

Implication of client's autonomy in duty of care as outlined by Daniels (2004)

- Promotion of client's decision making: Bearing in mind that clients who present themselves for medical or nursing management are thinking beings that have will, likes, dislikes and therefore should be allowed to exercise their autonomy
- Patients should therefore be allowed to participate in decisions about their healthcare so long their decisions do not jeopardize their recovery or deleterious to their health.
- Supports client's right to informed consent. The need to maintain the dignity, individuality and confidentiality of clients can not be over emphasized and so,
- Client's autonomy should be taken as priority during care considering their state of health.
- Client's autonomy demands that respect be accorded to the client by requesting for his/her consent before carrying out procedures.
- It helps the nurse to make decisions when, client's choice poses harm. In a situation where the choice of the client poses harms, it is believed that the nurse has the required skills and best judgment for the patient and should therefore act for the good of the patient.

Non-Maleficence

This principle simply denotes "duty to do no harm" or better put "duty not to do harm". The nurse is expected in her struggle to ensure that her client's life is saved, should not harm the client no matter how impossible it may seem to be. The use of technological advances for the management of the client should by no means harm the client. The harm in it generality covers physiological, psychological and socio-cultural harm. Nursing actions should give remedy effects, succor to cushion the effects of trauma on the sick individuals and not to inflict any pain no matter how minute the pain may seem to be. In the ICN

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elements "nurses and people" affirms that the nurse uses technology and scientific advances that must be compatible with safety, dignity and rights of the people (ICN in Anerado, 2002).

Implication to Nursing

As an implication, the nurses should therefore avoid deliberate harm, risk of harm and harm that occurs during performance of nursing cares. This therefore holds that:

- The nurse has to consider the degree of risk permissible in the performance of procedures around or on her clients.
- The nurse should determine whether the use of the technological advances provides benefits that out-weigh the risks to her clients.
- The nurse working in emergency situation should be calm, composed, demonstrate mastery of action to minimize harm to her clients.

Beneficence

This denotes showing kindness or doing good to people. In the scarcity of resources, the nurse is expected to do good to her clients and ensure that they are contented with the actions carried out. The nurse is, by this principle expected to provide good benefit to the clients under her care and apt to consider how best the client will be helped out of the identified problems.

By implication, the nurse should be a good critical thinker and always initiate ideas that will yield positive outcomes on the clients. It should be noted that where beneficence overtakes autonomy, paternalism develops i.e a situation arises where the nurse begins to do what is the best for the client regardless of the client's own determination (Daniels, 2004). In this situation, the nurse takes up "seeming" parental ideas on the patient's behalf. Intimacy formed between the nurse and the patient determines how deep paternalism develops when beneficence overrides autonomy.

Justice

Justice, as ethical principle deals with fairness and equity. This applies in both care rendered to clients and allocation of resources available for equitable management. There should be explicit display of "equal right and justice" in nursing care among the clients irrespective of ethnic, religion or race differences. An aspect of code of ethics for the National Association of Nigeria Nurses and Midwives holds that the nurse must always care without minding the clients' religion, creed or colour. Accepting that resources are very limited in the healthcare arena, the nurse should not be deterred in using good judgment in making most use of the available resources to enable substantial number of clients benefit from the limited resources.

Implications of justice to nursing include:

- Determining the order in which clients should be treated eg priority

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treatment for clients in severe pain.

- Determining the appropriate intervention that is cost-effective and to the benefit of many clients.

According to Daniels (2004), the application of distributive justice is complicated by the following problems.

- Not everyone is equal in every way.
- Occasion may warrant that one person should receive a greater or lesser share than another.

Case I

The president of your country was admitted in your hospital for treatment of Chronic Obstructive Pulmonary Disease and a farmer from your community has been on admission in the same ward for treatment of the same disease. The doctor in charge prescribed intramuscular Hydrocortisone injection 200mg start in case any of the patients develops difficulty in respiration. At a time, the two patients developed difficulty in respiration simultaneously. If this problem occurred at 2am and only one vial of injection hydrocortisone is available in the pharmacy stock, considering the principle of justice, beneficence and non-maleficence, to whom will you give the only injection Hydrocortisone knowing that the two lives are jeopardized simultaneously? This may place the nurse in an "ethical dilemma".

Veracity

Every individual, sick or well, desires to be told the truth in all ramifications irrespective of the situation on ground. Veracity means truthfulness, neither lying nor deceiving others (Daniels 2004). It is the duty owed by the nurse to her clients to tell the truth. Non-disclosure or partial disclosure of information and intentional lying are often adjudged to be good means of managing some dilemmas but this by no means justifies the nurse to lie to her clients. In telling the truth, **how much truth to tell** poses a problem thereby making the achievement of veracity very difficult. No matter how difficult it may seem to achieve veracity in difficult situations, the nurse must ensure that the best is done for the benefit of the client.

In certain decision-making issues like obtaining informed consent before surgical or invasive procedures and in the use of human beings for the conduct of research or drug trial, veracity must be implemented to the core. Where telling the patient the truth will lead to client's refusal to accept procedure or management, let the client withdraw but let the truth be told.

Case II

A donor of kidney to chronic renal failure patient comes to your hospital ready to donate his kidney to a recipient who has few hours to live if kidney is not transplanted. If partial insanity will occur as side effect of the drugs that will be given to the donor, will you tell the donor the whole truth considering the state

health of recipient and bearing in mind that the donor may decline if he knows the possible negative effect he will suffer after donating the kidney?

Confidentiality

Confidentiality is duty to respect privileged information in the discharge of professional duty among the nurses. This principle goes beyond the health care arena in its implementation. It is expedient that nurses maintain the privacy of the clients under their care.

Nurses by this principle should not discuss the health conditions and other information regarding their client, to their knowledge, with anyone who is not directly involved in such client's care. Principle of confidentiality guides the nurse in encouraging the client to share information about illness eg HIV/AIDS, TB, prosy etc. Confidentiality is a basic ethical principle that ensures client's privacy.

Principle of confidentiality does not permit a nurse to discuss the health problems of a client with a fellow nurse, in the same hospital who is not part of the care team or works on the same ward. Discussing the health problems or letting out information about a client must be done with high sense of judgment and should be done on the ground that the person to whom the information is relayed to will make positive contribution(s) to the client's healthcare (Daniels, 2004).

Case III

You boarded a public bus to a place with two nurses in the same bus who work in the hospital. If these two nurses open up discussion as colleagues in the bus to their consumption and it happens to be about a patient with Diabetic foot ulcer, or worse stills, a patient with HIV/AIDS who has been on admission in their hospital, how will you feel as a nurse in the same bus hearing them like other passengers? In no account should patient's health status be disclosed to another person without permission from the patient. When this is contravened, the patient may go for redress at the jury.

A dilemma may arise when a client chooses to keep confidential health information that places other people at risk. For example a HIV positive individual may decide to hide the information from his family members or spouse. This poses problem to the family and the society because the non-disclosure may lead to the patient infecting other people.

Case IV

You witnessed when hysterectomy was performed on a particular lady in your hospital due to infection following criminal abortion. This lady attends the same church with you and presently a brother is asking for her hand in marriage. This lady did not disclose this to the brother. The brother approached you as a trusted church member to tell you his intention to marry the girl and concluded by telling you that they will be joined together as husband and wife in a week time. Your

further discussion with the brother made you to know that the brother is the only child of his parents. What will you do? Veracity guides the nurse in case III and confidentiality in IV but do good and not harm.

The concept of confidentiality should not be toyed with irrespective of whom seeking for the information. This is evidence on the common inscription on client folders "confidential". Confidentiality in nursing goes beyond clients care. It is obtainable in the use of humans for the conduct of research and drug trial.

Fidelity

This is principle that denotes faithfulness and keeping promise. Fidelity constitutes ethical foundation of nurse-client relationship. The principle of fidelity requires that nurses keep promises made to their clients no matter how insignificant the promise may seem to be. Conflict between fidelity and justice raises ethical dilemma. It may be seen as deceit for a nurse to promise a patient that she will be coming on next shift to attend to his needs or answer some of the questions posed to her by the patient when the nurse knows that she will not report on duty at the time in question.

Case V

A cancer patient under severe pain requested that you give him the opioid analgesic which was formerly prescribed by his doctor to relieve his pain, but the doctor has warned that the drug should be discontinued due to serious emergent side effects of the drug on the patient. In your attempt to pacify the patient, you told him that you were coming to give him the drug. The patient kept waiting for you to bring the drug but you handed over to the nurses on next shift and sneaked out of the ward through unusual exit. Can this action by the nurse be justified?

Creating good therapeutic relationship with the clients assists in the management of ethical dilemmas that may arise in the application of ethical principles in client care.

Case VI

As a community health nurse in charge of a Community Health Post, you confidentially engaged into a discussion with a 14-year old teenage girl whom you suspected to be pregnant. Your suspicion turned to be a reality after simple pregnancy investigation. This teenager requested that you tell no body about including her parents. She confided in you that she was going to abort the fetus. You promised her to keep your discussion with her undisclosed.

Considering the age of this girl and possible complications that may accompany her planned criminal abortion, what will you do remembering that you should keep to the principle of confidentiality even as you maintain veracity and fidelity?

Ethical Dilemma In Nursing Practice And Informed Consent

ETHICAL DILEMMAS

It is wish to repeat that every female working in healthcare institution, dressed on white with or without cap, is not a nurse. Not every nurse is a nurse in all ramifications. Professional nurses are known for their ability to apply critical thinking in decision-making or judgment about their clients' care. In event of conflict of principles isolated in duty of care, ethical dilemmas arise which calls for the action of the "real" nurse who is constantly observed by her numerous "publics" as she does her professional jobs.

From the time knowledge spurred in nursing and its attendant professionalism in the 19th century, which was authored by the changing societal needs and dynamism in health profession, nursing ceased to be mere "obedient" to physicians' orders but requires critical thinking and application of knowledge of care to achieve better client outcomes. Critical thinking ability therefore remains the nurses' anchor in every condition that is liable to different individual's criticism.

Ethical dilemma often occurs when a critically thinking nurse keeps to the professional code of ethics in her day to day nursing actions. Dilemma is a battle of the mind between two moral principles. It is a situation that may involve apparent mental conflict between moral issues whose resolution may be viewed as morally unjust by so many. Resolution of ethical dilemma is based on ones values and what one believes.

Ethical dilemmas may arise in issues like;

- ☐ Decisions for abortion.
- ☐ Continuity of care when client rejects best available management modalities.
- ☐ Decision of client to maintain confidentiality of his disease even when it poses danger to so many others eg HIV/AIDS, TB, etc.

In event of ethical dilemma, morally accredited nurse must remember that the client's right, integrity and confidentiality should not be tempered with. This calls for the nurse's systematic approach to the existing problem considering all options for solving the dilemma bearing in mind to do good to the patient with regard to the persons involved in the patient's management. The nurse will often be caught in a dilemma involving what you ought to do (on the

basis of one ethical principle) and what you ought not to do (on the basis of another principle) (Anarado, 2002).

Accepting that there is no best answer to ethical dilemmas does not rule out the assertion that the nurse knows the best for her clients. In the opinion of Daniels (2004), ethical dilemma requires the nurse to do the following:

- **Assessment:** this involves critical study of the situation on ground. Understand the position of the patient, the relation and possibly that of other healthcare providers.
- **Analysis and Diagnosis:** having studied the situation, analyze the entire case at hand and identify the necessary outcomes (problems/prospects). This is a time of full understanding of the ethical dilemma involved.
- **Planning:** consider primitives of action from alternatives for resolving the dilemma noting consequences of each action bearing in mind to do the best for the client.
- **Implementation:** carry out selected moral action which may be criticized or rejected by many others including your colleagues.
- **Evaluation:** Access the outcome of the moral outcome taken. Determine the consequences and make your conclusions.

Case VI: - Extracted from Daniels (2004)

A client with chronic cancer pain has some strong analgesics which were prescribed by the doctor. He hid the drugs in his personal bag and takes them secretly any time the pain is very severe on him. One day you engaged yourself into therapeutic discussions with him and he revealed this act to you but pleaded that you should not tell his doctor. Before this time, his doctor has warned that the patient may develop severe liver problem due to the strong analgesic he has been taking and has reduced the dosage but since the reduction of the dosage the patient kept complaining of severe pain despite the administration of the medications. It was surprising to the nurses that the patient suddenly stopped complaining of severe pain until you learnt his secret act.

What will you do now that the patient has told you his secret bearing in mind that reporting him to the doctor will mean collecting the drug from him and he will come down with severe pains again and eventually lose confidence in you? Allow him to continue with the drug means encouraging him on self-medication and possibly engineering liver damage. If you decide to pretend as if you do not know about this as means of helping the patient, and another nurse or his doctor find out and the patient says that he told you about it earlier how will you feel?

INFORMED CONSENT

Ethical principle of respect demands that patients or subjects used for research study or drug trial should be given the opportunity to choose what shall or shall not

be done to them. Informed consent is the act of obtaining formal permission from the client to perform certain kinds of procedures/actions on him/her. Such procedures include invasive procedures like spinal tap (lumbar puncture) or surgeries.

Informed consent is a written document signed by the client or his/her relation prior to invasive procedure or surgical manipulation. For a client to give his/her informed consent means that he/she understands the reason(s) for the proposed actions, the benefits and risks that may result from the procedure and agrees to the treatment. Informed consent must be obtained before beginning any invasive procedure such as spinal tap, surgical procedures, administration of experimental drugs and using a client for research study.

To obtain informed consent the individual concerned must have adequate reasoning capability and good knowledge of all relevant facts at the time the consent is granted i.e. client is legally permitted to sign consent form if he/she is mentally sound and not a minor. The individual must be of sound mind. Impairment of reasoning and judgment may make it impossible for someone to give informed consent. For instance individuals with traces of the following are not expected to give informed consent:

- Persons with intellectual and emotional immaturity
- Individuals experiencing high level of stress
- Mental retardation (mild or severe)
- Persons suffering from mental illness
- People that are intoxicated (under the influence of alcohol or drug)
- Persons suffering from Alzheimer's disease
- Persons in coma.

Where an individual is considered unable to give informed consent, another person is generally authorized to give consent on his/her behalf. Parents, legal guardian of a child, care givers for the mentally-ill are permitted to give informed consent on behalf of their unable relations. In some circumstances instead of parents of a minor or legal guardian giving the informed consent, the child may be required to give informed assent. If for any reason a client refuses to sign the consent form but permits the healthcare provider to perform the procedure, this should be properly documented before the procedure commences.

Informed consent should meet the following criteria (Daniels, 2004)

- A. Consent document must be written in the language the client or guardian can understand.
- B. The document must delineate all possible risks and actions of the healthcare provider or researcher to minimize the risks.
- C. The consent document must list the benefits of the procedure to the client. If there is no known benefit at present, the consent document must make it known to the client.

- D. Any alternative to the procedure must be specified even if the only alternative is non-participation.
- E. The document must state that participation is voluntary and that the client can refuse to participate or withdraw from participation without having further health care withdrawn.
- F. A client who gives informed consent must be rational and must be told how to reach the healthcare provider or researcher performing the actions.

A TYPICAL INFORMED CONSENT FORM

Date _____

Name of Hospital _____
 Consent for medical procedure and
 Acknowledgment of receipt of
 information

In keeping with the Nigerian law, you are being asked to sign a confirmation or advisable that we have discussed your contemplated operation or medical procedure. We have already discussed with you the common problems or risks. We wish to inform you as completely as possible. Please read the form carefully. Ask about anything that you do not understand and we will be pleased to explain it.

1. I have authorized and directed Dr. _____ with associates or assistants of his choice to perform upon _____ the following surgical, diagnostic, or medical procedure _____ including any necessary or advisable anaesthesia.
2. In general terms, the nature and purpose of the operation or medical procedure _____
3. This procedure has been explained to me. Alternative methods have also been explained to me with the advantages and disadvantages. I am advised that though good results are expected, and the probability and nature of complications cannot be accurately anticipated and that therefore there can be no guarantee as expressed or implied either as to the result of the surgery or as to cure. The possible risks include death, brain damage, quadriplegia, paraplegia loss of organ, loss of arm or leg or disfiguring scars.
4. I authorize the administration of a blood transfusion and such additional transfusion as may be deemed advisable in judgment of the attending physician or his associates or assistants. It has been fully explained that blood transfusions are not always successful in producing a desirable result and that there is a possibility of ill effect, such as the transfusion of infectious hepatitis, or other diseases or blood impairment. It has also been explained that emergency may arise when it may not be possible to make adequate cross-matching tests and that immediate need may make it necessary to use existing stock of blood which may not include compatibility blood types.
5. I further authorize the doctors to perform any other procedure that in their judgment is desirable for my well being. I hereby authorize and direct the above named physician and associates or assistants to provide such additional services as they may deem reasonable and necessary including, but not limited to, the administration of any anaesthetic agent, or services of x-ray or laboratories and I hereby consent thereto.
6. I hereby state that I have read and understood this consent, all questions about the procedure or procedures have been answered in a satisfactory manner and that all blanks were filled to my signature.

Signature _____
 (Patient/Authorized person)

Witness _____

I certify that all blanks in this form were filled in prior to signature and that I explained them to the patient or representative before requesting the patient or representative to sign.

Signature _____
 (Above named physician to sign)

Adapted from Daniels 2004.

EXCEPTIONS IN THE RULE OF INFORMED CONSENT

The rule guiding informed consent is made flexible in the following situations:

Emergency: In emergency situations, the healthcare providers are believed to have sufficient knowledge, skill and were trained to do the best for the good of the client. In view of this, where it becomes impossible for informed consent to be obtained from the client probably due to the client's condition the healthcare provider can act with his/her best judgment for the client's good.

In minor (child): The consent of a minor is viewed as sufficient to carry out certain procedures on her eg treatment of sexually transmitted infections. This is because the health of the minor is at risk. Remember that this has nothing to do with the injunction that entering into contract with a minor with exception of contract for supply of necessity is wrong.

In cases of court order or legal authorization: If the client is a minor and the parent or legal guardian denies him/her treatment for life sustenance, the court may overrule the decision of the parent or legal guardian. With this, the healthcare provider disregards the parents/legal guardian and can go ahead to treat the minor.

Nurse should bear the following in mind about informed consent.

1. It is not the duty of the nurse to obtain informed consent from the client or be delegated to do so by the doctor in surgical or invasive procedures except where the procedure will be carried out by the nurse. This is because informed consent requires that the client be given clear and detailed information surrounding the procedure in question with likely positive and negative outcomes which the nurse may not proficiently know. The nurse is only expected to act as client advocate ensuring that the client signs the form while she witnesses the signing of the document.
2. On no account should a nurse assist a doctor/surgeon in any surgical/invasive procedure without ensuring that the client's informed consent has been obtained. In event of negative outcome that leads to litigation, the nurse must be included in the litigation suit and the jury will penalize the nurse for not reminding the doctor to do the necessary things expected of a reasonable prudent practitioner.
3. The nurse in her assertive nature as client's advocate is in the right position to notify the doctor when condition that may render signing of consent form invalid surfaces. Such conditions may include where client's condition may be threatened by delay else the formal practice must be maintained.

Legal Liability In Nursing

An obligation one has incurred or might incur through any act or failure to act is referred to as liability. When a nurse fails to perform the duties expected of her or performs the duties but short of standard expected of her, or in the course of performing her duty injures or harms her client, the client can initiate act of harm or injury against the nurse.

The nurse is therefore expected, in her discharge of duty, to demonstrate skill and actions expected of a reasonable nurse of equal standing. A Registered nurse is expected to professionally act in the same way another nurse of equal training will act in the performance of professional duties and nothing less than that. It is a clear statement that there is no excuse to be accepted for an individual to claim ignorant of existing law and so the common saying "*Ignorantia juris non excusat*". Clear understanding of the following legal issues in the professional practice arena is essential in nursing.

Negligence of Duty

Negligence is failure to act in a reasonable and prudent manner expected of a professional of equal training. It is intentional tort where by the nurse fails to meet the appropriate standard considering her level of training. Accordingly, the claim of negligence is based on the principle that everyone is expected to conduct himself/herself in a reasonable and prudent manner.

A nurse may be found to be negligent when carelessness undermines action such as

1. Wrong administration of medications.
2. Not observing clients vital signs.
3. Wrong application or utilization of techniques during procedures which may become harmful to the client.
4. Injury to the client due to unguided actions or non-close observation of clients which may result to accidental fall.
5. Errors in counting materials and instruments used in surgical procedures.

At any time a nurse, at her professional duty post, fails to implement skills that a reasonable prudent and careful person would use under similar circumstances, negligence is said to have been committed by the nurse. When such negligence results to injury it is known as malpractice (Daniels, 2004).

Malpractice

Malpractice is a situation where a professional person's wrong conducts improper discharge of professional duties, or failure to meet the standard of acceptable care that results in harm to another person (Daniels, 2004).

ESSENTIAL ELEMENTS TO PROVE NEGLIGENCE OR MALPRACTICE

To conclude that the nurse's action was negligence or malpractice the following must be established (Daniels, 2004):

1. **Duty:** It must be established that the nurse owed duty of care to the client based on the established nurse-patient relationship. It should be noted that the nurse forms implied contract with her client but this notwithstanding, there is need to prove that the nurse owes duty of care to the client who claims negligence of care by the nurse.
2. **Breach of duty:** There must be evidence that the nurse breached her duty to the client. In essence, it must be established that the nurse failed to care as she would have done.
3. **Injury:** It must be established that the client was injured, suffered harm or damage in the course of nurse's action.
4. **Causation:** It must be proved that the breach of duty legally caused the injury i.e the breach to standard of care lead to the damage or injury suffered by the client.

Negligence and the doctrine of Res Ipsa Loquitor

In Negligence suits, the jury may lack the scientific and technological knowledge to establish or determine the standard of care but the negligent act may lie within the Jury's common knowledge and experience (Daniels, 2004). In essence the professional standard needed to prove that there was negligent act is lacking since members of the jury do not have the professional competency to prove this. This notwithstanding, evidence before the jury remains the basis for determining that negligence occurred during the client's care. In this situation, the doctrine of Res Ipsa Loquitor will be applied by the Jury. This doctrine simply states that the "case speaks for itself".

For instance, a situation where an instrument such as mosquito artery forceps was forgotten in the abdominal cavity of a woman who went for abdominal surgery in hospital 'A' was discovered after some weeks or months and removed through another surgery in hospital 'B', the jury does not require any special or professional knowledge to prove that there was malpractice by the surgeon in hospital 'A' because the evidence is there to prove that there was malpractice/negligence.

Note that the nurses that assisted the surgeon in hospital 'A' during the surgery will be included in the suit. This is because the nurses failed to maintain a proper

count of the instruments to ensure that none was forgotten in the woman's abdominal cavity before the surgery was over.

Avoidance of negligence and malpractice suits requires that the nurse makes her stand very clear in all procedures ensuring that nothing is short of standard of practice irrespective of the fact that she is the primary person in charge of the procedure or simply assisting the doctor in the procedure.

Exception in the application on doctrine of Res Ipsa Loquitur exists where, there is "Novus actus interveniens volenti non fit injuria" or Contributory negligence (ie to which a man has consented to, he should not complain). This explains a situation where negligence occurs but the client contributed to the cause of the negligence. Take for instance, where a client was instructed by the nurse to lie on bed for thirty minutes after hemodialysis to minimize the risk of disequilibrium syndrome. If the client refuses to heed to the nurse's instruction and gets up from the bed without the knowledge of the nurse and in the process falls down and injures himself the nurse will be exonerated from negligence because the client contributed to his problem.

Negligence and Emergency Situation

Emergency is an unexpected and sudden event that must be dealt with urgently. Emergency is often a dangerous state where decision has to be taken quickly. Mistakes or act of negligence may not be totally avoided in some emergency situations. In view of this, in emergency situations, certain acts of negligence may be exonerated but this depends on the Jury handling the case and the after effect of the negligence.

If situation came up where a lady was brought to your hospital with perforated appendicitis and emergency appendectomy was carried out to save the life of the lady. If the nurse, in an attempt to save life, failed to scrub the lady's abdomen with iodine or methylated spirit as required by routine before the surgery was performed and the suture later resulted in skin infection which healed with scar tissue formation. If the lady sues the doctor and the nurse, the jury may argue that to save the life of the lady the doctor and the nurse acted reasonably. The decision of the jury may be viewed as reasonably good because the nurse and the doctor acted to save life but there must be witness to prove that the lady was brought in a critical condition and going through standard procedure. If preparing the lady may delay the intervention and lead to loss of life.

Criminal Negligence

A negligence where the act constitutes crime is where the unintentional act results to serious harm or death of the client. For instance where a nurse gives a child dosage of drug that is meant for an adult and the child

dies, a criminal negligency may be established.

Criminal charges against a nurse may lead to the nurse losing her job or disciplinary action by the state Licensing Board (in Nigeria Nursing and Midwifery Council of Nigeria). The disciplinary action may include the licensing Board revoking the nurse's license at the end of the trial, be suspended from practice, or made to pay penalty due to violation on non-standard practice.

From the example cited above, if there was prescription error by the medical doctor which led to the administration of the over dosage of the drug, the nurse will still be held responsible because she is expected to clarify any suspected prescription error before drug administration.

Measures to Prevent Negligence and Malpractice

- ☐ Adherence to standard of care at professional practice arena.
- ☐ Deliver competent health care to the clients at all times.
- ☐ Develop competent rapport with the clients.
- ☐ Be careful and objective in your documentations.
- ☐ Adhere to the institutional policies, procedure and guidelines (Daniels, 2004).

Tips On Nursing Process And Care Plan

NURSING PROCESS

Nursing process was born out of challenges faced by nursing in its professional development. It spanned from research into the professional practice as means of solving client, family and community health problems. Before the emergence of nursing process, nurses' participation in clients' assessment was relegated to the background as they were often made to observe clients and carry out doctors' instructions and hold their instruments or examination tools.

The expansion of nurses' role and quest for professionalism attainment could not spare nursing to remain behind the scene in client care arena. To successfully improve in the health care delivery with accountability, nursing process remains indispensable tool for every nurse all over the globe. In the opinion of Chitty (2005: 390), Nursing Process is a method of thinking critically about how to solve patient's problems in professional practice. She saw it as an outgrowth of scientific method and can be used as a framework for approaching almost any health related problem. Nursing Process tends to act as a tool used in exploring the individual client's problems as well as dealing with the identified problems.

Yura & Walsh (1983), described Nursing Process as a designated series of actions intended to fulfill the purposes of nursing to maintain the patient's wellness and, if his state changes, to provide the amount and quality of nursing care the situation demands to direct the patient back to wellness. They further added that "if wellness cannot be achieved then the purpose of nursing process is to contribute to the patient's quality of life, maximizing his resources as long as life is a reality". Nursing process as an organized approach to the identification of patient's nursing care problems and the utilization of nursing actions that effectively alleviate, minimize or prevent the problems being presented or from developing (Judith & Karen, 2006).

The nursing process is the framework for providing professional quality nursing care. It directs nursing activities for health promotion, health protection, and disease prevention and is used by nurses in every practice setting and specialty. Nursing process therefore is a tactical approach to client care which requires detailed collection of health data of the client, making clinical judgement about the client with the data and reviewing the outcomes against the judgement. It is a step by step method of rendering

individualized care to clients, family or the community beginning with assessment, and ending with evaluation which is specific to nursing profession. Nursing Process is not a book-oriented content but an approach to client needs that requires critical thinking approach to alleviate client's pains and suffering or attend to health needs of the individual, family and community. It requires critical thinking which is a cognitive process during which an individual reviews data and considers potential explanations and outcomes before forming an opinion or making a decision. It requires continual question of belief.

Critical thinking in nursing practice is a discipline specific reflective reasoning process that guides the nurse in generating, implementing and evaluating approaches for dealing with client care and professional concerns (Judith & Karen, 2006). Critical thinking requires that the nurse uses a systematic and logical process to address the challenges the nurse faces during patient care delivery.

Nursing Process and critical thinking are connected to one another or better put "inseparable". Nursing practice through nursing process cannot yield expected outcome without critical thinking in the process which is expected to involve Evidence Based Practice.

The Nurse, Nursing Process and Critical thinking

A nurse is an artful and scientifically-based certified person that assists an individual (sick or well) in the performance of those activities that lead to health or its recovery (or to a peaceful death), which the individual would have performed unaided if he has the necessary will, knowledge or strength. A nurse acquires knowledge and skill to carry out these functions through change in behavior gained from experience which is acquired from designated academic institutions. A nurse is therefore a carrier of knowledge of actions that could lead to health promotion, prevention and capacity to cure. In order-words, a nurse is a critically thinking being that has the welfare of others at heart. For a nurse to actualize her goal of care the application of nursing process remains indispensable and the major tool for the nurse. Nursing process enables the nurse to establish a goal-directed, individualized care for each client, family or community which is based on scientifically acceptable approach.

Critical thinking tends to look at what informs the nurses' decisions.

- How do nurses make decision about patient care?
- What do nurses rely on to help them in decision making?
- It is not a linear step by step process.
- It is a process acquired through hard work, commitment and an active curiosity toward learning.
- Decision making is the skill that separates the professional nurse from technical or auxiliary nurses.

- ☐ Not always a clear textbook answer.
- ☐ Nurse must learn to question, look at alternatives.
- ☐ Reflect on past experiences and previous knowledge.
- ☐ Listen to others point of view.
- ☐ Select best solution for improving client's health.

National group for the Classification of Nursing Diagnosis, now called North American Nursing Diagnosis Association International (NANDA -International) published its first list of Nursing Diagnoses in 1973 (ICN, 2001). The group (NANDA-I) identifies terminology and definitions that may be used as nursing diagnoses.

Nursing process has five phases or steps but in recent time, Nursing Process has shifted from its original five (5) phases to six (6) phases with additional Phase before Planning referred to as **Outcome Identification**. This is having the patient's expected response in view before planning and implementation of care plan. The outcome identification was an integral part of planning before the separation and many nurses are yet to see the need for this separation. This, in way, gives the nurse a direction on what responses are expected from the patient/client as a result of the interventions rendered.

In view of the above the steps goes thus: Assessment, diagnosis, outcome identification, planning, implementation and evaluation (ADOPIE instead of ADPIE)

Phase I

Assessment: This is the first phase in the nursing process. It is a time of data collection or information gathering about the individual patient, family or community. Data gathering include physiological, socio-cultural and environmental information about the patient, family or community. Nurses obtain two types of data about their patients: subjective and objective data.

Subjective data: Subjective data are primary data obtained from the patient. They are the symptoms felt by the patient due to the disease process. This may include patient's expression of pain, complaints of weakness, etc.

Objective data: These are data obtained by the nurse through observation and measures; physical examination, information from other health personnel such as doctor's clerking sheet or laboratory and other investigation reports including x-ray, ultrasound scan, etc.

Data obtained from the client/patient is also referred to as primary source of data. The data obtained through physical examination, family members and significant others are referred to as secondary sources of data while the data obtained through other health care providers, and medical records are referred to as tertiary sources, of data. In a nut shell, during the assessment phase, data are

obtained from primary, secondary and tertiary sources. The primary data are subjective while the secondary and tertiary data are objective.

Data collected from the patient are sorted or organized depending on the nurse's personal preference and the institutional method.

The following are very essential tools that the nurse uses in organizing the data.

- ☐ Abdella's 21 nursing problems,
- ☐ Henderson's 14 nursing problems,
- ☐ Yura and Walsh's human needs approach
- ☐ Gordon's functional health patterns.

Theories and frameworks that may also be useful in the organization of data include Pepleau theory of psychodynamism, Sister Calista Roy's theory, Dorothy Orem's theory, Florence Nightingale's Environmental theory, etc. The method the nurse chooses to organize the data should not undermine the importance of confidentiality of the collected data.

Patient can readily tell the nurse more about himself/herself, if only he develops "trust" and "intimacy" on the ground that nurse is essentially good to confide on. If the nurse applies the ethical principles of confidentiality very well more information will be got from the patient to aid his clinical management.

Phase II

Nursing Diagnosis: Nursing diagnosis is otherwise called problem identification. The data generated during the assessment must be clustered or grouped to enable identification of specific client problems(s). The client/patient is focused on to aid good identification of problem(s) for effective, individualized nursing care. Knowledge from biological sciences, social sciences and nursing enables nurses to analyze relationships among the various pieces of patient's data (Chitty, 2005:396).

Models and theories applied in Nursing will be of great important to the nurse in problem identification and prioritizing patient's problems/needs. For example, Abraham Maslow's Hierarchy of needs places *physiological needs* before *safety and security needs* which comes before *love and social belonging needs*; then *self esteem* before *self actualization*. A nurse has to in effect, think about patient's oxygenation, body temperature, elimination and nutrition before social deviation, and role reversal. Making nursing diagnosis therefore requires the application of critical thinking and best judgement.

In the assertion of Gordon (1976; 1299), Nursing Diagnosis is the actual or potential health problems which nurses, by virtue of their education and experience, are capable and licensed to treat. North America Nursing Diagnostic Association (NANDA, 1990), defined Nursing Diagnosis as a clinical judgment about individual, family or community responses to actual or potential health problems/life process which provides the basis for selection of nursing

interventions to achieve outcomes for which the nurse is accountable. Nursing Diagnoses are reviewed by NANDA-I from time to time and the nurse should be conversant with changes made for efficient utilization of the same in client care. NANDA-I approved nursing diagnoses consists of five components as enumerated below.

- **Label:** concise term or phrase that names the diagnosis.
- **Definition:** clearly delineates meaning and helps differentiate from similar diagnoses.
- **Defining characteristics:** clusters of observable cues or inferences (signs or symptoms).
- **Risk factors:** Factors that increase vulnerability to an unhealthy event.
- **Related factors:** Factors that proceed, are associated with or related to the diagnosis (NANDA, 2003 in Chitty, 2005; 397).

Accurate diagnosis is very important as inaccurate diagnosis can lead to loss of time, wastage of resources and may endanger the patient (Lunney, 2001). Gordon in 1987 developed a format used to write the diagnosis abbreviated as PES (Problem, Etiology, Signs and symptoms).

- P:** *Problem:* represents NANDA-I label. For example Hyperthermia, Ineffective airway clearance, Fluid volume excess, etc.
- E:** *Etiology:* this spells out the cause of the problem which begins with "related to" i.e. the factors that can lead to the problem. From the previous "ineffective airway clearance related to the presence of mucus secretion in the airway", the ineffective airway clearance was due to the presence of mucus secretion in the airway.
- S:** *Signs and symptoms* (defining characteristic): This may be concluded with "as evidenced by" or "as manifested by".

From the previous "ineffective airway clearance related to the presence of mucus secretion in the airway as evidenced by the patient's gasping for air" denotes that the nurse concluded that the patient was experiencing problem with gaseous exchange because she saw the patient gasping for air.

Have a good look at this nursing diagnosis and compare it with the first example. "Hyperthermia related to invasion of micro-organism at the renal pelvis, as evidenced by patient's body temperature reading of 39°C".

Identification of problem should be done in conjunction with the patient and prioritizing the identified problem will take cognizance of patient's socio-cultural background except in life threatening situations.

NB: Nursing diagnosis is different from medical diagnosis in the sense that:

1. Medical diagnosis focuses on the disease process but nursing diagnosis identifies patient's immediate problem which he experiences as a result of the disease process, that is, the patient's response to the illness, injury or threat. For example a patient diagnosed of plasmodiasis may

experiencing hyperthermia which is the concern of the nurse and this forms the nursing diagnosis.

2. Nursing diagnoses are simply the patient's problems which the nurses are legally permitted to manage. For example a patient diagnosed of cancer may be experiencing pain which forms the nursing diagnosis and concern of the nurse.

Knowing that nursing diagnoses differ from medical diagnoses, the nurse should remember that the inclusion of the following following in stating nursing diagnosis of a patient will make a mockery of the diagnosis.

- "due to"
- "altered"-out dated in use
- "as a result of"
- "following"

Identify the errors in the following Nursing diagnoses

- i. Ineffective airway clearance due to presence of mucus in the airway as evidenced by patient's ballooning of cheeks".
- ii. Altered body comfort (pain) due to surgical incision as evidenced by patient's complaints.
- iii. Anxiety following congested airway as evidenced by patient's facial expression.

Phase III

Outcome identification

This phase was initially part of planning. It involves formulation of measurable goals and outcomes which act as standard for evaluating nurse's accomplished purpose. Outcome identifications are prioritized based on the severity of the identified client's problems. The most life-threatening conditions requiring immediate attention are taken care of before other conditions that are less life threatening. For instance, where a client has high fever and at the same time experiencing severe pain, the fever will be seen as more life threatening than the

Criteria for measuring the outcome must be very clear to enable good understanding of when goal was achieved. In essence, a clear statement of expected client's response after nursing actions must be made. For instance, it is expected that after exposing, fanning and tepid-sponging a child with high fever, the body temperature will reduce. So the outcome to measure will be reduced body temperature".

Phase IV:

Planning

Planning is referred to as goal setting. It is the phase when the patient and the

nurse identify actions that will assist in reducing patient's experiences. It entails defining the objective(s) to achieve based on the identified problem (diagnosis). Planning should be patient oriented and thus the use of the phrase "patient will" or "patient will be able to" or "patient will show signs of", etc.

The use of such phrase(s) delineates the fact that the patient plays active role in the entire process of care except where situation does not permit. Such exceptional situation empowers the nurse as patient's advocate, to make decisions for the patient using the best of her knowledge and understanding of the identified problem(s).

The steps enumerated below will guide the nurse in planning.

- i. Set priority among the identified problems.
- ii. Establish the outcome criteria
- iii. Identify intervention strategies
- iv. Examine alternative actions and result(s) that may likely occur.
- v. Determine nursing actions necessary to manage the problem.

In writing the Nursing Care Plan (to be discussed later), every plan must have measures (not less than three) to achieve it. A good plan for a patient with Hyperthermia may read, "Patient's body temperature will reduce by 0.5 within 4 minutes of nursing actions. Planning can be short term or long term depending on the client's experiences. A patient with chronic pain due to cancer should have a long term plan for pain reduction while a patient with pain following surgical incision should have short term planning/goal.

Nursing Care Plan

Nursing Care Plan is drawn for individual client at the planning level. It is a purposeful client centered approach to identification of problem(s), setting goals, taking necessary actions to achieve the set goals and evaluating the actions against the set goals. Comprehensive Nursing Care Plan provides an avenue for reaching expected outcome on identified client, family or community health problem(s). It is a set of actions the nurse will implement to solve nursing diagnoses identified through nursing assessment.

Characteristics of Nursing Care Plan

1. It is holistic in focus, and is based on the clinical judgment of the nurse using assessment data collected from a nursing framework.
2. It is based upon identifiable nursing diagnoses, (actual, risk or health promotion)
- It reflects clinical judgments about individual family or community experiences/responses to actual or potential health problem/life processes.

3. It focuses on client-specific nursing outcomes that are realistic for the care recipient.
4. It includes nursing interventions which are focused on etiologic or risk factors of the identified nursing diagnoses.
5. It is a product of a deliberate systematic process.
6. It relates to the future.

Elements of the Nursing Care Plan

- It consists of nursing diagnoses
- Defining characteristic (subjective and objective data that support the diagnosis),
- related factors or risk factors,
- Expected outcome/goals and nursing interventions.

Phase V

Implementation:

This is a time of performance of nursing actions or orders to assist the patient gain relief from pain and the experiences resulting from the disease process. It is also aimed at preventing complications or achieving a health status that will enable the patient to live a normal or near normal life.

Implementation of care plan (intervention) depends widely on the diagnosis and the set goal (plan). It involves the application of the art and science implicit in the definition of nursing through critical thinking and application of best judgment to do the best for the patient. It may involve teaching, counselling, delegation of duty to other nurses and proper documentation of actions performed. Implementation of planned care, in a nutshell, is the application of different measures to take care of the patient's experiences in the disease process which were identified as problems.

For instance, a patient with hyperthermia may require measures like-exposure, fanning, tepid sponging and administration of prescribed anti-pyretics to enable reduction in the body temperature. These are actions that will help to solve the identified patient's problem (hyperthermia).

NB: Where collaborative nursing diagnosis is made, it requires collaborative planning of care and collaborative implementation of care plan. For instance, a patient that is suffering from post-partum haemorrhage may require collaborative actions of different health professionals working with the nurse to save the patient's life.

Phase VI

Evaluation:

Evaluation is a systematic determination of a subject's merit worth and significant, using criteria governed by a set of standards. Evaluation gives room

to reflect on the actions/measures taken and assists in the identification of future changes (Rossi, Lipsey & Freeman, 2004). The progress of the patient is monitored continually against standard and the set goal. A simple question that will explain this is "was the set goal achieved?"

Evaluation can be formative or summative evaluation when the patient care is going on and evaluation when the nursing actions/orders have been completed respectively. Evaluation is also carried out at any time the patient's condition changes.

Evaluation, as the 5th and last phase in the nursing process helps to determine whether the patient will be reassessed for change in measures implemented in the care or to conclude the care/management. It is a critical phase because once the patient's problems remain, it means that the plan of care will also be revisited or there is a problem that is yet to be identified which calls for re-assessment.

Looking at a patient with "hyperthermia" again, the evaluation report may read "patient's body temperature reduced by 0.5°C after 30 minutes of nursing actions." It is worth mentioning that evaluation must be carried out before the expected time of positive result.

NB: At the planning phase, the goal of the nurse was that patient's body temperature will reduce by 0.5°C within 45 minutes of nursing actions; the evaluation phase will no longer have the word "within" but "after", and time has to be a time before the stipulated time of goal achievement. For instance if the goal said "within 45 minutes" the evaluation will read after 30 minutes or thereafter and not after 45 minutes.

NURSING CARE PLAN

Planning of care for patients requires that the nurse writes care plan for each patient according to the identified patient's problems. Nursing care plan is individualized and the development of appropriate care plan lies on the nurse's ability to think critically. The science in nursing plays a great role at this point in time as the consideration of the holistic nature of each client is paralleled with flexibility, logic and creative ability of the nurse for better outcome at possible shortest time frame.

Nursing care plan is drawn in tabular form under the following sub-headings:

1. Nursing Diagnosis
2. Nursing Objective or goal
3. Nursing orders/Action/Interventions
4. Rationale/Scientific principle
5. Evaluation

Nursing Diagnosis:

This is the identified client problem which evolved from assessment comprising

of history taken and observation. The information generated from primary and secondary data along with the client's cues are clustered to formulate nursing diagnosis. There is no limit to the number of problems to be identified at a time but the etiology of the disease process and the clinical manifestations determine the number of problems a patient may present at a time.

Most essential in the identification of patient's problems is to prioritize them according to how life threatening they are to the patient. Nursing diagnosis is simply referred to as identified patient's problem. Identified nursing problems are not static but fluctuate according to the individual's response to management. In view of this, it is essential to note that a patient presenting with hyperthermia may after few minutes have ineffective airway clearance as a priority problem. In stating the patient's problem, the nurse looks at the PES as guide. A patient with respiratory disease such as asthmatic attack may present with difficulty in gaseous exchange which can be stated like this; "Ineffective airway clearance related to mucus secretion as evidenced by the use of accessory muscles of respiration on inhalation."

A patient with CVA may also present with ineffective airway clearance which can be written thus; "Ineffective airway clearance related to presence of mucous in the airway as evidenced by patient's ballooning of cheeks during exhalation"

Nursing Objective:

This is setting a goal on the identified problem. It is simply, stating expected outcome from actions to be taken on the patient before the commencement of the actions. Each problem (diagnosis) is attended to with the aim of achieving positive outcomes. Setting objective requires that the nurse puts the patient into consideration and works with him or the relations where the patient's condition may not lean him to full participation in his care.

In view of the above, objective should be patient oriented, specific, measurable, achievable, and realistic with time bound. For example, if a patient with pyelonephritis is experiencing pyrexia (high body temperature) of 39.2°C, the nurse identifies this as problem and states as follows: "Hyperthermia related to invasion of microorganism at the renal pelvis as evidenced by thermometric reading of 39.2°C". The objective can read thus: "Patient's body temperature will reduce by 0.5°C within 45 minutes of nursing actions".

Looking at this objective, it is specific, to the desired outcome, it is measurable i.e. if the body temperature drops by 0.5°C, it will be evidenced in the thermometer reading, it can be achieved through different intervention measures and it is realistic. The time bound set to achieve this is within 45 minutes the nurse begins her nursing actions.

Objective should not be ambiguous and does not lean itself to difficult interpretations. It is worth noting that appropriate nurse-patient therapeutic relationship enables the nurse to involve the patient in his care thereby leading to proper formulation of patient oriented objective.

Nursing Order/Action/Intervention

This is action statement to take care of patient's identified needs. They include the different measures utilized to meet the needs of the patient which are stated using action verbs. For instance, a client with pyelonephritis who is experiencing pyrexia as stated above will have the goal of reduction in his body temperature achieved through the following nurses' actions:

- "expose the patient"
- "fan the patient"
- "tepid sponge the patient"
- "administer doctor's prescribed antipyretic e.g paracetamol 1000mg tds"

These actions will help to reduce the patient's body temperature by 0.5°C within 45 minutes.

Action to be taken requires the nurse's understanding of the aetiology of the disease process, the management protocol, the effect of the different actions in reducing the fever and the ability of the nurse to reason critically as to what will be the best for the patient to achieve desired positive outcome.

Wrongly executed actions lead to wastage of time and resources which impinge negatively on the profession.

Note that it is wrong for a student to use personal pronouns like "I will tepid sponge patient". Always remember to use "action verbs" such as "tepid sponge", etc.

Rationale/Scientific Principle

This simply denotes the reason(s) for every action/measure taken to solve the patient's identified problem(s). Any action without a reason should not be initiated from onset. It is the reason for an action that calls for the execution of the action. This implies that outcome can be visualized by the nurse following the understanding he/she has about the disease process and the various patient's present experiences.

For example, the actions taken to ensure reduction in body temperature of the patient with pyelonephritis who is experiencing pyrexia may have the following rationale:

- **Exposure of patient:** Helps to reduce body temperature by convection.
- **Fanning of patient:** Also helps to reduced body temperature by convection.
- **Tepid sponging:** Helps to reduce body temperature by conduction, convection and evaporation.

- **Antipyretics:** Act on the temperature regulating centre in the hypothalamus to reduce body temperature.

Understanding the rationale for each action will enable the nurse to initiate the action. Supposedly, all actions have rationale behind them but all actions may not have scientific principles behind them. To avoid the error of writing "**Scientific Rationale**", students are encouraged to write only "**Rationale**" as this is acceptable.

Evaluation

This is measure of goal/objective against standard or outcome criteria. For instance, the different actions taken to reduced fever on the patient with pyelonephritis are expected to yield positive outcomes. The only way to confirm this is by checking the patient's body temperature using thermometer after a specified period of time, say, 30 minutes. If the body temperatures reduced from 39.2°C to 38.7°C, after a time, it means that the goal was met. In this case the nurse may state a follow: "Patient's body temperature reduced by 0.5°C after 30 minutes of nursing actions as evidenced by thermometric reading of 38.7°C."

Do not make the mistake of writing or saying that "patient's body temperature was reduced by 0.5°C. This is an error and indication that the objective was not patient oriented. A patient oriented objective involves the patient in every process of care ranging from assessment to evaluation.

Note that the time frame in the objective was within 45 minutes, so the evaluation will be done before the elapse of 45 minutes. If the evaluation is done after 45 minutes it gives impression to the assessor that the goal was not met within the time frame.

Evaluation determines the next action plan in the clients care. Where the goal was not met, re-assessment begins with setting of new goals and action plans else the client/patient deteriorates increasing the time of stay on admission and cost of management which lays burden on the patient, friends, family members and the community.

Note: 12 Common errors to avoid while writing care plan

1. Wrong Nursing diagnosis leads to wrong setting of objectives, wrong actions which lead to wastage of resources and long stay of the patient on admission. When a wrong diagnosis is made by a student no mark will be awarded to the student for the diagnosis and every other aspect of the care plan may possibly not be considered for marks because wrong problem was identified.
2. The diagnoses must be stated according to priority i.e according to how life-threatening they are to the patient. For instance a child that is running high body temperature and at the same time experiencing pain should be managed for pyrexia before the pain. But where pain exists with imbalanced nutrition, pain should be taken care of before the imbalanced nutrition.

3. Avoid the use of the word "Normal" in the writing of care plan. For example, do not say "patient's body temperature will come down to normal" or in your evaluation "patient's body temperature reduced to normal".
4. Do not use the phrase "will be allayed" because it makes your objective non-patient oriented. For example if you say "patient's anxiety will be allayed; it means that your patient will not be involved in the process or actions that will help in the reduction of the anxiety. It is better to say "patient will show signs of reduction in anxiety" or "patient will demonstrate signs of reduction in anxiety".
5. Stating of nursing diagnosis in the following form is obsolete, "Altered body comfort (pain):....." or "ineffective air-way clearance (actual) or (potential):....." visit NANDA website to download recent NANDA labels.
6. There should be use of "within" in the objectives and "after" in the evaluation. Students make mistakes like this in evaluation "patient's body temperature reduced by 0.5°C within 30 minutes of nursing actions".
7. Never use "due to" rather, relate the problem to the etiology of the disease. Due to is used in medical diagnosis so do not say "hyperthermia due to presence of micro-organism at the renal pelvis".
8. Do not relate any label to the disease condition. For example "pain (acute) related to disease condition". It is better to relate it to the aetiology (cause) effect (eg. "pain (acute) related to muscle stiffness".
9. Remember that whatsoever you relate the label to, the same will be where you will focus your attention in your actions. For example if a patient with Glomerulonephritis comes down with anxiety as a result of change in the colour of urine from amber to cola-coloured, irrespective of your different actions to reduce his anxiety, once the colour of the urine remains cola-coloured, the anxiety remains with him. It therefore means that your effort above all, will be what actions to take that will lead to change in the colour of his urine from cola-colour to amber.
In the same vein, a patient with disturbed body image related to swollen face remains with the diagnosis until the swollen on the face disappears and gains understanding of the reason(s) for the swollen or can psychologically live with it.
10. It is often better to use the gerund of the action verbs at the Rational column. For instance, in tepid-sponging a patient with hyperthermia, instead of saying "this will"....., "It is better to say "tepid-sponging will....." or "tepid sponging helps to..."
11. It is better for students to avoid exposing their identity in examination conditions. So do not write your name or sign the signature column in your answer script. This may attract disqualification.
12. Right placement of answer scripts for writing of care plan should not be overemphasized. Draw your care-plan with your answer script vertical but avoid writing in the examiners' margins noted "do not write here" or relate instruction.

Achieving Success In Examinations And Attempting Examination Questions

HOW TO APPROACH EXAMINATION QUESTIONS

Education is making investment in you and students who realize this will not spare their time but to aspire for profit of their investment as years go by.

We owe a lot to great thinkers of old and pioneers of different professions including the pioneers of nursing profession. Reflecting on efforts of few of these great achievers like Michael Faraday, Hippocrates, and Florence Nightingale, one wonders how primitive the world would have remained if not for invention of electricity, development of modern medical and nursing practices.

These people and numerous others gave enough time to critical thinking and development of the mind which yielded some of the good things we enjoy in our generation. They were not interested in remuneration or the certificate to acquire but to better the lot of the future generation which turned to reality as we witness in that present. The evolving scientific emancipations may be seen as building on the scientific discoveries of yesteryears. Most of the scientific discoveries were made without intention of profit-making but to satisfy curiosity of the great thinkers and better the lots of humanity.

Education involves teaching/learning activities that assist individuals to develop critical thinking and reasoning ability. It is an avenue to discover the hidden treasures in people and for the development of the mind. Achievement of this is measured through tests and/or examinations.

It is quite a pity that the enrolment into schools in the present generation is no longer enrolment to develop the mind but to get hold of certificate(s). This mind set has ruined the academia to some extent as enrollees are interested in receiving their certificates without trying to know whether the knowledge they came to seek have been acquired or gained. This is a time when students read/study just for sake of passing examination. There is no integration of knowledge to the extent that the trained professionals practice like the auxiliaries and critical reasoning that informs good decision-making in client care is gradually petering out. Nevertheless, time is now to re-think because good grades that can not be equated with proficiency in practice amounts to nothing.

Good approach to academic pursuit is determined by the philosophy one has about his/her life and the academic goal set by the student. A determined student should see academic challenges as stepping stones to the ladder of success and actualization of life's goals. This is my opinion because success is not achieved at platter of gold and everything is possible to a willing mind.

A student that will succeed in examinations must develop the ability to make constructive argument, analyze and critique every academic information from instructors, teachers or focus group. Such a student will not depend solely on information got from others including teachers on subject matter but goes beyond learning from sources that are common to others. For instance a student that will be outstanding in Medical Surgical Nursing will be well versed in Anatomy, Physiology and Biochemistry and will go beyond using one textbook and the teachers lecture note to study the different topics taught by the teacher.

In the opinion of Northedge (2007: 89), studying involves a student having dialogue with his/her chosen subject and then takes in new idea and information and reply by putting the idea and information to use. Continuing, he states that the basic ways of engaging in the dialogue are through reading, listening, speaking and writing. Zeroing to this assertion entails that an academically successful student must be a good reader and reading here can be interpreted to mean study which refers to examining what the student reads in detail. Listening skill developed through learning and practice and a good listener is known to speak little but gives more time to reflect what he/she hears.

Speaking in this context refers to the ability to speak to an audience in polished and articulated manner. A good student chooses his words and is eloquent and can easily extrapolate public speeches. A brilliant student is often measured by his ability to construct meaningful sentences void of grammatical errors.

Suffice me to state that the above make a student that will have good approach to examination. It is a common saying that examination is not a good test of knowledge but we must recall that it is an avenue to evaluate the student and the teacher. In other words, examination is a way of having an idea whether teaching and learning have taken place. So a good student must prepare for examination and plan to successfully pass every course he/she registers with higher grade.

Things to note in achieving examination success

- 1. Start planning how to pass each course from the day you receive the course outline or course content.
- 2. Do not leave any new topic taught by the teacher beyond 24 hours before reading it.
- 3. If possible, read each topic before the teacher teaches it.
- 4. Always jot salient points down as the teacher teaches. This suggests that

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the student must try by all means keep to constant attendance to fixed lectures.

Be apt to ask questions in the class but where this poses problem as a mark of inferiority complex, the student should cultivate the habit of asking his fellow students at the end of the lecture or meet the lecturer/teacher but with no attachment of intimacy.

Pay attention to the teacher's illustrations during the lecture period because they are used to emphasize subject matters in the topic of discussion.

Student should manage his/her time well by preparing personal time table which will put into consideration every activity for each day including rest periods. (See Appendix I)

Develop the habit of carrying out your daily activities according to your time table. Where deviation will occur, let it be minimal.

Ensure that you do every assignment given by the teacher to the class or to your group either as individual or group assignment. Some assignments form part of course assessment and so must be taken seriously.

Engage yourself into group discussion which must not take bulk of your time. A discussion group should not be less than three (3) students but not more than eight (8) students. In my opinion, if it is less than three students, it will be turned into story telling or gossip meetings and if it comprises more than eight students, it may metamorphose into unnecessary argumentative and debate group with many non-active participants.

Take every quiz serious as quiz form part of the assessment for each course. In some schools, quiz/test or assignment constitute 30% of the total course scores and so must be taken seriously.

Give yourself time for recreation. "All work and no play makes Jack a dull boy". Your recreation should be done with minimal mental activities at a time.

When you are too tired to keep studying and you cannot give yourself to sleeping, try to ease yourself by learning how to draw some diagrams from your textbooks or learning pathophysiologies of diseases with flow charts. Always study with your jotter. You can use a very big note book as jotter with a line ruled at the centre of each page to make it different from your note book. This will be very useful to you for revision when the examination draws nearer.

As you study or read each topic, try to formulate questions and answer them.

Read/study difficult topics over and over and look for somebody that will put you through such topics.

Form the habit of going to bed earlier and wake up in the morning to read. It is more benefitting to go to bed by 11pm and wake up by 4am to continue your study undisturbed than studying deep into the night.

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- ❑ Do not use sustenance that enhances activities to keep awake at night or an attempt to study better. Such sustenance's include coffee, cola and some drugs like ICD and Permoline or other activity enhancing substances. These substances can lead to hangover the next day thereby disorganize you for a whole day.
- ❑ Finally, cultivate the habit of reading with concentrating for many hours with your dictionary (English, Medical or Nursing) handy to avoid reception of disjointed information.

On the day of Examination:

- ❑ Arrive at the venue of the examination not later than 30 minutes before being checked in to the examination hall.
- ❑ Stop reading when it is due for you to be checked in to the examination hall. There may be tendency to become confused over what you read at this time.
- ❑ Ensure that you are not deceived to copy formulae or seemingly difficult topics even mnemonics on your laps, palm or piece of paper. This could be very dangerous and can lead to expulsion from the school thereby ruining your academic goal.
- ❑ When seated in the examination hall, do not talk to any student but remain calm and meditative. If you wish, pray at this time.
- ❑ Listen carefully to the invigilators instructions and comply as appropriate.
- ❑ Remember to write your examination number at the space provided for this in your answer booklet.
- ❑ Try as much as you can to read through all questions in the essay question paper before deciding on which ones to attempt or on the one you know best. For instance, if you are required to answer three (3) out of five (5) questions, select the ones that best appeal to you and answer them. But where you are not given such opportunity, start with the number that you feel that is easier for you or you know best before the ones that are more difficult.
- ❑ Do not waste so much time on one question despite how well you can supply the answers. It is better to give equal time to all questions. For instance, if you are given five essay questions to answer in 3 hours it is better to use a maximum of 35 minutes to answer each number. This is not a rule but a guide of the mind remembering that less difficult questions will likely consume lesser time than the difficult ones. It is better to attempt the difficult questions last. It is better to enter the examination hall with your wrist watch to help you check the time flow as you go on.
- ❑ If the instruction reads "attempt all questions," it is expected that the student must attempt all the questions. For instance if a student is required to attempt five essay questions and due to improper time management or lack of knowledge of a whole number (let's assume number 3) the student

fails to attempt all the questions as instructed, that student stands the risk of failing that examination irrespective of his score from the four(4) he/she attempted. This assertion is dependent on the body that is examining the student e.g Nursing & Midwifery Council of Nigeria.

- ❑ Objective questions should be answered according to the instruction. Objective question could be simple select (multiple choices), filling the gap or space, matching of items in two columns or writing the full meaning of abbreviations. The student must answer as specified by the instruction. Some times two or more options, in simple select questions, may look like possible correct options. This does not for any reason mean that the student will choose two options as being correct instead of one. Simply re-read the question with better understanding and select one option.
- ❑ Always maintain good speed but accurate with your points so that you will have a time to go through the answers you supplied before handing your answer booklet to the invigilator or examiner. This helps to identify some mistakes in the answers.

A guide to Oral/Practical Examination

Nursing is art and science and therefore requires critical thinking, proficient approach to procedures and recognition of action to take in the course of carrying out a procedure.

A student is expected to demonstrate proficiency in any procedure she is being examined on. In view of the above the student should have his procedure manual handy throughout the period of training.

- ❑ No procedure is said to have been completed when the instruments/materials used for the procedure have not been discarded or returned to where they are kept.
- ❑ Make sure that you are sure of the procedure your examiner asks you to carry out before going to set the tray or the trolley.
- ❑ Student must appear neat and cheerful before the examiner(s) and the client's.
- ❑ Obtain consent from the patient before setting the tray or trolley for any procedure or better put, inform your patient about a procedure you intend to carry-out on him/her before setting a tray or trolley. If the patient is unconscious, still tell him what you want to do but get the consent from the relations.
- ❑ If the procedure requires an assistant, set the trolley/tray before asking for permission to get an assistant. Student should request for an assistant through the ward head or any person deputizing for the ward head in her absence.
- ❑ Approach your assistant cheerfully and be cordial in requesting for him/her to assist you. Put her in the know of what you would want him/her assist you to do. Use sentence like, "kindly assist me to - ---", "may you assist me in - ---", "please assist me to - ---".

- ❑ Do not assume that your assistant is knowledgeable about the procedure, so there is need to teach your assistant or discuss the procedure with her as two of you get along. This may not be so during wound dressing. Occasion may demand that you simply instruct your assistant eg. During wound dressing you may want your assistant to remove the old dressings, simply instruct him/her on what to do.
- ❑ Remember to maintain the privacy of your patient and allow minimal exposure of the body parts in the course of the procedure.
- ❑ Talk with your patient as you go on with the procedure but always remember the principle of confidentiality and veracity so that you be very careful about what you tell your patient or the promise you make to him/her.
- ❑ When the procedure is over make your patient comfortable and discard used instrument/materials. It is only when used instrument/material are discarded or returned to appropriate place(s) that the procedure is said to have been completed.
- ❑ Ensure that every procedure is recorded in the Nursing chart at the end. This is useful for continuity of care, litigation purposes and Nursing Audit.
- ❑ Sometimes materials and instruments for procedures may not be enough and this will call for improvise of the missing materials or instrument. The student must inform the examiner when such occurs.
- ❑ In case the student forgets to include any instrument/material in the tray or trolley but remembers it when the procedure is about to commence, he/she can take permission from the examiner to collect the forgotten item. The student tray or trolley must be left under the care of a prudent person and the forgotten material or instrument must be brought in a tray and not with bare hand.
- ❑ Show remorse for any error committed in the course of the procedure but try as much as you can to minimize errors.
- ❑ Remember to wash your hands at the beginning and at end of every procedure.
- ❑ Always discuss with your patients in the language or dialect they understand.

APPROACH TO QUESTIONS

The manner of approach to questions is dependent on how well the student knows the subject matter.

Experience gained over years of assessing student's shows that most students approach to examination questions gives idea to the examiner how well the student knows the subject matter.

Writing Pathophysiology of a disease

Pathophysiology is simply a merge of two words "pathology" and "physiology" but is not direct interpretation. It discusses the effect of a disease in a defective

anatomical part. It tells one what happens in the body or a particular structure in a pathologic state. Well written pathophysiology spines from good knowledge of anatomy of the affected part of the body including its function(s) and the clinical manifestations of the disease and possible complications. It goes through definition, causes, clinical manifestations with possible complications. It is not written with subheadings. It is a continuous writing and should not be ambiguous but can be written in paragraphs.

Eg: Brief description of pathophysiology of acute Glomerulonephritis (Bright's disease)

Glomerulonephritis is a non-infectious disease in which one or two kidneys are affected and partially destroyed by an inflammatory process originating as an immune response on the glomeruli of the kidney(s) due to prior infection by Beta-haemolytic streptococcus or upper respiratory tract infection. In event of Beta-haemolytic streptococcal infection, antigen-antibody complexes are formed which are trapped in the glomeruli stimulating inflammatory process which injures the kidneys with complete or incomplete obstruction. This leads to reduced filtration and subsequent oliguria. The rupture of some glomeruli permits the escape of blood cells and casts into the cloudy urine. The porous membrane permits the escape of albumin that appears in urine (albuminuria). As normal excretion of fluid has been affected due to defective glomeruli, there is retention leading to oedema which appears at the periorbital and ankle areas first. The kidney(s) can no longer excrete nitrogenous wastes in the blood leading to the elevations of urea and creatinine in the blood. Leucocytes may escape from the capillaries. This in conjunction with loss of protein leads to reduction in immunity and patient is placed at risk for infection.

Persistent damage to the kidneys and high level of urea and creatinine in the blood will lead to kidney (renal) failure.

A critical look at the pathophysiology will identify the following.

- 1) Definition of Glomerulonephritis
- 2) Cause/Predisposing factors
- 3) Clinical manifestation(s)
- 4) Complication

This still boils down to the fact that the student must have good understanding of the anatomy and physiology of the affected structure to enable him/her understand the dysfunction(s) that may occur with likely complication(s) when the structure is not taken care of in its disease state.

Writing short notes

The approach to short notes is dependent on whether the question is about a disease, procedure or topical issues in nursing eg Millennium Development Goals, Health Sector Reform, etc.

Writing short note on diseases

Steps in writing short note on diseases goes as below.

1. Definition of the disease
2. Causes(s)/ predisposing factors
3. Types/classification incidence (if necessary)
4. Clinical manifestations
5. Management, both medical and Nursing with emphasis on nursing management)
6. Likely complications (if any)
7. Preventive measures (if any)

The total score for this may not be more than 5 marks but that does not mean that the student will simply give information about the disease condition and hope to gain the 5 marks.

Writing short note on procedure

Steps to guide the student include

1. Define /explain the procedure
2. State the aim or objective(s) of the procedure
3. Discuss the nurses responsibilities before, during and after the procedure step by step
4. State the complications that may result or occur following the procedure.

Writing short note on current issues

Topical issues do have nursing implications and therefore require that the student will show good understanding of the issue in question.

Steps to demonstrate this include

1. Definition of the issue
2. Objective of the topical issue
3. Approach to the topical issue
4. Advantages
5. Disadvantages

Examples

Write short note on the following

- A. Epilepsy
- B. Urinary catheterization
- C. Drug Revolving Fund

A. Epilepsy

Definition: This is a convulsive attack due to disorders of electrical activity of the brain. It is a chronic disorder characterized by recurring seizures of unknown cause

Cause(s)/Predisposing factors

The cause is unknown but is attributed to;

- ☐ Intracranial pathological conditions such as increased intracranial pressure.
- ☐ Brain damage associated with head injury
- ☐ Cerebral oedema.
- ☐ Intracranial space occupying lesion.
- ☐ Intracranial haemorrhage.
- ☐ Infection that resulted to brain damage.

Incidence: It affects all sex, age and race.

Types:

- i. Grand mal epilepsy
- ii. Petit mal epilepsy
- iii. Jacksonian epilepsy
- iv. Temporal lobe epilepsy

Grandmal Epilepsy is a major motor or tonic-clonic seizure. It may occur without a warning sign but an aura or unpleasant sensation such as headache, tension may precede a fall. At the fall of the patient tonic phase of convulsion sets in, followed by the clonic phase, after which the patient goes into unconscious state. The tonic spasm of the muscles causes rigidity and distortion of the body with an involuntary cry (epileptic cry) due to sudden contraction of the thoracic and abdominal muscles leading to forceful exit of air through spastic glottis. There is temporary arrest of respiration with cyanosis. Jaws and hands become fixed and eyes opened with pupils dilated. These last up to 60 seconds. Subsequently, there is regular jerky movement (convulsion) which marks the clonic phase. Secretion of saliva which may be frothy occurs. There may be incontinence of urine and if the tongue was bit during spasm, bleeding occurs. This stage lasts for few seconds and victim regains consciousness.

Victim may become confused and falls into deep sleep for several hours. On waking up, he may become disorientated, complain of headache, fatigue or manifest epileptic automatism.

Petitmal Epilepsy

This is simple momentary loss of consciousness. It may not be noticed by external person but the client/victim only and may not require nursing care.

Jacksonian Epilepsy

This may manifest as symptoms of cerebral lesion like confusion momentary forgetfulness, etc. Only close observation of the onset of convulsion and course of the attack can lead to good diagnosis.

Temporal lobe Epilepsy

This is characterized by hallucination, with memory loss. It may be secondary to temporal or parietal lobe disease.

Management (Grand Mal epilepsy)

The goal of medical and nursing management of epilepsy is to prevent harm to the patient during seizure and prevent further convulsions.

- ☐ When patient falls down, remove all harmful materials around.
- ☐ Do not touch the patient at the tonic phase.
- ☐ During the clonic or convulsion phase, gently use padded spatula to prevent the patient from biting off the tongue, lips or oral mucosa.
- ☐ Loose tight clothings around the neck when clonic phase begins to subside, turn patient to one side to promote drainage of mucus and prevent aspiration.
- ☐ Observe patient for duration of seizure and incontinent.
- ☐ When patient sleeps off, do not disturb him.
- ☐ When he wakes up, give supportive care.
- ☐ Be careful to control his actions in case epileptic automatism occurs.

Drugs of choice in the treatment of epilepsy include

- ☐ Phenytoin sodium
- ☐ Paraldehyde
- ☐ Diazepam
- ☐ Phenobarbitone

Complications

Injury e.g biting off the tongue, lips, etc.

Patient may fall into harmful object and gets injured.

Prevention of epileptic attacks

- ☐ Predisposing conditions should be treated appropriately.
- ☐ Patient should always take his drugs to prevent attacks.
- ☐ Patient should not climb trees or stay in high altitude.

B. Urinary tract Catheterization

Definition: This is a process by which catheter is introduced into the bladder through the urethral meatus to enable exist of formed urine from the bladder.

- **Aim:** This aims at relieving patient of formed, retained urine in the urinary bladder.
- It is an avenue to obtain urine specimen for investigation.
- It is used to relieve patient of pain in obstructive situation that does not allow exit of formed urine in the bladder eg Benign prostatic Hyperplasia (BPH).
- It is carried out on women in labour to empty the bladder and allow easy delivery of the baby.
- It is also carried out in incontinent patients to prevent soiling of the bed and subsequent bed sore.

Before the procedure:

- ☐ Inform the patient about the procedure and obtain his/her approval.
- ☐ Set the catheterization tray.

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- ☐ Screen the patient.
- ☐ Take the tray to the patient bed side.
- ☐ Reassure the patient.

During the procedure

If the nurse is simply going to assist the doctor or urologist, do the following

- ☐ Offer the doctor sterile gloves.
- ☐ Open the catheter container (in sealed bag)
- ☐ As the doctor picks the sterile catheter, get the KY jelly and press the tube to release the quantity needed for the procedure.
- ☐ As the doctor introduces the tubing, reassure the patient.
- ☐ Draw 5-10 mls of sterile water with syring and hand it over to the doctor for ballooning of the catheter.
- ☐ Connect the tubing to the uribag or
- ☐ In some cases, a kidney dish may be used to collect the first voided urine before connecting the tubing to the uribag.

After the procedure

- ☐ Measure the first voided urine record the volume in fluid chart and discard.
- ☐ Make the patient comfortable.
- ☐ Remove used materials in the tray.
- ☐ Remove the screen
- ☐ Care for the catheter till it is removed.
- ☐ Keep strict intake and output chart.

Complications:

UTI's

Excessive removal of urine at initial time may lead to negative pressure.

Drug Revolving Fund (DRF)

This is a system whereby fund is provided to the hospital pharmacy to purchase drugs from registered pharmaceutical companies at cheapest price to make it affordable and available to patients.

Aim:

- ☐ To stamp out the "out of drug syndrome" which was the bane of public hospitals.
- ☐ To promote the production of drugs by indigenous pharmaceutical companies.
- ☐ Provision of essential drugs at lowest possible cost

Method of operation (Modus operandus)

The pharmacy unit or department of the hospital is given a start-up fund to purchase an initial supply of drugs (seed stocks) which are sold at cash and carry basis. From this seed stock, the pharmacy department builds its capacity of securing drugs dispensed in the hospital.

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An accounting system is set up to ensure continuity of the system. In the state, the state Ministry of Health controls the DRF through the state financial controller. He releases the fund and monitors the activities that go on in the system including purchase, disbursement and management of the fund.

Advantages

- DRF helps to reduce unavailability of drug in the hospitals
- It helps to reduce stocking the pharmacy unit with expired drugs as only essential drugs are procured through this fund.
- Indigenous pharmaceutical companies are patronized through the system.
- Highly expensive drugs that are not prescribed often are not stocked in the pharmacy stores.

ANSWERING QUESTIONS ON NUTRITIONAL MANAGEMENT OF DISEASES

Students may be required to discuss nutritional management of diseases. This helps to understand how well they can apply knowledge of nutrition/dietetics in the management of patients. To deal with this effectively, the student has to show good understanding of the disease in question and the aim of using nutritional measures in the care of a patient with such disease or health problem. Common diseases that demand application of nutrition in conjunction with medical-nursing approaches include:

- Diabetes mellitus
- Hypertension
- Liver cirrhosis
- Hepatitis
- Peptic ulcer
- Diverticulitis
- Inflammatory bowel disease
- Congestive cardiac failure
- Renal failure, glomerulonephritis, etc.

The best approach is for the student to take the steps below.

1. Definition/explanation of the disease in question.
2. Aim of nutritional management of the disease in question.
3. The application of nutrition in the management of the disease condition.

The questions to ask include:

"Are there food substances to increase in quality, reduce in quantity, or avoid completely?" The answer to the above questions will enable the student gain direction on the way forward. The clinical manifestations of the disease as another guiding principle to knowing the nutritional approach to the disease process. For instance, a patient that is suffering from Acute Renal failure will

Oedema, proteinuria and weakness, by implication requires reduction in the quantity of salt in the diet to prevent increase in the fluid retention. Protein is lost in urine but because the patient cannot eliminate urea which is produced from deamination of protein due to defective kidneys, excess protein should not be given to him. Intake of excess protein will lead to deamination which increases the level of urea in the blood worsening the patients' problem. It is better to maintain level of protein in the diet in such a way that it will not affect the individuals immunity but once the kidneys begin to heal, protein can be increased to boost immunity.

ANSWERING QUESTIONS THAT REQUIRE THE STUDENT TO DIFFERENTIATE OR COMPARE TWO ISSUES OR DISEASES.

The best approach is to use table with sub-headings. This method enables the student to remember missing items that are yet to be differentiated or compared. Generally questions on short notes, nutritional management of diseases, comparison/contrasting of issues and discussion of procedures are better written with sub-headings. This gives better co-ordination, better direction and makes way for better remembering of points to discuss.

TEST YOURSELF

The answers to the question 1 below are not correct in all ramifications. Identify the different places where errors occurred as they will help you to refine your answers in related cases. Read through the questions and answers and critique the same.

Question 1

Mr. Joaz, a 50-year old farmer was brought to your hospital in a semi-comatous state, history of generalized spasm on minimal touch, an injury sustained at the farm two weeks ago which was treated at a nearby patent medicine shop. After series of physical examinations and body temperature observation which read 39.5°C, the doctor on duty made a diagnosis of Tetanus.

- A. What is Tetanus? 2marks
- B. List 6 other clinical manifestations of Tetanus not mentioned above 3 marks
- C. Briefly described the pathophysiology of Tetanus 4marks
- D. Identify three (3) nursing problems of Mr. Joaz and utilize the nursing care plan to solve the identified problems in order of priority. 9mks
- E. Enumerate the likely complications the patient will suffer if he is not properly managed 2 marks

Answer

- A. Tetanus is a neurologic condition caused by *Clostridium tetani* (a spore forming microorganism) and is characterized by generalized spasm and muscle rigidity.

B. Other clinical manifestations of Tetanus are;

- i. Rigidity of muscles
- ii. Risus sardonicus
- iii. Stress incontinent of urine and faeces
- iv. Photophobia
- v. Headache
- vi. Fever

C. Pathophysiology of Tetanus:

When clostridium tetani gains entrance into the body through a wound, it incubates. After the incubation period the clostridium tetani germinates into vegetative bacilli but only in the anoxia injured tissue.

The toxin has affinity to the anterior horn cell that controls the skeletal muscles. Stimulation leads to spasm which first occurs at the jaw with sustained contraction giving it the name "lock jaw". The jaw becomes locked ie patient will not be able to open his mouth. Other muscles follow in succession with slightest noise or stimulation voice sound or banging of door bringing up spasm which is often generalized. The following manifestations occur

- ☐ The motor muscles become irritated and spasm occurs.
- ☐ Slight noise like banging of the door triggers spasm
- ☐ Patient develops pseudo-smile
- ☐ Opisthotonos position becomes evident as the extensors become more retracted. This lead to back pain and possible asphyxiation. Patient may die of excessive fatigue due to constant muscle spasm or asphyxia.

D. Nursing problems of Mr. Joaz with Tetanus

1. Imbalanced nutrition related to locked jaw as evidenced by patient's weight.
2. Hyperthermia related to reset of temperature control center in the hypothalamus.
3. Deficient self-care related to the disease condition as evidenced by patient's inability help himself.

E. Likely complications of tetanus

1. Asphyxia
2. Respiratory arrest
3. Deep vein thrombosis
4. Dysphagia
5. Temporal incontinent of urine and stool
6. Fracture (during the process of spasm)
7. Mental confusion (temporal)

NURSING CARE PLAN OF A PATIENT WITH TETANUS

SN	Diagnosis	Objective	Action	Rationale	Evaluation	Sign
1	Imbalanced nutrition related to locked jaw as evidenced by patient's weight.	Patient will gain weight within 2 weeks of nursing action.	- Pass Ng tube. - Serve adequate nutritive fluid diet through the Ng tube. - Weigh patient daily.	- To feed patient easily - To improve patient's nutrition - To determine patients improve in nutrition and weight gain.	patient gained 1kg after 2 weeks of nursing actions	✓
2	Hyperthermia related to reset of temperature control centre in the hypothalamus.	Patient's body temperature will be reduced to normal within 1 hour of nursing actions.	- Expose patient - Switch on fan - Tepid sponge patient - Give prescribed antipyretic eg paracetamol 1000mg tds.	- Exposing patient helps to reduce body temperature through conduction - Fanning helps to reduce body temperature through convection. - Tepid sponge helps to reduce body temperature through conduction and convection - Antipyretic reduce fever by acting on the temperature regulating centre in the hypothalamus.	Patient body temperature was reduced to normal within 30 minutes of nursing action	✓
3	Deficient self care related semi conscious state as evidenced by patient's inability to help himself.	patient will develop self care within 5 days of nursing actions.	- Assist patient in self care activities until he is able to do so eg - Feed patient - Bed bath patient - Encourage patient to carryout self care activities when he becomes conscious.	- Assisting patient in self care activities like feeding and bathing help to keep him groomed until he is able to do so and nursed. - Encouraging the patient on self care when he is conscious helps him to develop self care.	Patient developed self care after 4 days of nursing actions.	✓

Critiquing Answers to Question 1 (A-D)

- A. Tetanus is not caused by clostridium tetani but the toxin secreted by clostridium tetani. The definition is therefore not correct in all ramifications.
- B. The question stated that student should mention only the clinical manifestations that were not mentioned in the question, so fever should not be included because it was mentioned in the framing of the questions.
- C. Writing of the Pathophysiology of Tetanus was started wrongly. There was no definition of the disease. There is need for the definition of the disease before jumping into what happens after incubation. The continuity was fairly good but no good connection between the dysfunction and the clinical manifestations. There was enumeration of the pathophysiology. There was no logical conclusion to complication.

D. Enumeration of the patients' problems was not properly arranged according to priority. Hyperthermia would have taken priority to imbalanced nutrition.

Looking at the nursing care plan.

Objective to the imbalanced nutrition was not stated to be either "less than" or "more than" body requirement. Either of this which would have been "less than", will say much about the goal to achieve in a client with imbalanced nutrition.

A patient's weight may increase following oedema so it is not a good goal to achieve here but can be an evidence for balanced nutritional status. Weight loss was evidence that the patient had nutritional imbalance and so the goal will aim at patient maintaining a nutritional balance or not developing malnutrition.

The approach to rationale in the first diagnosis (imbalanced nutrition): less than body requirement was also wrong. It is better to use the gerund of the action verb. For instance rationale for second action would have read, adequate nutritive diet prevents malnutrition or adequate nutrition prevents malnutrition.

Considering the evaluation of the cares in the first diagnosis, it is evident that the objectives were not met at stipulated time in the objective. This is because the evaluation was after 2 weeks which could be after 2 weeks and 2 days. It will be better to evaluate at a time before the stipulated time in the objective. For instance, it would have been better if the evaluation reads "patient gained weight after one week, 3 days, etc, provided it was before the elapse of 2 weeks."

A critical look at the second identified problem, shows that there was no evidence of fever indicated and the etiology of the disease was not made very clear. The problem would have read "Hyperthermia related to reset of temperature control centre in the hypothalamus secondary to presence of microbial activity in the body as evidence by body temperature reading of 39.5°C"

Objective to this second diagnosis was patient oriented objective and was stated with the word "normal" the phrase "will be reduced" removed patient from being part of key players in reducing body temperature. Non involvement of patients in their management poses problem in patient's management and great challenge in achieving the objective/goal of care.

The third identified problem "deficient self care" was not specified. NANDA-I label requiring that "Deficient self care demands that specification should be made ie whether it is hygiene deficient, feeding, etc."

Finally the student signed signature at the signature column. It is expected that examinee remains anonymous to the assessor especially in Nursing and Midwifery Council of Nigeria (N&MCN) examinations.

There may be some other problems to identify from the above student's response to other stated questions. Substantially any student that can always, remember to limit the above mistakes stands chances of having better score.

Corrections in a-d of the critiqued question

Answer

A. Tetanus is a neurologic condition caused by ^{toxins secreted by} clostridium tetani and is characterized by generalized spasm and muscle rigidity. Tetanus is not directly caused by clostridium tetani but by the toxin secreted by the microorganism.

B. Other clinical manifestations of Tetanus not mentioned in the question are;

- i. Rigidity of muscles
- ii. Risus sardonicus
- iii. Stool incontinent and faeces
- iv. Photophobia
- v. Headache

C. Pathophysiology of Tetanus:

Tetanus is a neurologic condition caused by the toxin secreted by clostridium tetani and is characterized by generalized spasm and muscle rigidity. When clostridium tetani gains entrance into the body through a wound, it incubates. After the incubation period the clostridium tetani germinates into vegetative bacilli not only in the anoxia injured tissue. The tetanus bacilli multiply locally but does not spread or invade nearby tissues. Toxins from the bacilli known as tetanus toxin (a neurotoxin) enters the blood stream and from there reaches the central nervous system. The toxin has affinity to the anterior horn cell that controls the skeletal muscles. Stimulation leads to spasm which first occurs at the jaw with sustained contraction giving it the name "lock jaw". The jaw becomes locked ie patient will not be able to open his mouth. Other muscles follow in succession with slightest noise or stimulation voice sound or banging of door bringing up spasm which is often generalized.

- Patient develops pseudo-smile
- Opisthotonos position becomes evident as the extensors become more retracted. This lead to back pain and possible asphyxiation. Patient may die of excessive fatigue due to constant muscle spasm or asphyxia.

Nursing problems of Mr. Joaz with Tetanus

1. Hyperthermia related to reset of temperature control center in the hypothalamus as evidenced by thermometric reading above normal range (39.5°C)
2. Imbalanced nutrition: less than body requirement, related to locked jaw as evidenced by patient's loss of weight.
3. Deficient self-care (hygiene) related to semi-conscious state as evidenced by patient's inability to help self.

TEST YOUR SELF WITH THESE QUESTIONS BY PROVIDING ANSWERS TO THEM

QUESTION

Mrs. Jill, a 50 year old woman was brought to your hospital by her husband with complaints of weakness, purple striae at the buttocks and thighs with sudden change of face to round-bloated looking. The doctor on duty conducted series of investigations and made a diagnosis of Cushing's syndrome.

- What is Cushing's syndrome? 1 mark
- Outline four (4) likely investigations that assisted the doctor to reach at the diagnosis 2 marks
- Briefly describe the pathophysiology of Cushing's syndrome 3 marks
- Identify in order of priority, three Nursing problems of Mrs. Jill and utilize the Nursing care Plan to solve the identified problem 9 marks
- The Doctor prescribed Caps Ampiclox for Mrs. Jill to take for 10 days. Write short note on the prescribed drug using the guideline below.
 - Groups (ii) Mode of action (iii) Dosage (iv) Side effect(s) (v) Nursing responsibilities. 5 marks

Answer

A. Cushing's syndrome is a rare disorder that occurs due to excessive secretion of corticoids by the adrenal cortex. It is characterized by truncal obesity, moon face, buffalo hump, low resistance to infection and weakness.

B. Investigations that assisted the doctor to reach at the diagnosis of Cushing's syndrome.

- Blood cortisol level estimation in the morning and later in the day.
- Computed Tomography Scan (C-T Scan) of the adrenal glands.
- Magnetic Resonance Imaging (MRI).
- Chest x-ray.
- Dexamethazone suppression test.

C. Pathophysiology of Cushing's syndrome.

Cushing's syndrome is a condition that follows excessive secretion of corticosteroids due to overproduction by the adrenal glands.

In event of excessive production of corticosteroids by the adrenal cortex there is distortion of normal feedback mechanism that controls the function of the adrenal cortex and the usual diurnal pattern of cortisol hormone secretion is lost. This leads to excessive protein catabolism, gluconeogenesis with abnormal distribution of fats especially at the trunk. This brings about truncal obesity to the patient. There is atrophy of lymphoid tissues which predisposes the individual to infection due to reduction in immunity.

Glucose tolerance is decreased with resultant, hyperglycaemia, and weakness occur. Purple striae occur at the buttocks, abdomen and thighs due to increased fragility of blood vessels and atrophy of skin.

Increase abnormal deposition of fats at the trunk lead to buffalo hump, with round bloated-looking face (moon face) with thin wasted limbs. Calcium is mobilized from the bone leading to osteoporosis especially in the vertebrae with the patient complaining of backache.

Excessive secretion of mineralocorticoids results in electrolyte, fluid and acid base imbalances. Hypernatraemia and fluid retention may lead to hypertension and oedema. Hypokalaemia may lead to extreme weakness and cardiac dysfunction. Female patients develop hairs on the face, deepened voice, breast atrophy, amenorrhoea and enlargement of clitoris due to production of a androgen.

A. Nursing Problems of patient with Cushing's syndrome.

- Deficient self care related to weakness as evidenced by patient's inability to carry out self care activities.
- Fluid volume excess (oedema) related to water and sodium retention as evidenced by patient's weight gain.
- Risk for infection related to reduced immunity secondary to protein catabolism and atrophy of lymphoid tissues.

E. Ampicillin capsule

Group - Penicillin group of antibiotic

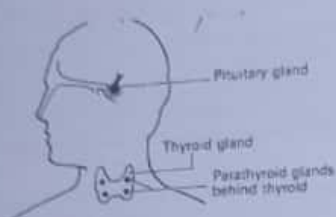
- Mode of action: Stops the synthesis of bacterial cell wall leading to bacterial cell death.
- Dosage 500mg orally qds
- Side effects - Diarrhoea, constipation, hypersensitivity to active ingredient rashes.
- Nursing responsibilities
- Administer without missing time interval.
- Monitor patient for hypersensitivity reactions.
- Serve drug with care to patients suffering from liver or renal diseases.
- Withhold drug if rashes are excessive on the patient.

QUESTION

Mrs. Dane, a 45 year old woman was admitted in your ward with medical diagnosis of Cushing's syndrome.

- With a well labelled sketch, show the locations of the Endocrine glands in the body. 5 marks
- What is Cushing's syndrome? 2 marks
- List eight (8) clinical features of Cushing's syndrome 4 marks
- Identify three (3) nursing problems of a patient with Cushing's syndrome and utilize the Nursing Care plan to solve the identified problems. 9 marks

Answer



A. Sketch of a human being showing the location of: Pituitary gland, Thyroid gland, Adrenal glands, Parathyroid glands, Thymus gland, ovaries in females and testes in males.

B. Cushing's syndrome is a disorder characterized by hypersecretion of cortisol and androgen and sometimes aldosterone which could be due to excess production of ACTH for unknown reason (Seeley, Stephen Tate, 1992: 567). Also refer to answer to question 1A

C. Clinical Manifestations of Cushing's syndrome

1. Truncal obesity
2. Moon face
3. Buffalo hump
4. Arms and legs are usually slender.
5. High blood pressure
6. Oedema
7. Purple striae on the buttocks, thighs and abdomen.

8. Weakness and easily tired
9. Women may develop secondary male characteristics eg. Hirsutism, deepening of voice, breast atrophy, amenorrhoea and clitoris enlargement.
10. Osteoporosis
11. Mood alterations-distress and depression.
12. Loss of libido but women may have increase sex drive.

D. In addition to Answer to Question "1D", the following are likely problems to be identified.

1. Disturbed body image related to change in physical appearance eg moon-face as evidenced by patient's constant viewing of self over the mirror.
2. Anxiety related to unknown outcome of the disease condition as evidenced by patient's facial appearance.
3. Deficient knowledge related to disease process, prognosis and management as evidenced by patient's numerous questions on the disease process, prognosis and management.

Nursing Care Plan of a Patient with Cushing's Syndrome

SN	Diagnosis	Objective	Order/Action	Rationale	Evaluation	Sign
1	Disturbed body image related to change in physical appearance eg moon-face as evidenced by patient's constant viewing of self over the mirror.	Patient will develop positive self image within 5 days of Nursing actions.	- Discuss the cause of the disease with the patient and make her to do away with misconception about the disease. - Encourage her to accept herself as she is since there is nothing to do to change it overnight. - Remove every visible mirror from the patient's place of care. - Encourage patient with the same problem who have positive self image to visit her.	- Understanding the cause of the disease and removal of misconception about the disease enables the patient to develop positive self image. - Encouragement will enable the patient to develop positive self image. - Removal of mirrors from the patient will make her to stop viewing herself over the mirror thereby enabling her to stop perceiving how her face looks, thereby assisting in developing positive self image. - Visitation by patients with the same condition who have developed positive self image will give her positive acceptance of self.	Patient developed positive self image after 5 days of Nursing actions.	

2	Anxiety related to unknown outcome of the disease condition as evidenced by patient's facial expression.	Patient will demonstrate steps of reduction in anxiety within 3 days of Nursing actions.	- Advise patient close to another patient who is recovering from the same or related condition. - Create anticipatory therapeutic relationship. - Encourage patient to verbalize her feelings about the disease. - Involve her in her care.	- Admitting her close to a recovery patient with the same or related condition instill hope and reduce anxiety. - Therapeutic relationship enable patient to be involved in her care thereby reducing anxiety that surrounds her care. - Verbalizing her feelings about the disease will enable the nurse to know the areas to lay emphasis while discharging her mind about the disease which helps to reduce anxiety. - Involving her in her care makes her to know more about the disease and reduce anxiety.	Patient demonstrated signs of reduction in anxiety after 3 days of nursing actions as evidenced by her freely discussing the condition with her visitors.
3	Deficient knowledge related to disease process, prognosis and management as evidenced by patient's incorrect questions about the disease process, prognosis and management.	Patient will show good understanding of the disease process, prognosis and management within two days of nursing actions.	- Find out what the patient knows about the disease. - Teach her the causes, prognosis and management of the disease. - Encourage her to ask questions about the disease and answer the question correctly. - Ask her to explain the truth she understood from what you taught her about the disease.	- Knowledge of what the patient knows about the disease enables the nurse to know where to start teaching her about the disease. - Teaching her the causes, the prognosis will enable her to have knowledge about the disease. - Her questions and appropriate feedback from the nurse will enable her to develop knowledge of the disease process, prognosis and management. - Her explaining the truth she understood from the nurse's teaching will enable the nurse to know areas of emphasis that will make the patient have good knowledge of the disease.	Patient showed good knowledge of the disease process, prognosis and management after 24 hours of nursing action.

QUESTION

Mrs. Jumatu, a 47 years old petty trader reported to your hospital with complaints of epigastric pain which is often accompanied by heartburn. Usually the pain persists until she eats food. The doctor on duty made a diagnosis of peptic ulcer.

A. With the aid of a well labelled diagram describe the human stomach.

- Diagram 1 mark
- Labelling 1 mark
- Description 4 marks

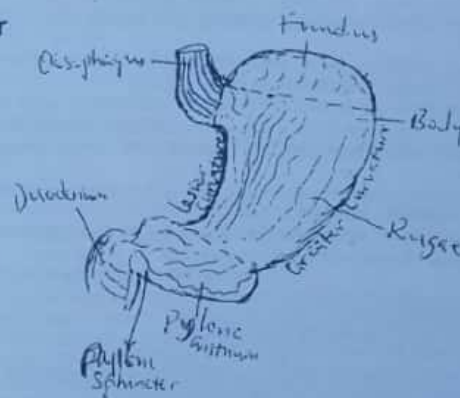
B. What is peptic ulcer? 1 mark

C. Enumerate four(4) predisposing factors/causes of Peptic ulcer. 2 marks

D. Mention four(4) clinical manifestation of Peptic ulcer. 2 marks

E. Identify three (3) Nursing problems of Mrs. Jumatu, and utilize the Nursing care plan to solve the identified problems. 9mks

Answer



The stomach

Ai. Good diagram of the stomach that clearly shows the following features:

i. The greater curvative

- The lesser curvature
- The fundus
- The body
- The pylorus
- The cardiac sphincter
- The pyloric sphincter

iii. Description of the stomach

- ☐ It is a "J" shaped dilated portion of the alimentary tract.
- ☐ It is situated in the epigastric, umbilicus and the left hypochondric region of the abdominal cavity.
- ☐ It is continuous with the oesophagus at the cardiac sphincter superiorly, and with the duodenum at the pyloric sphincter inferiorly.
- ☐ The size of the stomach is dependent on the food content.
- ☐ It has two curvatures the lesser curvature is short and at the posterior aspect while the greater is at anterior aspect.
- ☐ Just before the pyloric sphincter, the lesser curvature curves upwards to complete the "J" shape.
- ☐ The stomach is divided into three parts, the fundus, the body and the pylorus (antrum).

- ☐ It has two sphincters, the cardiac sphincter guards the opening from the oesophagus while the pyloric sphincter guards the opening into the duodenum.
- ☐ The stomach is made up of four distinct layers, the peritoneum, the muscle layer, the sub-mucous membrane and the mucosa.
- ☐ The peritoneum is the outer-most covering, the muscle layer has three muscle fibres (the longitudinal, oblique and circular muscle fibres). The sub-mucous membrane contains nerves and blood vessels with lymphatics.
- ☐ The mucous membrane is thrown into rugae which iron out depending on the food content in the stomach.
- ☐ Blood supply-gastro-epiploic and gastric arteries (left and right)
- ☐ Venous drainage-corresponding veins drain into the portal vein.

Functions

- ☐ It acts a reservoir of food.
 - ☐ Churning of food occurs in the stomach.
 - ☐ It produces HCL which helps in food digestion.
 - ☐ It contains intrinsic factor-B12 which helps in erythropoiesis.
 - ☐ It assists in onward movement of food in the GIT.
 - ☐ Digestion of food occurs in the stomach.
 - ☐ Absorption of substances like alcohol occurs in the stomach mk for each point.
- B. Peptic ulcer refers to the excoriation or erosion of the mucosal walls of the stomach, duodenum and the oesophagus. It is a circumscribed ulceration of the mucous membrane penetrating through the muscularis and mucosa and occurs in areas exposed to acid and pepsin.
- C. Predisposing factors to peptic ulcer infection with helicobacter stress which increases secretion of hydrochloric acid.
- ☐ Ingestion of excess caffeinated beverage.
 - ☐ Excessive smoking
 - ☐ Excessive alcohol intake
 - ☐ Familial tendency
 - ☐ Chronic use of non-steroidal anti-inflammatory drugs eg Aspirin, Ibuprofen
 - ☐ Gastritis.
- D. Clinical manifestations of peptic ulcer:
- ☐ Epigastric pain exacerbated by hunger
 - ☐ Heart burn
 - ☐ Tenderness on gentle pressure to the stomach.
 - ☐ Sour regurgitation.
 - ☐ Melaena due to bleeding from the erosion site (vomiting- in severe cases)
 - ☐ Weakness.

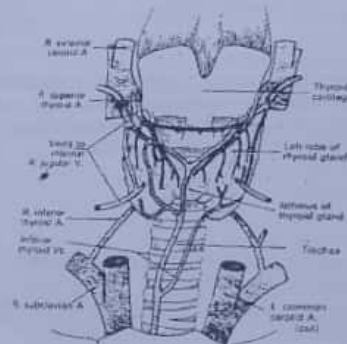
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- E. Nursing Diagnoses
1. Pain (acute) related to the effect of gastric acid on damaged mucosal lining as evidenced by patient's complaints.
 2. Imbalanced nutrition: less than body requirement related to vomiting as evidenced by patient's loss of weight.
 3. Anxiety related to unknown outcome of the disease conditions as evidenced by patient's facial expression.
 4. Deficient knowledge related to the cause, prognosis and management of the disease.

QUESTION

1. Mrs. Dave, a 54years old petty trader came to your hospital with obvious swelling at the neck region, complaints of hoarseness of voice and pain during swallowing. After investigations, the doctor on duty made a diagnosis of Endemic goiter and booked her for Thyroidectomy in two weeks time.
- A. Utilizing a well labelled diagram describe the thyroid gland.
- | | |
|------------------|------|
| i. Diagram | 1mk |
| ii. Labelling | 1mk |
| iii. Description | 3mks |
- B. What is endemic goiter? 1mk
- C. Enumerate four (4) investigations that assisted the doctor to reach at the diagnosis? (2mks)
- D. Briefly describe the Pathophysiology of endemic goiter. (3mks)
- E. Identify two pre and one post operative nursing problems of Mrs. Dave and utilize the Nursing care plan to solve the identified problems.

A



Answer

(i) & (ii)
Position of the thyroid gland and associated structure

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A.
(iii)

Description of thyroid gland

Definition: Thyroid gland is an endocrine gland that is located at the neck region.

Position: Located at the neck region at the level of 5th, 6th and 7th cervical vertebrae and 1st thoracic vertebra.

Shape: It is like butterfly in shape, having two lobes (left and right).

Structure: Thyroid gland is highly vascular with fibrous capsule surrounding it. It consists of two lobes, one on each side of the thyroid cartilage and upper cartilaginous ring of the trachea.

Blood supply: Superior and inferior thyroid arteries supply blood to the gland. Venous drainage: Thyroid vein.

Functions of the thyroid gland.

The thyroid gland secretes thyroid hormones

- 1) Thyroxine (T₄)
- 2) Triiodothyronine (T₃)
- 3) Calcitonine

Iodine from ingested food helps in the formation of these hormones especially T₃ & T₄ and the body, concentrates it and releases it on daily basis for body use.

The release of hormones of the thyroid gland is assisted by the presence of thyroid stimulating hormone (TSH) through stimulation by thyroid releasing hormone from the hypothalamus.

T₃ & T₄ help to increase Basal Metabolic Rate.

They also regulate metabolism of carbohydrate, protein and fats.

These hormones are essential for growth and development.

The gland stores and releases iodine on daily bases at 120ng.

Calcitonine reduces calcium level in the body.

B. Simple or endemic goiter is a goiter which is neither associated with increased or decreased thyroid hormonal activity nor is neoplastic. It is an asymptomatic condition of the thyroid gland with swelling in the neck. It occurs when there is normal or insufficient iodine for the production of thyroid hormones T₃ & T₄.

C. **Four investigations that assisted the doctor to make diagnosis of Endemic goiter.**

1. Palpable lesion that is fixed and hard at the neck region.
2. Estimation of serum thyroxine level to confirm normal thyroid function.
3. X-ray of the chest and thoracic inlet to determine the degree of tracheal deformity due to pressures.
4. Radioactive isotopes scan to determine the anatomy and position of the goiter.
5. CT scan.
6. Estimation of serum thyroid stimulating hormone (TSH).
7. Ultrasound scanning to distinguish thyroid mass from a cystic swelling.

D. **Pathophysiology of endemic goiter**

Endemic goiter is an asymptomatic condition of the thyroid gland with swelling at the neck region which often results from insufficient iodine intake. In an event of reduced iodine in the body the ability of the thyroid gland to produce its hormones (T₃ & T₄) reduces.

To this effect, there is excessive stimulation of the gland by thyroid stimulating hormone (TSH) from the anterior pituitary to produce its hormones. This excessive stimulation will lead to the enlargement of the glandular tissues with obvious painless swelling in the neck region. The enlargement may lead to tracheal compression which results in dyspnoea. Further compression may affect the pharynx leading to dysphagia. Where recurrent nerve is affected, voice becomes hoarse.

E. **Pre-operative Nursing problems of Mrs. Dave**

1. Pain (acute) related to compression of the pharynx and trachea by the enlarged gland as evidenced by patient expression of pain during swallowing.
2. Imbalanced Nutrition: Less than body requirement related to dysphagia as evidenced by patient's inability to finish plate of meal served.
3. Anxiety related to unknown outcome of the surgery as evidenced by patient's facial expression.

Post Operative nursing problems of a patient that went for thyriodectomy

1. Ineffective airway clearance related to the effect of anaesthesia as evidenced by the presence of secretions air the airway.
2. Pain (acute) at incision site, related to surgical incision as evidenced by patient's complaints.
3. Risk for haemorrhage related to surgical manipulation of blood vessels around the gland.
4. Risk for infection elated to surgical incision.

PRE-OPERATIVE NURSING CARE PLAN OF A PATIENT WHO WAS BOOKED FOR THYROIDECTOMY

S/N	Diagnosis	Objective	Order/Action	Rational/Scientific	Evaluation
1	Pain (acute) related to compression of the pharynx by the enlarged gland evidenced by patient's facial expression during swallowing.	Patient will show signs of reduction in pain within 7 days of nursing action.	• Serve patient soft diet. • Use diversional therapy on the patient e.g. turning on preferred TV or Radio station or use of print media. • Serve prescribed analgesics e.g. paracetamol 1000mg tds. • Serve prescribed anti-thyroid drugs e.g. Carbimazole 30-60mg tdy.	• Soft diet poses easily down the GI tract thereby reducing pain during swallowing. • Diversional therapy removes the patient's concentration on the pain thereby giving temporal relief of pain. • Analgesics reduce pain by acting on the pain centre and increasing the patient pain threshold. • Anti-thyroid drugs help to bring the gland to euthyroid state thereby reducing the compression at pharynx and reducing the pain.	Patient demonstrated signs of reduction in pain as evidenced by her not complaining of pain during swallowing after five days of nursing care/treatment.
2	Anxiety related to the unknown outcome of the surgery, evidenced by patient's nervousness throughout the surgery.	Patient will demonstrate signs of reduction in anxiety within 2-3 days of admission.	• Admit patient close to close to another patient that has undergone the surgery or related surgery. • Create nurse/patient therapeutic relationship and involve her in her care. • Answer patient's queries about the surgery tactfully. • Use diversional therapy e.g. watch or TV/Radio or give readable news print. • Serve prescribed sedatives e.g. diazepam.	• Admitting patient to another patient that is recovering from the surgery instill hope on the patient and reduces. • Nurse/patient therapeutic relationship makes patient to feel at relaxed thereby reducing anxiety. • Answering her question tactfully helps to build hope and confidence on the patient thereby reducing anxiety. • Sedative induces sleep and reduces anxiety.	Patient showed signs of reduction in anxiety after two days of admission evidenced by reduction in her question about the surgery.

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POST-OPERATIVE NURSING CARE PLAN OF A PATIENT WHO WENT FOR THYROIDECTOMY

S/N	Diagnosis	Objective	Order/Action	Rational/Scientific	Evaluation
1	Ineffective airway clearance related to the effect of anaesthesia, evidenced by the presence of airway and the mouth.	Patient will maintain patent airway throughout the period he is under the effect of anaesthesia.	• Lie patient supine with her head turned to one side. • Maintain artificial air-way in-situ. • Suction secretions using suction machine or manual suctioning apparatus pm.	• Supine position with head turned to one side aids free flow of secretions without blocking the airway. • Artificial air-way allows easy exchange of air and prevents asphyxiation. • Suctioning helps to remove the secretions thereby allowing the air-way to remain patent.	Patient maintained patent air-way throughout the period he was under the effects of anaesthesia.

QUESTION

Research is indispensable in the professionalism of every profession and maintenance of standard of practice.

A. What is Research in Nursing? (1mk)

B. Write short note on the following.

- Abstract
- Background of the study
- Limitation of the study
- Hypothesis (2mks each)

C. Discuss the ethical considerations in the use of human beings for conduct of research. (8mks)

D. Referencing is integral part of academic research and APA method is the preferred method in Nursing. Reference the book named below using APA method. Textbook of Research by Andrew C. Chukwu, 2nd edition published in Owerri by Franklin publishers in 2012. (3mks)

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Answer

- A. Research in Nursing is an empirical investigation into a problem identified in nursing profession to proffer solution. It is an attempt to provide a total or partial solution to a problem identified in nursing to achieve quality nursing care, education and management.

B. (i) Abstract

- This is concise form of research which gives a reader clear information about the research.
- It is written with single line spacing and not paragraphed.
- It does not exceed 300 words.
- It is found before the introduction.
- Abstract contains:
 - Intro
 - Aim/Purpose
 - Methodology
 - Results
 - Conclusion/Recommendation
 - 4-6 keywords
- The research topic
- The scope of the study
- The population and sample size for the study.
- The design used for the study.
- The study area
- Sampling technique.
- Major findings from the study and recommendations.

Content

- Intro
- Aim/Purpose
- Methodology
- Results
- Conclusion/Recommendation
- 4-6 keywords

(ii) Background of the study

- This is the first subtopic in chapter one of research.
- It expresses clearly the fundamental idea or motive underlying the identified problem under study.
- The research must start from a reasonable point and progress logically.
- Ideas should not be disjointed and must flow freely.
- The variables in the topic of study must be explored in the background.
- The researcher must show comprehensive knowledge of the research area in the background.
- Background may start from global view of the subject matter and narrowed down to the community level or vice versa.
- Background of the study should not be too lengthy.
- Background is like clearing the land for farm work.
- In a nut-shell, it gives background information on the topic.
- It ends by showing that there's a gap that needs to be closed.

ii) Limitation of the study

- This discusses the problems in the course of conducting the research.
- It involves the hindrance or factors that delayed or lead to poor conduct of the research.
- Such problems may include; geographical coverage of the study area, the attitude of the study population or respondents/subjects used for the research, problem reaching the respondents due to their daily schedules. It may also be time or financial constraints.

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- Limitations may result to poor conclusion of research.
- It may affect generalization of findings and acceptance of findings of the research.

(iv) Hypothesis

- This is a guess statement that needs to be verified for truthfulness or falsehood. Assumption taken to be true for the purpose of scientific argument.
- It is a tentative statement made by the researcher that requires to be proved through data generation and analysis.
- It can be used instead of research objectives and questions.
- There are two types of hypotheses (null and alternative)
- Academic research hypothesis are mostly stated in null form.
- Null hypothesis is a statement of no difference between two variable.
- When null hypothesis is rejected following analysis of generated data, the alternate is accepted.
- When a null hypothesis is rejected when it should be accepted, it is referred to as type 1 error.
- When null hypothesis is accepted when it should be rejected, it is type II error.

C. Ethical considerations in the conduct of research using human beings.

- **Consent** Basically there should be freedom to consent by the subjects i.e they should not be forced to be part of the investigation.
- **No bias** Selection of the subjects must be carried out without bias.
- The researcher must explain the hazards or risks that may occur in the course of the research to the subjects.
- Where the hazards outweigh the benefits, the research should not be carried out.
- Benefits (financial and others) from the research must be made known to the subjects.
- Participants have the right to bow out at any point in time, i.e the subject can discontinue at will and this will by no means affect the subject's medical care/management.
- **Confidentiality** Confidentiality of information supplied by the subjects must be maintained.
- The subjects should have good communication channel with the researcher and the researcher's supervisor.
- The researcher must be competent in the field he/she is conducting the research.
- Any reported negative effect of the study on the subject must be taken care of by the researcher except it was stated otherwise at the commencement of the study.
- The findings of the research must be communicated to the subjects.

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D. Referencing a book

Chukwu, A.C. (2012). Textbook on Research, (2nd Ed). Owerri: Franklin Publishers.

QUESTION

A. A good nurse manager applies principles and elements of management to ensure better management of the resources under him/her.

Write briefly on the following:

- i. Planning
- ii. Directing
- iii. Unity of command
- iv. Stability of tenure
- v. Subordination of individual interest to organizational interest (3mks each)

B. Discuss the emergency management you will render to a 200 level Medical student who suddenly started bleeding from the nostrils during Biochemistry lecture in your school (5mks)

Answer

Planning

- i. Planning is an essential function of a nurse manager for organizational goal achievement.
 - It is selecting in advance, What to do, how to do it, and who will do it.
 - It is selecting in advance future possible courses of action from alternatives.
 - It is a means of assessing a situation on ground, forecasting the future and developing an action plan on how to move forward in order achieve organizational goals.
 - In planning, objectives to be achieved are set as priority and means of achieving the objects is determined with regards to the available resources.
 - Planning involves asking fundamental questions like:
 - Where are we now?
 - Where do we want to go?
 - How are we going to get there?
 - Answers to these questions will enable the manager and his subordinates to have direction for successful achievement of the organization goal

ii. Directing

- This is managerial function that is aimed at guiding human resources actions in the organization.
- It is the most difficult function of the manager because management of human resources is often difficult.

- It guides the manager to initiate and move subordinates to action according to planned objectives of the organization.
- The manager gives orders and instructions would/could be written or oral to the subordinates.
- Expectations of the organization are clearly communicated to the subordinate.
- The manager manages the workers and the work through effective communication, motivation and proper leadership styles.
- The nurse manager directs better where conducive environment exists in the organization.

iii. Unity of command

- This is one of the principles of management asserted by Henry Fayol.
- It is principle of one employee, one boss.
- It states that responsible to and receives directives from only one, boss, his immediate supervisor.
- Receiving instructions from multiple supervisors create confusion, conflict, poor execution of action, low morale and duplication of function and these are avoided through unity of command.
- It prevents the subordinate from receiving conflicting command, on the same subject from different superior officers.
- The nurse on duty must be answerable to one superior officer at a time.

iv. Stability of tenure

- This is another principle of management by Henry Fayol.
- He asserted by that there should be time for workers to stay and progress in their job.
- He advocates for rotation of staff in the organization but this should not be too frequent.
- This enables the workers to develop proficient skills in their different job descriptions.
- Third will lead to high productivity in the organization.
- Too long stay of the staff should not be preventing the job getting over the head of the staff.
- In nursing, the nurses should be allowed a good period of time in a unit before being transferred to another unit.
- Nurse should not be allowed to remain on one ward or unit longer than turn over of work.

v. Subordination of individual's interest to organizational interest.

- Henry Fayol advocates that personal interest of the staff should be subordinated for organizational interest.
- Managers must define ways of reconciling diversities in the interest for individual and the organization.

- ☐ Laziness, selfishness and weakness among staff is discouraged.
- ☐ This does not prevent identifying the problem of his subordinates and treating them with empathy even when this will affect the organizational goal achievement.
- ☐ A nurse on duty should not abandon her duty post and go to the market or bank. When there are other times to do so.
- ☐ The interest of the institution must surpass that of the staff.
- ☐ A nurse where is sick or whose must be considered by the manager to consult a doctor even if she is on duty.

- B. Emergency management of a Medical student who suddenly started bleeding from the nostrils.**
Bleeding from the nostrils is called epistaxis and may occur due to severe headache or excessive pressure in the tiny blood vessels of the nose.

Management:

- ☐ Tell the bystanders to leave the venue.
- ☐ Reassure the student.
- ☐ Make her to sit in an arm-chair position on a chair.
- ☐ Encourage her to stop blowing the nose.
- ☐ Raise her head up and forward, then pinch the nostrils and encourage her to breathe through the mouth.
- ☐ Clear blood from the nostrils.
- ☐ Check her vital signs.
- ☐ Put ice block or cold compresses at the bridge of the nose.
- ☐ Keep reassuring the student.
- ☐ If possible get adrenalin 1:1000, soak it in a cotton wool and pack the nostrils with it.
- ☐ Refer her to the school medical centre for as quick as possible or to a hospital.

QUESTION

Mr. Dikko, a 65 year old man was brought to the OPD Unit of your hospital with complaints of inability to void, pain at suprapubic region and obvious restlessness. After physical examination, the doctor on duty made a diagnosis of benign prostate hypertrophy and booked him for prostatectomy.

- A. Describe the act of micturition (4Mks)
- B. List four(4) investigations that might have assisted the doctors in making the diagnosis (2 Mks)
- C. Identify two (2) preoperative and one post operative nursing problems Mr. Dikko will likely suffer from and utilize nursing care plan to solve the identified problems (9 Mks).

- D. One of the drugs prescribed for Mr. Dikko after the surgery was Ibuprofen. Write short note on the drug using the guideline below

- i. Group
- ii. Mechanism of action
- iii. Dosage
- iv. Side effects
- v. Nursing responsibilities

Answer

A. Act of Micturition

When about 200-400mls of urine have accumulated in the bladder, the autonomic nerve fibres in the bladder wall, which are sensitive to stretch, become stimulated. This stimulator sends impulse to the brain in adult to create awareness of the desire to micturate.

This conscious arousal brings a return action to the bladder for contraction and at the same time relaxation of the internal sphincter.

The relaxation of the internal sphincter is followed by the relaxation of the external sphincter and outlet of the formed sphincter and outlet of the formed urine from the bladder through the urethra to the exterior.

Desire to micturate can be suspended for a while in adult.

In infants only spinal reflex action is needed to achieve micturition.

Four (4) investigations necessary for the diagnosis of BPH

1. Digital Rectal Examination
2. Palpation, Percussion and Auscultation of the bladder.
3. Urodynamic flow studies
4. Cystoscopy
5. Cystourethrosocopy
6. Serum creatinine level estimation
7. Serum urea level estimation
8. Urine analysis, culture and sensitivity
9. Prostate specific Antigen (PSA) test
10. International Prostate Symptoms score (IPSS)
11. Clinical manifestation/history

C. Two pre-operative problems of a patient with BPH

- i. Acute pain (supra-pubic) related to accumulation of urine in the urinary bladder secondary to enlarged prostate gland, evidenced by palpable bladder and no urine voided.
- ii. Anxiety related to unknown outcome of disease and surgery or inability to void urine as evidenced by patient's facial expression.
- iii. Risk for infection related to indwelling catheter

See care plan below

Post-Operative Problems of a patient with BPH

- Ineffective airway clearance related to the effect of anaesthesia
 - Deficient self-care related to temporal confinement on bed secondary to surgery.
 - Risk for urinary incontinent related to possible nerve damage or catheterization
 - Risk for infection related to surgical incision.
- See care plan below

PRE-OPERATIVE NURSING CARE PLAN OF A PATIENT WITH BPH

SN	Diagnosis	Objective	Order/Action	Rationale	Evaluation
1	Acute pain (supra-pubic) related to accumulation of urine in the urinary bladder and distension secondary to enlarged prostate, evidenced by palpable bladder and no urine voided.	Patient will verbalize reduction in pain within two hours of nursing actions.	- Assessing patient's level of pain. - Open nearby up and let the patient hear the sound of the running water. - Stroke the inner portion of the patient's thigh. - Set catheterization tray and assist the urologist as the doctor in catheterizing the patient.	- Assessing patient's level of pain helps the nurse to determine improvement in the management. - Having the sound of running tap stimulates the desire to initiate voiding will when achieved relieves the pain. - Stroking the inner thighs stimulates voiding which if achieved relieves pain. - Setting catheterization tray and assisting in catheterization helps in emptying the bladder thereby relieving the pain.	Patient verbalized reduction in pain after 1 hour of nursing actions.
2	Anxiety related to unknown outcome of the disease and surgery evidenced by patient's numerous question about his condition.	Patient will demonstrate signs of reduction in anxiety within 8 hours of nursing care.	- Advise patient close a patient that is recovering from the same condition. - Create nurse-patient therapeutic relationship and involve him in his care. - Implement all necessary management protocols that will enable him empty his bladder. - Answer his question properly.	- Admitting him close to a patient that is recovering from the same condition instills hope in him and reduces anxiety. - Creating therapeutic relationship and involving him in his care builds confidence in him and helps to reduce anxiety. - Implementing management protocols helps to relieve patient of the pain and reduces anxiety. - Answering his question properly helps to remove misconceptions about the condition thereby reducing anxiety.	Patient demonstrated signs of reduction in anxiety as evidenced by his freely discussion with his visitors.

POST OPERATIVE NURSING CARE OF A PATIENT THAT WENT FOR PROSTATECTOMY

SN	Diagnosis	Objective	Order/Action	Rationale	Evaluation
1	Ineffective air-way clearance related to the effect of anaesthesia	Patient will maintain patent air-way throughout the period he is under the effect of anaesthesia.	- Place patient on supine position with head turned to one side. - Ensure that air-way apparatus is intact. - Suction accumulated mucus, if any. - Monitor patient VS/ especially respiration	- Turning patient head to one side prevents the tongue from falling backward and blocking the airway. - Air way apparatus enable patent air-way. - Suctioning of mucus prevents blockage of air way patients. - Monitoring patients respiration/vital sign helps to detect when airway occlusion occurs.	Patient maintained patent air way throughout the period he was under the effect of anaesthesia.

D. Short note on Ibuprofen

- Group - Non-steroidal anti-inflammatory drug.
- Mechanism of action-stops the synthesis of prostaglandin thereby reducing pain.
- Dosage - 400-800mg tds
- Side effects-esophageal varice, peptic ulcer, prevents blood clotting.
- Nursing responsibilities
 - Do not serve the drug on empty stomach
 - Do not give to patient with peptic ulcer.
 - Serve drug with milk if necessary
 - Do not serve with other NSAID like Aspirin

QUESTION

Mr. Zungwen, a 72year old man was brought to your hospital with history of inability to pass urine and obvious supra-pubic tenderness.

The doctor on duty made an impression of Benign Prostatic Hyperplasia.

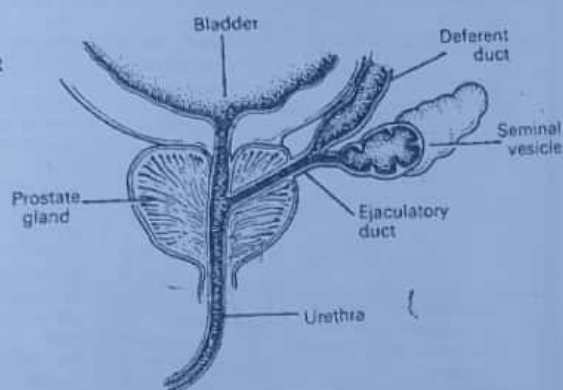
- A. Draw a well labeled diagram showing the position of the prostate gland and associated structures and describe the prostate gland.

Diagram 1mk
Labeling 1mk
Description of prostate gland - 3mks

- B. What is Benign Prostatic Hyperplasia? 1mk

- C. List four(4) investigation that will help the doctor to reach a deficit diagnosis (2mks)
 D. Briefly described the pathophysiology of Benign Prostatic Hyperplasia (3mks)
 E. Identify two (2) pre-and one post-operative nursing problems of Mr. Zungwen and utilize the nursing care plan to solve the identified problems. (9mks)

ANSWER



- A. Diagram that shows the position of the prostate gland and associated structures.

Labelling (inclusion of the following)

1. Urethra
2. Bladder
3. Deferent duct
4. Seminal vesicle
5. Prostate gland
6. Ejaculatory duct

iii. Description of prostate gland

- Prostate gland is an accessory gland for reproduction.
- It is located at the neck of the urinary bladder at the first segment of the urethra.
- It is near the size and shape of a walnut measuring about 4cm in length and 2cm in width.
- It surrounds the prostatic urethra at the base of the bladder.
- It also surrounds the ejaculatory ducts

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- The gland is composed of an indistinct smooth muscle capsule and numerous smooth muscles. Partitions that radiate inward downward the urethra
- Covering these muscular partitions is a layer of columnar epithelial cells that form saccular dilations into which the cells secrete prostatic fluid
- 20-30 small prostatic ducts transport these secretions into the prostatic urethra. (1/2 mk each, 6 points = 3mks)

- B. Benign Prostatic Hypertrophy is a non cancerous enlargement or hyperplasia of the prostate gland. It is a condition in which the prostate gland contains more than the usual number of cells.

C. Investigations to help confirm diagnosis in BPH.

- Physical examination - Digital Rectal Examination
- Transrectal ultrasound scan of the prostate
- Urodynamic flow studies
- Cystoscopy
- Cystourethroscopy
- Serum creatinin level estimation
- Blood urea level estimation
- Prostate specific Antigen

D. Brief description of pathophysiology of BPH

Benign Prostatic Hypertrophy is a non-cancerous enlargement of the prostate gland. When the gland enlarges, it extends upward into the bladder leading to gradual compression of the urethra. This leads to dribbling of urine or incomplete emptying of the bladder. As the compression continues, there is retention of urine in the bladder which may lead to urinary tract infection subsequently; the retention will lead to hydronephrosis. Prolonged obstruction can lead to kidney stone and in conjunction with hydronephrosis brings about renal failure.

Nursing Diagnoses of a patient booked for prostatectomy (pre-operative Diagnoses)

1. Acute pain (at supra-bubic region) related to retained urine in the bladder as evidenced by patient's restlessness and suprabubic tenderness and no urine voided.
2. Anxiety related to unknown outcome of the surgery as evidenced by patient's facial expression.
3. Knowledge deficit related to the cause and management of the disease, as evidenced by patient's numerous questions about the cause and management of the disease.

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NURSING CARE PLAN OF A PATIENT WITH BPH

SN	Nursing Diagnosis	Objective	Order/Interventions	Rationale/Scientific principle	Evaluation
1	Acute pain (suprapubic) related to retained urine in the bladder evidenced by patient's restlessness and suprapubic tenderness and no urine voided.	Patient will show signs of reduction in pain within 30mins to 1 hour of nursing actions	- Assess the level of pain on the patient. - Set tray for catheterization. - Assist the urologist or doctor during the procedure. - Allow patient to assume any comfortable position. - Re-assess the level of pain.	- Assessing the level of pain will enable the nurse to know when the pain reduces. - Setting the tray and assisting the doctor in catheterization helps to reduce delay in draining the retained urine. - Catheterization helps to evacuate urine from the bladder thereby reducing the pain. - Comfortable position to the patient helps to reduce pain momentarily. - Re-assessing the level of pain helps to determine whether the pain reduced following the nurse's actions.	Patient showed signs of reduction in pain after 45 minutes of nursing actions.

POST-OPERATIVE DIAGNOSES OF A PATIENT THAT WENT FOR PROSTATECTOMY

1. Ineffective air-way clearance related to the effect of anaesthesia
2. Pain related to surgical incision evidenced by patient's groaning
3. Deficient self care deficient (hygienic) related to confinement on bed secondary to surgery as evidenced by patients inability to carry out actions.
4. Risk for infection related to aseptic management of incision site
5. Risk for sexual dysfunction related to structural changes secondary to surgical manipulations

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POST-OPERATIVE CARE PLAN

SN	Nursing Diagnosis	Objective	Order/Intervention	Rationale/Scientific principle	Evaluation
1	Ineffective airway clearance related to the effect of anaesthesia	Patient will not show any sign of airway obstruction throughout the period he is under the effect of anaesthesia	- Check vital signs especially respiration. - Maintain the airway-in situ. - Suction mucus pm. - Turn the patient's head to one side while lying supine.	- Checking the vital signs especially respiration will help to ensure that there is patency of the airway. - Airway in situ will help to ensure patency of the airway. - Suctioning of mucus ensures removal of obstruction from the air passage. - Turning patient's head to one side while lying supine helps to prevent the tongue from blocking the airway.	Patient did not show any sign of airway obstruction through the period of he is under the effect of anaesthesia
2	Deficient self care (hygienic) related to confinement on bed secondary to surgery.	Patient will develop self care within 2-3 days after the surgery	- Bed bath patient until he is able. - Assist him in oral care until he is able to do so. - Encourage him on early ambulation	- Bed bathing will help to maintain personal of the patient. - Assisting him in oral care helps ensure personal hygiene and prevent halitosis. - Early ambulation will help the patient to develop self care	

Nursing diagnoses = 1/2mk, Nursing Objective = 1/2, Nursing Order = 1mk, Rationale/Scientific principles = 1/2mk, Evaluation = 1/2mk

QUESTION

Mr. Damlan, a 68year old man was brought to the emergency unit of your hospital in unconscious state obvious hemiplegia, aphasia, weakness and speech disturbances.

On further examination, the doctor on duty noted a blood pressure of 200/110mmHg and made an impression of cerebrovascular accident (CVA)

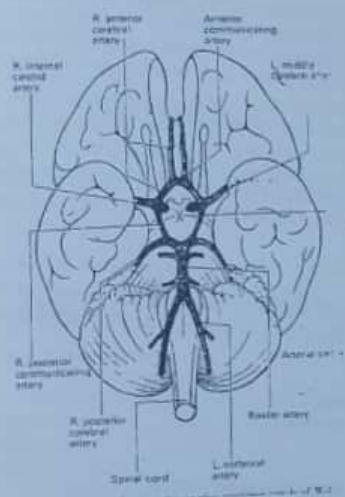
- A. Without description, sketch blood flow in the brain (circle of Willis)
Diagram - 1mk

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ii. Labeling-2mks

- B. What is cerebrovascular accident (CVA)? – 1mk
 C. Briefly describe the pathophysiology of stroke – 4mks
 D. Identify three major nursing problems of Mr. Damian according to the priority and utilize Nursing care plan to solve the identified problems 9mks
 E. Enumerate the likely complications Mr. Damian will suffer from if he is not properly managed – 2mks

Answer A



Sketch of circle of willis

B. CVA or stroke is clinical syndrome that accompanies a sudden and sometimes severe attack of weakness that affects one side of the body due to total or partial interruption of blood flow in the brain. It is an interruption of blood supply to a part of the brain and development of neurological deficit which occurs as after effect of haemorrhage or thrombosis of the blood vessels of the brain.

C. Pathophysiology of CVA

CVA is a medical emergency that results from interruption in oxygen supply to the

brain due to presence of cerebrovascular lesion. In even of thromboembolic disease due to arteriosclerosis in the major cranial arteries leading to vascular occlusion, or cerebral haemorrhage following excessive vascular presence, anastomotic channels from other arterial territories may restore the brain. A reduction in oxygen perfusion brings about compensating mechanism to maintain oxygenation thereby preventing apparent clinical effect on the patient. If blood flow increases again, function returning to normal and patient is said to have Transient Ischaemic Attack (TIA).

However, if flow fails further, it may lead to a level where cell death begins. Hypoxia leads to reduction in ATP supply which affects the membrane pump with influx of sodium (Na^+) and water into the cell (cytotoxic oedema), and release of excitatory neurotransmitter glutamate, into the extracellular leads to the opening of membrane channel permitting influx of calcium and more sodium into the neurons. This activates the intracellular enzymes that complete the destruction process. Inflammatory mediators are released by microglia and astrocytes which leads to the death of all cells in the area of maximum ischaemia. The absence of ATP leads to anaerobic production of lactic acid with consequent fall in tissue pH. Outcome depends on the competence of the circulatory homeostatic mechanism severity and duration of reduction in blood flow.

Lack of O_2 and glucose affect the efficiency of the neurons leading to irritability and disruption in nerve occurs at the opposite side of the body affected in the brain. If infarction occurs in the hypothalamic area near the fourth ventricle, changes occurs which interferes with temperature regulation and antidiuretic hormone production which affects urinary output. Basal metabolic rate rises due to increase temperature affecting metabolism.

B. Nursing Problems of Mr. Damian with CVA

1. Ineffective airway clearance related to presence of secretions in the airway, evidenced by patient's ballooning of the cheeks during exhalation
2. Hyperthermia related to the involvement of thermoregulatory centre in the brain, evidenced by high thermometric reading above 37.2°C .
3. Deficient self care (bathing, toileting) related to paralysis, evidenced by patient's inability to move his hands effectively.
4. Imbalanced nutrition; less than body requirement related to unconscious state evidenced by patient's inability to swallow.
5. Risk for development of pressure sore related to immobility and incontinent of urine.

See care plan

C. Complications of CVA

1. Constipation
2. Incontinent
3. Renal insufficiency

4. Pressure sore
5. Injury due to fall
6. Infection eg. Hypostatic pneumonia.

QUESTION

Write short note on the following

- A. Anorexia Nervosa
 - B. Nasogastric feeding
 - C. Sinusitis
 - D. Hydrocephalus
- (5mks each)

Answer

A. Anorexia Nervosa

Definition:

This is a eating disorder classified under psychoneurosis. It is a condition that is characterized by dissatisfaction with body weight and thinner body size. The individual achieves severe weight loss by strict dieting, excessive exercise, laxative abuse or vomiting. It mostly affects female young adults and adolescents.

Signs and Symptoms

- ☐ Lack of desire to eat
- ☐ Patient looks emaciated
- ☐ Menstrual cessation
- ☐ Indulgence in physical exercise

Nursing Management

- ☐ Emotional support around meal times
- ☐ Monitoring of weight
- ☐ Patient education on meal planning, skills to resist social pressure
- ☐ Relaxation skills
- ☐ Quantity, quality and pattern of intake of food.
- ☐ Drugs that stimulate appetite is given as prescribed eg. T. cypron/cypri-gold 1 daily

B. Naso-gastric Feeding

This is defined as feeding a patient by the use of naso-gastric tube. Only fluid diet can be fed through this method.

Indications

- ☐ Unconscious patient
- ☐ Cases of mouth and throat surgery
- ☐ Patient with bulbar paralysis

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- ☐ Weak infants that are unable to suck eg preterm baby.
 - ☐ Psychiatric patients who refuse to eat eg in severe depression
- In Naso gastric feeding the tube can be retained insitu for feeding or be passed each time the patient needs to be fed.

Method

- ☐ Pass the tube, aspirate to be sure that it is in the gastrum.
- ☐ The feed must be warm eg. Temperature of 38°C.
- ☐ There is need for privacy so screen the bed
- ☐ A 20cc syringe is used to introduce the feeding with the patient in sitting up or semi-reccumbent position and the tube raised higher above the level of patients head.
- ☐ The quality of feed given must be recorded with the time of feeding.
- ☐ Flush the tube with water after feeding and serve medication, when due, at the same time.
- ☐ Reassure patient and make him comfortable.

Complication

If the tube is not tested to make sure that it is in the gastrum feed may be introduced into the respiratory tract and this will lead to aspiratory pneumonia.

C. Sinusitis

Definition: This is an acute or chronic inflammation of the sinus cavities resulting from infection or allergies. It is often accompanied by rhinitis or runny nose.

Signs and Symptoms

- ☐ Feeling of heaviness
- ☐ Discomfort
- ☐ Headache
- ☐ Mild fever is present when infection
- ☐ General malaise

Management

Management is symptomatically carried out

- ☐ Give prescribed anti-inflammatory drug eg. Aspirin 300mg tds, Prednisolone 5mg tds.
- ☐ Fluid intake is increased
- ☐ Serve balanced diet
- ☐ Adequate rest is encouraged

D. Hydrocephalus

Definition: This is enlargement of the skull due to abnormal collection of CSF

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around the brain or in the ventricles. It can be congenital or acquired from inflammation of the meninges during infancy.

Causes/Types

If blockage is in the foramina of the 4th ventricle or cerebral aqueduct, there is accumulation within the ventricle and is called internal hydrocephalus. If the CSF accumulates in the subarachnoid space, it is called external hydrocephalus.

Pressure is applied to the brain due to this accumulation, compressing neural tissues and causing brain damage.

Severe enlargement of the head equally occurs.

Management

- Give supportive care.
- Drainage tube (shunt) may be placed between the brain ventricles and one of the body cavities to eliminate the high internal pressures. However these shunts may provide route through which infection can be introduced into the brain and shunt must be replaced as the person grows.

QUESTION

Discuss your responsibilities as a nurse in the under-listed procedures

- A. Haemodialysis
- B. Oxygen administration
- C. Lumbar puncture
- D. Indwelling catheter (5 marks each)

Guideline with exception of D

- i. Before the procedure
- ii. During the procedure
- iii. After the procedure

Answer

A. Nurse's responsibilities in haemodialysis

Haemodialysis is a procedure carried out on renal patients to remove accumulated toxic wastes in the blood such as creatinine thereby assisting the kidney functions to return to normal.

Before the procedure

- Explain the purpose of the dialysis to the client and family.
- Possible visit to the dialysis unit to show the patient and introduce him to the unit staff if his condition permits to allay anxiety.
- Client is positioned comfortably on bed or reclining chair with the exposure of the artero-venous fistula or shunt.
- Observe patient's vital signs and the weight and record to form a

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baseline data.

- Laboratory result for PCV, electrolyte, blood urea and creatinine levels as well as clotting time are collected and kept for the doctor.
- Get prescribed drugs ready.
- Reassure patient

During the Procedure

- Observe the vascular access site for signs of infection, haematoma or infection.
- Observe the patient's skin for dryness, tugour, oedema.
- Keep monitoring the vital signs according to schedule.
- Monitor rate of flow, blood clotting and heparinization.
- Monitor the machine for normal functioning.
- Follow doctor's instruction strictly and report any deviation from normal to the doctor.
- Administer all prescribed drugs due in the procedure time except anti-hypertensives.

After the Procedure

- Clamp the line and allow the blood in the circuit to return to the client.
- Remove or disconnect dialysis line and reconnect the cannula, in case of a shunt.
- Apply dressing with pressure to prevent bleeding.
- Observe the area for a while.
- Return used instruments, wash and sterilize as appropriate.
- Make the patient comfortable.
- Check the vital sign and weight.

Clients on regular dialysis should be educated on:

1. Diet
2. Care of the artero-venous fistula or shunt.
3. Signs and symptoms of worsening condition and to return to the hospital when such occurs.

B. Nurse's responsibilities in oxygen administration

Oxygen is artificially administered to patients with tissue hypoxia which may occur due to respiratory disease, anaemia or cerebrovascular diseases.

Before the Procedure

- Check the arterial O₂ level
- Check the O₂ cylinder to ensure that there is oxygen and the cylinder is in good functional state.
- Inform the patient about the procedure
- Set the requirements

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- Ensure that there is no naked light within the vicinity
- Place patient on a sitting up position.
- Provide privacy
- Put caution "No smoking or carrying flame of fire about".
- Clean clients nostrils with cotton buds or ask client to blow his nose if he is able
- Reassure client

During the Procedure

- Regulate the flow
- Lubricate the nasal catheter and insert it into the nostril making sure that it passes along the floor of the nose.
- Fasten the catheter with strapping
- Fasten the excess tubing to the proximal end of the bed
- Reassure patient as the procedure goes on.
- Monitor effect of therapy by observing patient's response.

After the procedure

- Disconnect tubing/catheter
- Clean patient's nostril
- Make patient comfortable
- Remove equipment to designated places observe client for complications and report if any.

C. Nurse Responsibilities in Lumber puncture

Lumber puncture or Spinal tap is a procedure carried out to obtain cerebrospinal fluid for examination and analysis. It can also be done as a means of giving intra-thecal injection such as spinal anaesthesia to the client.

Before the Procedure

- Explain the procedure to the patient to reduce his anxiety
- Assemble required equipment.
- Ensure that consent form is signed by the client or his/her relation.
- Help to position patient on a couch or edge of the bed with his back towards the doctor
- Instruct patient to flex the thigh and legs as much as possible to create space between the spinous process especially at L3 - L4 & L4 - L5.
- Provide privacy

During the Procedure

- Assist the patient to maintain the position to prevent injury
- Describe the procedure as it goes to the patient.
- Reassure him as the procedure goes.

After the Procedure

- Apply sterile dressing at the puncture site
- Instruct patient to lie prone 2-3 hours to allow separation of dura from the arachnoid matter.
- Monitor for complications
- Encourage patient to increase fluid intake
- Send the specimen to the lab without delay.
- Encourage patient to rest for a long period of time (between 6 - 24 hours).

D. Nurse's responsibilities to a client on indwelling catheter

- The nurse should observe strict hand washing or scrubbing before and after attending to the patient.
- Do bd vulva or perineal meatus swabbing and rotation of the catheter to prevent it from sticking to the urethral wall.
- Encourage patient to drink fluid liberally except where his/her condition does not permit that.
- Ensure that the catheter does not kink.
- The catheter should be fastened on the patient's anterior thigh i.e. patient should not lie on the catheter.
- Change catheter prn, 5 days - 2 weeks.
- Encourage patient not to pull the catheter.
- Serve prescribed urinary track antiseptic eg Nitrofurantion
- The uribag should not touch the floor.
- Empty the uribag when it is three quarter full.
- Keep strict fluid intake and out-put recording.

NB: Answer to the question on indwelling catheter may not require the application of the instruction on **before, during and after.**

QUESTION

Write briefly on the flowing drugs using the guideline below

- Tramadol
- Coartem
- Amoxillin
- Azidothymidine (Zidovudine)
- Doctors prescription reads
"Give injection Gertamycin 40mg 7 hourly x 5/7
If the drug comes in one Ampule of 80mg in 2ls, how many Ampules will the Nurse collect from the pharmacy shop?"

Answer

- Tramadol
(i) Group- Opioid analgesic.

- (ii) Mechanism of action: Stops moderate to severe pain such as post-operative and amputation pains.
- (iii) Indications-Management of severe acute or chronic pain, painful diagnostic measures, post-surgery pain, pain associated with fractures, cancer, dislocation and osteoarthritis.
- (iv) Dosage-50-100mg bd.
- (v) Side effects-constipation, drowsiness, confusion, respiratory depression, euphoria, nausea and vomiting.
- (vi) Nursing responsibilities
 - Do not give with alcohol because it potentiates the effect of alcohol.
 - Do not give with sedative hypnotics or psychotic drugs.
 - Avoid concurrent administration with Mono Amino Oxidase inhibitors.
 - Give with caution in pregnancy and lactation.
 - Do not give to patients who have demonstrated hypersensitivity to tramadol.

B Coartem

- (i) Group-Multiple drug resistant Antimalaria, Amyl alcohol.
- (ii) Mechanism of action:- Stops the growth and multiplication of the falciparum by destroying their cell wall.
- (iii) Indication: Malaria especially falciparum malaria.
- (iv) Dosage-4 tablet twice daily x3 days.
- (v) Side effects:- Abdominal pain, diarrhea, rashes, headache, vomiting, cough.
- (vi) Nursing responsibilities
 - Severe drug with food to enhance absorption.
 - Do not give in concurrent with mefloquine derivative.
 - Do not give to individual with hypersensitivity to the active ingredients.

C Amoxicillin

- a. Group-Penicillin group of antibiotics.
- ii. Mechanism of action-stops the synthesis of bacterial cell wall.
- iii. Indications- urinary tract infection, Enteric fevers e.g Typhoid fever, Respiratory tract infections, otitis media, Bronchitis, etc.
- iv. Dosage-250-500mg 12 hourly or 8 hourly.
- v. Side effects - skin rashes, constipation or diarrhea, hypersensitivity reactions, nausea and vomiting.
- vi. Nursing responsibilities
 - Do not administer to patient with hypersensitivity to penicillin.
 - Do not administer at the same time with antacids.
 - Give with caution in patients with renal diseases.

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Administer with probenecide to reduce the rate of excretion through the renal system

Azidothymidine (Zidovudine)

- (i) Mode of Action:- it is an antiviral drug, a transcriptase inhibitor that interferes with the DNA synthesis that is needed for viral replication.
- (ii) Indication:- Antiviral drug used in the management of patients with HIV positive pregnant woman.
- (iii) Dosage:- 100-200mg 4 hourly or 1-2mg/kg/dose every 4 hours.
- (iv) Route of administration:- Orally

- (v) Side effects:- Anaemia, neutropenia and Leucopenia, severe headache, low platelet count, neurotoxicity, cholestatic jaundice, myopathy, weakness.
- (vi) Nursing Responsibilities:-
 - Advise patient to take drug 30 minutes before or one hour after a meal with a glass of water.
 - Observe patient for any adverse effect as it may require discontinuation eg. Anaemia.
 - Patient to adhere to 4 hourly dosage schedule. Alarm clock may be necessary.

NB: Zidovudine is neither a cure for HIV infection nor does it prevent the risk of transmission of HIV to others.

F. Doctor's prescription

Injection Gentamycin 40mg 8 hourly x 5/7
1 Ampule = 80mg - 2mls.

Solution

Daily dosage = 40mg x 3 = 120 mg.
Dosage for 5/7 = 5 x 120 = 600mg
Total dosage = 600mg.
Total Ampule = X
1 Ampule = 80mg
X Ampule = 600mg.

$$600\text{mg} \times 1 \text{ Ampule} = 80\text{mg} \times X \text{ Ampule}$$

$$\frac{\text{Total Ampule (X)} - 600 \times 1}{X} = 80$$

$$X = \frac{600 \times 1}{80} = 7.5 \text{ Ampule} \approx 8$$

$$\frac{1}{2} \text{ Total Ampule to be collected (x)} = 8 \text{ Ampules.}$$

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8 Ampules of Gentamycin will be collected from the pharmacy for master Uche.

QUESTION

Mrs. Dave a 54 year old petty trader reported to your hospital with obvious swollen at the anterior neck region and mild pain during swallowing. The doctor on duty made a diagnosis of endemic goiter and booked her for thyroidectomy in two weeks time.

Answer

- Draw a well labeled diagram showing the position of the thyroid gland and the associated structure.
- List four (4) investigation that assisted the doctor in reaching the diagnosis. 2mks
- Briefly describe the pathophysiology of endemic goiter
- Identify (2) pre-operation and one post operative nursing problems of Mrs. Dave and utilize the nursing care plan to solve the identified problems in order of priority. 9mks

Position of the thyroid gland and associated structure

See previous sketch.

- | | | |
|---|---|----------------------------|
| A | = | External carotid artery |
| B | = | Inferior thyroid vein |
| C | = | Thyroid cartilage |
| D | = | Left lobe of thyroid gland |
| E | = | Isthmus of thyroid gland |
| F | = | Trachea |
| G | = | L. Common carotid artery |
| H | = | R. Subclavian artery |
| I | = | R. Lobe of thyroid gland |

B. Four investigations that assisted the doctor to make diagnosis of Endemic goiter.

- Palpable lesion that is fixed and hard at the neck region.
- Estimation of serum thyroxin level to confirm normal thyroid function.
- X-ray of the chest and thoracic inlet to determine the degree of tracheal deformity due to pressures.
- Radioactive isotopes scan to determine the anatomy and position of the goiter.

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- CT scan.
- Estimation of serum thyroid stimulating hormone (TSH).
- Ultrasound scanning to distinguish thyroid mass from a cystic swelling.

C. Pathophysiology of endemic goiter

Endemic goiter is an asymptomatic condition of the thyroid gland with swelling at the neck region which often results from insufficient iodine intake. In an event of reduced iodine in the body the ability of the thyroid gland to produce its hormones (T3 & T4) reduces.

To this effect, there is excessive stimulation of the gland by thyroid stimulating hormone (TSH) from the anterior pituitary to produce its hormones. This excessive stimulation will lead to the enlargement of the glandular tissues with obvious painless swelling in the neck region. The enlargement may lead to tracheal compression which results in dyspnoea. Further compression may affect the pharynx leading to dysphagia. Where recurrent nerve is affected, voice becomes hoarse.

D Pre-operative Nursing problems of Mrs. Dave with Endemic goitre

- Pain (acute) related to compression of the pharynx and trachea by the enlarged gland as evidenced by patient expression of pain during swallowing.
- Imbalanced Nutrition: Less than body requirement, related to dysphagia as evidenced by patient's inability to finish a plate of meal served.
- Anxiety related to unknown outcome of the surgery as evidenced by patient's facial expression.

Post Operative nursing problems of Mrs. Dave

- Ineffective airway clearance related to the effect of anaesthesia as evidenced by the presence of secretions in the airway.
- Pain (acute) at incision site, related to surgical incision as evidenced by patient's complaints.
- Risk for haemorrhage related to surgical manipulation of blood vessels around the gland.
- Risk for infection related to surgical incision.

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PRE-OPERATIVE NURSING CARE PLAN OF A PATIENT BOOKED FOR THYROIDECTOMY

S/N	Diagnosis	Objective	Order/Action	Rational/Scientific	Evaluation
1	Pain (acute) related to compression of the pharynx by the enlarged gland evidenced by patient's facial expression during swallowing.	Patient will show signs of reduction in pain within 7 days of nursing action.	• Serve patient soft diet. • Use diversional therapy on the patient e.g. tuning to preferred TV or Radio station or use of print media. • Serve prescribed analgesics e.g. paracetamol 1000mg tds. • Serve prescribed anti-thyroid drugs e.g. Carbimazole 30-60mg b.i.d.	• Soft diet passes easily down the GI thereby reducing pain during swallowing. • Diversional therapy removes the patient's concentration on the pain thereby giving temporal relief of pain. • Analgesics reduce pain by acting on the pain centre and increasing the patient pain threshold. • Anti-thyroid drugs help to bring the gland to euthyroid state thereby reducing the compression at pharynx and reducing the pain.	Patient demonstrated signs of reduction in pain as evidenced by her not complaining of pain during swallowing after five days of nursing care/treatment.
2	Anxiety related to the unknown outcome of the surgery, evidenced by patient's numerous questions about the surgery.	Patient will demonstrate signs of reduction in anxiety within 2-3 days of admission.	• Admit patient close to close to another patient that has undergone the surgery or related surgery. • Create nurse/patient therapeutic relationships and involve her in her care. • Answer patient's queries about the surgery tactfully. • Use diversional therapy e.g. switch on TV/Radio or give readable news print. • Serve prescribed sedatives e.g. diazepam.	• Admitting patient to another patient that is recovering from the surgery instill hope on the patient and reduces anxiety. • Nurse/patient therapeutic relationship makes patient to feel at relaxed thereby reducing anxiety. • Answering her question tactfully helps to build hope and confidence on the patient thereby reducing anxiety. • Sedative induces sleep and reduces anxiety.	Patient showed signs of reduction in anxiety after two days of admission evidenced by reduction in his question about the surgery.

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POST-OPERATIVE NURSING CARE PLAN OF A PATIENT THAT WENT FOR THYROIDECTOMY

S/N	Diagnosis	Objective	Order/Action	Rational/Scientific	Evaluation
1	Ineffective airway clearance related to the effect of anaesthesia, as evidenced by the presence of secretion in the airway and the mouth.	Patient will maintain patent airway throughout the period she is under the effect of anaesthesia.	• Lie patient supine with her head turned to one side. • Maintain artificial airway in-situ. • Suction secretions using suction machine or manual suction apparatus pm.	• Supine position with head turned to one side aids free flow of secretions without blocking the airway. • Artificial airway allows easy exchange of air and prevents asphyxiation. • Suctioning helps to remove the secretions thereby allowing the airway to remain patent.	Patient maintained patent airway throughout the period she was under the effects of anaesthesia.

QUESTION

Mrs. Tezzy, a 52 year old petty trader came to your hospital with obvious swelling at the neck region, complaint of hoarseness of voice, and pain during swallowing. The doctor on duty made a diagnosis of Endemic goiter and booked her for thyroidectomy in two weeks time.

- A. Utilizing a well labeled diagram describe the thyroid gland
- Diagram - 1 mark
 - Labeling - 1 mark
 - Description - 3 marks

B. Describe the necessary preparations the nurse will carryout on Mrs. Tezzy before the surgery on the following areas.

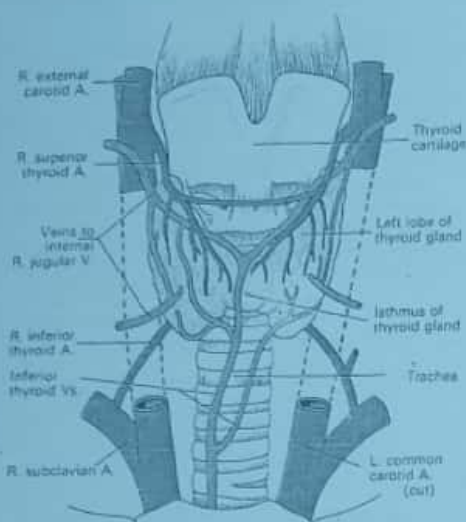
- Pre-Operative Medication - 2 marks
- Pre operative teaching - 2 marks

C. Identify two (2) pre-operative and one post operative Nursing problems of Mrs. Tezzy and utilize Nursing care plan to solve the identified problems - 9 marks.

D. Enumerate two complications that Mrs. Tezzy may suffer after the surgery. - 2 marks

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ANSWER



Ai. Diagram of the thyroid gland.

ii. Labeling in the following inclusions

1. Thyroid cartilage
2. Left lobe of thyroid gland
3. Right and Left subclavian artery
4. Right lobe of thyroid gland
5. Trachea
6. Isthmus of thyroid gland
7. Right and Left common carotid artery
8. Right and Left superior thyroid artery
9. Inferior thyroid vein

Any 3 inclusion = 1/2 mk; 6 inclusion = 1mk

iii. Description of thyroid gland

Definition: Thyroid gland is an endocrine gland that is located at the neck region.

Position: Located at the neck region at the level of 5th, 6th and 7th cervical vertebrae and 1st thoracic vertebra.

Shape: It is like butterfly in shape having two lobes (left and right)

Size: It measures 5cm long and 3cm wide

Structure: Thyroid gland is highly vascular with fibrous capsule surrounding it

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consists of two lobes, one on each side of the thyroid cartilage and the cartilaginous ring of the trachea. The lobes are joined by the isthmus in front of the trachea.

Blood supply: Superior and inferior thyroid arteries supply blood to the gland.
Venous drainage: Thyroid vein.

Functions of the thyroid gland

The thyroid gland secretes thyroid hormones

1. Thyroxine (T4)
2. Triiodothyronine (T3)
3. Calcitonine

Iodine from ingested food helps in the formation of these hormones, especially T3 & T4 and the body concentrates it and releases it on daily basis for body use.

The release of hormones of the thyroid gland is assisted by the presence of thyroid stimulating hormone (TSH) through stimulation by thyroid releasing hormone from the hypothalamus.

- T3 & T4 help to increase Basal Metabolic Rate.
- They also regulate metabolism of carbohydrate, protein and fats.
- These are essential for growth and development.
- The gland stores and releases iodine
- Calcitonine reduces calcium level in the body.

B. Necessary preparations for thyroidectomy

i. Pre-operative medication

Aims of pre-operative medications include

- ☐ To help bring the gland to euthyroid state
- ☐ Reduce the risk of hemorrhage
- ☐ Render the gland firm before surgery

Measures to achieve these

1. Lugol's iodine - 5 drops tds or potassium iodine 60mg tds
2. Propranolol - 80mg orally 6 hourly for 4-7 days - to control some β_1 adrenergic stimulations that may occur in thyrotoxicosis
3. Carbimazole - 30-60mg dly
4. Diazepam 5mg nocte, 2mls - 1ml for aim and 1ml for any 2 drugs

ii. Pre-operative teaching

Aim: to prevent stress on the incision site after surgery. Teach the patient how to support the neck with the hands.

Method:

Raise the elbows and place hands behind the neck to provide support and reduce strain and tension on the neck muscles and surgical incision.

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C. Pre-Operative Nursing Diagnoses

1. Pain (acute) related to compression of the pharynx by the enlarged gland evidenced by patient's expression of pain during swallowing.
2. Imbalance Nutrition: Less than body requirement related to dysphagia evidenced by patient's inability to finish a plate of meal served.

NB: Evidenced could also be by patient complaints.

3. Anxiety related to the unknown outcome of surgery evidenced by patient's numerous questions about the surgery.

NB: Evidence can also be by patient's facial expression.

Post-Operative Nursing Diagnoses

1. Ineffective airway clearance related to the effect of anesthesia, evidenced by presence of secretions at the airway.
2. Pain (acute) at incision site, related to surgical incision evidenced by patient's complaints.
3. Risk of haemorrhage related to surgical manipulation of the blood vessels.
4. Risk for infection related to surgical incision.

NB: Revisit answer to question 13C for nursing care plan

D. Complications of thyroidectomy

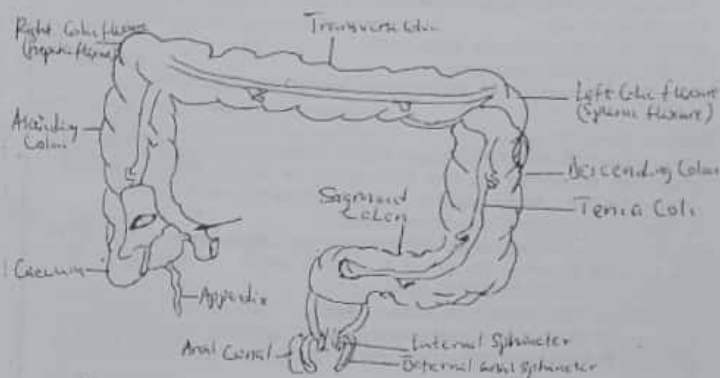
- Haemorrhage
- Infection
- Trauma to the laryngeal nerve and subsequent loss of voice
- Scar tissue formation at the neck
- Tetany

QUESTION

1. Mrs. Kuun, a 35 year old woman was admitted into the surgical ward following accidental ingestion of corrosive (acid) and had colostomy performed.

- a. With a well labeled diagram describe the large intestine (5mks)
- b. Define colostomy (1mk)
- c. List four (4) other indications for colostomy (2mks).
- d. Outline the three types of colostomy (3mks).
- e. Utilizing the nursing care plan identify and solve in order of priority one (1) pre-operative and two (Post operative nursing diagnosis of Mrs. Kuun (9mks)

Answer



The Large Intestine

Description

- Begins at the caecum in the right iliac fossa and terminates at the rectum and anal canal.
- Length: It is about 1.5 metres long and 7.5cm wide.
- Situated in the right iliac fossa deep in the pelvis.
- It forms an arch round the coiled-up small intestine.
- It is divided into caecum, ascending colon, transverse colon, sigmoid colon, rectum and anal canal.
- It is made up of four (4) layers, viz: adventitia or outer covering, muscle layer, submucosal layer and mucosa lining.
- The muscle layer is made up of three (3) bands collectively called taenia coli.
- Blood supply-arterial is mainly by the superior and inferior mesenteric arteries while venous drainage is superior and inferior mesenteric veins.
- Function is mainly absorption of digested materials.

b. Definition of Colostomy

Colostomy is an artificial opening (stoma) in the large intestine (colon) brought to the surface of the abdomen for the purpose of evacuating the bowel. (1mk)

c. Four other indications for colostomy

- Imperforate anus
- Injury or abnormality to the spinal cord
- Cancer of the colon and rectum.
- As temporary measure to relieve obstruction of the large bowel.

d. Types of Colostomy

- Loop colostomy-where a section of the colon is brought out onto the abdomen supported on the surface by a rod or bridge, opened longitudinally on anterior wall and mucosal edges sutured to the skin
- Single loop colostomy. Has only one stoma.
- Double barrelled colostomy. The proximal and distal loops of the bowel are sutured together for about 10cm (4inches) and both ends are brought up onto the abdominal wall.
- Divided colostomy: Consists of two edges of bowel brought out into the abdomen but separated from each other. The opening from the digestive or proximal ended is the colostomy while the distal end is the mucous fistula.
- Wet colostomy: Where urine and faeces are passed through one opening done especially after transplantation of the ureters (1mk each x 6 = 6mks)

e. Pre-Operative nursing diagnosis

- i. Pain related to tissue compression evidenced by verbalization and facial expression of patient.
- ii. Anxiety related to impending surgery and diagnosis of cancer, evidenced by patient's facial expression.
- iii. Imbalanced nutrition less than body requirement nausea and anorexia evidenced by patient's refusal to eat.
- iv. Deficient knowledge about surgical procedure and pre-operative preparation evidenced by patient asking too many questions.

Post-Operative nursing diagnosis

- i. Pain acute related to surgical intervention evidenced by verbalization.
- ii. High risk for infection related to possible contamination of abdominal cavity evidenced by increased pulse rate and pyrexia.
- iii. Risk for impaired skin integrity related to irritation of the peristomal skin by effluent.
- iv. Body image disturbance related to abdominal stoma change on faecal elimination and control of odour.

PRE-OPERATIVE NURSING CARE PLAN OF A PATIENT BOOKED FOR COLOSTOMY

S/N	Nursing Diagnosis	Objective	Nursing order/Intervention	Scientific principle/Rationale	Evaluation
1	Anxiety related to impending surgery and diagnosis of cancer evidenced by patient's facial expression.	Patient will show signs of reduction in anxiety within 24 hours of nursing intervention.	1. Educate patient about the surgery and diseases condition. 2. Encourage patient to verbalize his fears and concerns. 3. Listen attentively to patient's complaints and questions. 4. Be honest in producing information in response to patient's complaints. 5. Introduce patient to another patient who had the same surgery and is progressing well.	1. Information eliminates doubt and fear and allays anxiety. 2. Verbalizing fears reveals areas of concern and misconception and creates opportunity for the misconception to be corrected thereby reducing anxiety. 3. Listening helps the nurse to know areas to lay emphasis thereby reducing the pain. 4. Information helps to improve her understanding and also builds confidence thereby reducing anxiety. 5. This helps to build her hope and confidence thereby reducing anxiety.	Patient showed signs of reduction in anxiety after 20 hours of nursing actions.

POST-OPERATIVE NURSING CARE PLAN OF A PATIENT WITH COLOSTOMY

S/N	Nursing Diagnosis	Objective	Nursing order/Intervention	Scientific principle/Rationale	Evaluation
1	Pain (acute) related to surgical incision as evidenced by patient's verbalization.	Patient will verbalize reduction in pain within 40 minutes of nursing action.	1. Receive patient on a well made bed without creases. 2. Assist patient to assume a comfortable position. 3. Engage patient in a quiet conversation. 4. Give prescribed analgesic example injection pentazocine 30mg 6hourly	1. A well made bed is comfortable to the body. 2. Comfortable position removes strain and tension on the incision site thereby reducing pain. 3. Conversation with the patient helps to divert patient's attention from the pain thereby reducing pain psychologically. 4. Analgesic on the medulla to increase pain threshold and reduce pain.	Patient verbalized reduction in pain after 30minutes of nursing action.

Body image disturbance related to abdominal stoma, change in physical appearance and control of urine	Patient will express acceptance with the new body image before discharge.	1. Encourage patient to express her concerns regarding the new body image that resulted from the surgery. 2. Be honest in providing information in response to patient and survey possibilities that will help to improve body image. 3. Arrange for her to talk to other patients that have undergone colostomy and are progressing satisfactory.	1. Expressing concerns within reveal areas of gap in knowledge and misconceptions and create the opportunity for the gap to be filled thereby enabling patient to accept self image. 2. Inform to improve her understanding and gives encouragement to the patient thereby improving self image. 3. This helps in instill hope thereby reducing body image disturbance	Patient accepted the new body image evidenced by less worry new body image.
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QUESTION

Communication is an essential skill in nursing practice.

- Define communication
- Describe with illustration the five (5) elements of communication process
- Describe verbal and non verbal communication with examples of each
- What is the importance of communication in nursing practice
- Enumerate the important

Answer

- Definition of communication**
Communication is concept or state of information between two or more people. It is a basic component of human relationship and instance of information transfer.

b. Elements of Communication

(i) Sender

- Sender or encoder of a message.
- Source-encoder
- A person or group or idea for communicating
- Encoding involves selection of specific signs or symbols example language or word to use, how to arrange to words tone of voice, gesture.
- Examples should be given.

(ii) Message

- What is usually said or written or demonstration that accompanies the words and how the message is transmitted.

(iii) Channel

- The medium used to convey the message.
- Must be appropriate for the message.
- Must help make intent of the message.
- Face to face telephone, writing, tape or recording, radio, TV.
- Non channel of touch.

(iv) Receiver

- The listener or the person that receives the message.
- The decoder.
- The interpretation of the sender's intent-giving meaning to the message.
- The process of decoding may be affected by the knowledge, experience and social-cultural background of both the sender and receiver.
- Ineffective communication and misinterpretation can occur at this level of the process.

(v) Response/feedback

- The message that the receiver returns to the sender.
- Also called feedback.
- Can be verbal, non verbal or both
- Feedback helps the sender to correct it gives assurance to the sender that information has been disseminated

c. Explanation of verbal and non-verbal communication

- Verbal communication uses the spoken or written word.
- Conscious people choose words to use.
- Words to be used depend on individuals, culture, socio-economic background, age and education.
- When choosing words consider pace, intonation, simplicity, clarity, relevance etc.

Examples: Face to face, phone calls, memo, letters etc.

Non verbal communication

- Also called body language
- Includes gestures, body movements, use of touch, physical appearance example adornment.
- Expresses feelings more than words.
- Less consciously controlled than verbal communication.
- Either reinforces or contradicts verbal communication.

- Observing and interpreting client's non-verbal behaviour is an essential skill in nursing practice.
- It involves noting the physical appearance of patient, gait, facial expression, gesture, smiling, shaking of hands, etc.
- d. Importance of communication in nursing practice.
 - It is skill needed in each phase of the nursing process.
 - Important when caring for clients who have communication problems.
 - Enhances relationship between clients, nurses and other health care professionals.
 - Helps in evaluating desired client outcomes.
 - Used to gather data during nursing assessment.
 - To express caring and comfort.
 - To teach or persuade a group (eg continuing education) or clients.
 - Used during health education or counselling.
 - Enhances quality client care etc.

Question

Master Ade, a 10 year old boy was admitted into the paediatric ward of your hospital with a diagnosis of bronchial asthma.

- List six (6) investigations that may be carried out on this patient? (3mks)
- Describe the mechanism of respiration (5mks)
- Discuss the nursing and medical management of Master Ade who is admitted in the ward with diagnosis of bronchial asthma (9mks)
- State six (6) possible complications of bronchial asthma. (3mks)
- Outline the nursing management of Master Ade from admission till discharge.

Answer

A. Investigations on Bronchial Asthma

- History taking
- Physical examination-inspection, palpation, and auscultation.
- Skin sensitivity test.
- Pulmonary function test.
- White blood cell count-total and differential.
- Chest x-ray.
- Sputum for culture and sensitivity.
- Inhalation sensitivity test.
- Arterial blood gas analysis

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- Serum immunoglobulin levels
- Erythrocyte sedimentation rate

B. Mechanism of respiration

- The mechanism of respiration is the process by which the lungs expand to take in air, then contract to expel it.
- It is performed by the respiratory muscles, the diaphragm during relaxation, the intercostals muscles during exertion and the sternocleidomastoid and the abdominal muscles assist during severe exertion or bronchial obstruction.
- It involves 3 cycles-inspiration, expiration and pause at the rate of 16-18 times per minute.

Inspiration or drawing in of air

- The diaphragm contracts, elongates the thoracic cavity and descends into the abdominal cavity causing negative pressure in the lungs.
- The intercostal muscles raise the ribs and increase the diameter of the thorax, air rushes from positive pressure in the alveoli to oxygenate the body tissues.
- Inspiration is an active process.

Expiration or the expulsion of air

- The diaphragm and the intercostals muscles relax.
- The lungs recoil and decrease in size.
- Air is expelled from higher pressures (the lungs) to lower pressure in the atmosphere.
- Expiration is a passive process in ordinary breathing and active in forced or laboured breathing when muscular efforts is involved.
- This is the period of rest after expiration then the cycle begins again

C. Nursing diagnoses of patient with bronchial asthma.

- Ineffective airway clearance, related to increased thick mucous secretions as evidenced by the use of accessory muscles of respiration.
- Impaired gas exchange related to broncho-constriction and mucosal oedema as evidenced by cyanosis.
- Anxiety related to laboured breathing, as evidenced by patient's facial expression.
- Impaired tissue perfusion (peripheral), related to increased heart beat and laboured respiration as evidenced by cyanosis.
- Activity intolerance related to fatigue and increased airway resistance.
- Imbalanced nutrition less than body requirements related to increase energy expended with breathing as evidenced by patient's loss of weight.

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Complications of bronchial asthma

- Status asthmaticus.
- Spontaneous pneumothorax
- Atelectasis
- Bronchiectasis
- Chronic bronchitis
- Respiratory failure
- Hypoxia

A. Nursing management of Ade with bronchial asthma

Admission

- Admit in a well ventilated corner of the paediatric ward or emergency unit.
- Place him in a comfortable up right position.
- Remove tight clothing's
- Ensure adequate bed rest.

Psychological care

- Reassure client and relatives
- Answer their questions as much as possible

Observation

- Observe temperature, pulse and respiration of master Ade.
- Maintain a regular vital signs observations one hourly, until client is stable. Breathing pattern should be monitored closely.
- Observe the general condition of the client.
- Observe his countenance and reassure when necessary.

Medical management:

- Give injection aminophylline IV slowly inj. Hydrocortisone 100-200mg iv stat inj. Ventolin diluted in H₂O IV slowly.
- Inhalation of salbutamol (ventolin)
- Administer oxygen 2-4 litres (if need arises).
- Give cough expectorant (if prescribed).
- Keep accurate records of administered drugs.
- Watch for the side effects of these drugs.

Investigations

- Sensitivity test
- X-ray of the chest
- Arterial Blood gas analysis
- Bacteriological examination of the sputum for possible isolation of microorganism

Diet

- Serve his food attractively and small quantity at a time.
- Encourage him to eat at least 2/3 of the food served him.
- I.V dextrose may be used in severe cases.

Care of bladder and bowel

- Serve him with urinal or bedpan as requested especially during the stage of attack.
- Keep strict intake and output record of fluid including vomitus.

Health education

- Education master Ade and relative on precautionary measures examples avoidance of dirty environment, exposure to cold etc.
- Treat chest infections adequately.
- To eat balanced diets

Advice on discharge

- Discharge client when his condition is stabilized.
- Advise him and relatives in keeping regular checkups.
- To take his drugs regularly.

QUESTION

Mr. Duke, a 44 year old bus driver was admitted into your ward with a diagnosis of fractured left femur as a result of road traffic accident. He is on traction waiting for the application of a plaster cast.

- Define traction (1mk)
- List any four (4) types of traction (2mks)
- Describe the nursing management of Mr. Okon who is on traction (8mks)
- Discuss the physiology of bone healing (5mks)
- Outline the care of this patient following the application of a plaster cast. (4mks)

Answer

- Definition of traction: Traction is the application of a pulling and counter pulling force to a part of the body to minimize muscles, spasm, align and immobilize fractures, reduce deformity and to increase space between opposing surfaces.

OR

The exertion of a pulling force to a fractured bone or dislocated joints to relieve muscle spasm, maintain proper position and facilitate healing.

b. Types of traction

- Skeletal traction

- Fixed traction
- Balance or sliding traction
- Skin traction
- Buck's extension traction
- Russell's traction (mixed traction).
- Dunlop's traction

c. Nursing Management of a patient on traction position

- Proper position of patient is to be maintained.
- There should be minimal/limited movement of patient.
- Explain to patient the need for maintaining a proper position for effective wound healing.
- Use accessories to prevent foot drop, example foam or foot board.

Observation - observe the following:

- Skin of affect site for sensitivity/circulation.
- Proper placement of the traction
- Patient's comfort or pain.
- Pin site for infection

Exercise

- Passive and active exercise should be encouraged.
- Flexing and extending of the feet (with caution).

Nutrition/diet

- Ensure a balanced diet (rich in protein) to promote bone healing.

Elimination

- Assist patient to make use of bed pan and urinal needs.
- Maintain neat environment.

Diversional therapy

- To prevent patient's feeling of boredom example provision of television, radio, newspaper.

Physiology care

- Bed bath, pressure areas, oral

Health education

Educate patient not to pull the traction or scratch the skin

d. Physiology of bone healing

Most fractures heal through enchondral ossification. When the bone is

injure, the bone fragments, are not merely patched together with scar tissue, instead, the bone regenerates by undergoing several stages of fracture healing processes.

1. Haematoma formation: With a fracture, the body's response is similar to that of injury to body tissue. There is bleeding into the injured tissue spaces and formation of fracture haematoma.
2. Inflammation and revascularization: The injured area is invaded by macrophages (large white blood cells), which debride the area. Inflammation, swelling and pain are present. The inflammatory stage lasts days and resolves with a decrease in pain and swelling. Within 5 days, the fractures haematoma undergoes organization. Fibrin strands form within the clot, creating a net work for revascularization and invasion by fibroblasts and osteoblasts.
3. Encourage gradual weight bearing and activities.
4. Encourage liberal fluid intake to prevent formation of renal calculi.
5. Pad cast edges to prevent skin abrasion.
6. When oedema subsides, look into the cast with flash light to check for skin changes to detect easily signs of infection of tissue breakdown and report accordingly.
7. Note and report any signs of neurological deficits-numbness, tingling and loss of sensation.
8. Avoid use of sharp instrument to scratch the skin within the cast to avoid tissue breakdown/infection.

QUESTION

Mrs. Adie, a 45 year old woman was brought to the OPD of your hospital with history of abdominal pain, tenderness, slight weight loss, anorexia and severe jaundice. Following investigations a diagnosis of Viral Hepatitis was made she was admitted into female medical ward for management.

- a. With a well labelled diagram describe the macroscopic structure of the liver.

i.	Diagram	-	6mks
ii.	Labeling	-	1mk
iii.	Description/function	-	4mks
- b. State six(6) possible investigations that may be carried out to confirm her diagnosis (3mks)
- c. Utilizing the nursing care plan identify and solve in order of priority three(3) nursing problems of Mrs. Adie
- d. Mention two (2) nutrients that are indicated and two (2) that are contra-indicated in Mrs. Adie's condition. 2marks

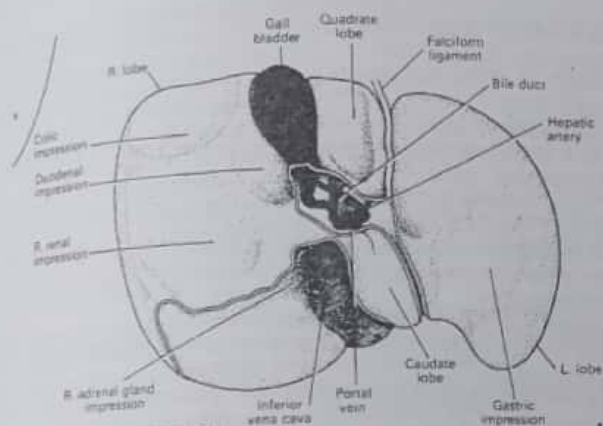


Figure 9.30 The liver — posterior view.

A. 1 & 2

1. Description of the liver

2. Definition: The largest gland in the body weighing 1.25kg (approximately)
3. Position: Upper part of the abdomen, right hypochondrium, part of epigastrium and part of left hypochondrium.

Organs in association: Superiorly and anteriorly and the anterior abdominal wall.

Inferiorly: stomach, gall duct duodenum. Hepatic flexure of the colon and the adrenal gland.

Posteriorly: Oesophagus, inferior vena-cava, aorta, gall bladder, and diaphragm.

Structure: Its enclosed in a elastic capsule with folds of peritoneum forming supporting ligament and also attaches it to the inferior surface of the diaphragm. Has four lobes; left, right, Caudate and quadrate, Right aortic fissure found at the posterior aspect which transmit various structures e.g. portal vein, hepatic artery, nerve fibres, lymph vessels, right and left hepatic duct.

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Blood supply: Hepatic artery and portal vein venous drainage-hepatic vein.
Functions

- Deamination of protein
- Detoxication of drugs
- Storage of ADEK
- Metabolism of fat
- Metabolism of protein
- Provision of heat
- Secretion of bile
- Storage of iron, copper, glycogen
- Breakdown of erythrocyte and defense against microbes

B. Six investigations for diagnosis of Viral Hepatitis

- i. History taking
- ii. Palpation to reveal enlargement and tenderness
- iii. Blood sample reveal hepatitis A & B antigens/antibodies
- iv. Serum enzyme tests i.e. Increases SGOT.
- v. Urine specimen for increased urobilinogen. Or bilirubin
- vi. Stool tests reveal positive hepatitis A virus (mark each)
- vii. Ultrasound scan
- viii. MRI

- 1D Two nutrients indicated-vitamin (B₁₂), CHO (glucose)
- Two nutrients contra-indicated -fat (oil) and protein (amino acid)

Other Nursing diagnosis include,

- 1) Imbalance nutrition related to anorexia, evidenced by patient's inability to finish a plate of meal served.
- 2) Pain (acute) related to enlarged liver, evidenced by patient's verbalization.
- 3) Hyperthermia related to resetting of hypothalamus evidenced by thermometric reading of 39.4°C
- 4) Anxiety related to unknown outcome of the disease condition evidenced by patient's facial expression.
- 5) Risk for spread of infection.

QUESTION

Discuss the underlisted drugs using the provided guideline

- A Frusemide (Lasix)
- B Pethidine

5marks each

Guideline

- Group
- Mode of action

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- Indication(s)
- Route/dosage
- Side effect(s)
- Nursing Responsibilities

Prescription for Gentacin

- D. Provided with stock of Gentamicin, 80mg in 2mls, and required to give a two year old baby intra-muscular injection Gentacin 20mg 8hourly x 5/7
How many ampoules will you collect from the pharmacy department for the child?

ANSWER

A. Frusemide (LASIX)

Group-diuretic

Action - It inhibit the sodium reabsorption in the proximal and distal tubules and ascending loop of Henle.

Indication- CCF, pre-blood transfusion syndrome, severe oedema.

Route- orally, IM IV.

Dosage-40-120mg in divided doses, 20mg IM IV

Side effect - Nausea, weakness, temporal hypotension

Nursing responsibilities

- maintain intake and output chart
- observe patient closely
- administer potassium tablet

B. Pethidine

Group - Narcotic

Action-exerts analgesic antispasmodic and selective effects

Indication - serve pain in labour

Route-orally, IM IV

Dosage 50-100mg orally, 25-100mg IM, 25-50mg IV

Side effects-may cause hypotension when given IV

Nursing responsibilities

- Advice patient not to operate machine.
- Monitor for dependency
- Do not give with sedative
- Monitor patient for drowsiness

C. Prescription Gentacin 20mg 8hourly x 5/7

1 Ampoule = 80mg = 1/2mk

Daily dose = 200mg x 8hours (3 times) 1/2mk

Five days = 60mg x 5 = 300mg 1/2mk

1 Ampoule = 80mg - 2mls 1/2mk

X Ampoule = 300mg 1/2mk

300mg = 80 x 1/2mk

X = 300/80 1/2mk

= 3.75 Ampoules 1/2mk

Equivalent to 4 Ampoules of Gentacin will be collected from the pharmacy department for the child. 1/2mk

QUESTION

Mr. Duke, an undergraduate, suddenly started disobeying his parents at home and lecturers at the University. He was found to be abusing alcohol.

- Define "alcoholism" 1mk
- List (6) potential substances of abuse 3marks
- Identify (4) main criteria for establishing a diagnosis of dependence 2marks
- Briefly describe the process of becoming drug dependent 4marks

- A. State how Mr. Ade can be managed in the ward from admission to discharge 10mk

ANSWER

A. Definition of Alcoholism

The term Alcoholism is used to describe a state in which excessive indulgence in alcohol has become harmful to the individual's physical or mental health, to his interpersonal relations or to his social or economic functioning.

ii. Six potential substance of abuse are:

- (i) Indian hemp (ii) Alcohol (iii) Cocaine (iv) Morphia (v) Amphetamine (vi) Sedatives (vii) Barbiturates (viii) Hypnotics

iii. Four main criteria for establishing a diagnosis of dependence are:

- Tolerance: Increase dose is necessary to produce same effect.
- Dependence: Psychological and physical dependence on the drugs. Psychological dependence exist when the drug is so central to the persons thought, emotion and activities that is difficult to using it.
- Craving: An overpowering desire or need to continue taking the and to obtain it by all means.
- Withdrawal syndrome: This is an unpleasant withdrawal effects which occurs if its use is stopped abruptly may produce shivering, stomach cramps, agitation, tremors, convulsion, delirium and diarrhea etc.

- Tolerance
- Dependence
- Craving
- Withdrawal syndrome

TBCW

IV Process of becoming drug dependence

- 1 Introduction/exposure to the substance: This is the first individual comes in contact with the substance.
- 2 Experimentation: This is when the individual wants to determine the effects of the substance/drugs.
- 3 Continuous use of the drug/substance: Having known the effects i.e. pleasurable effects, the patient now continues with the use of the drug to achieve expected effects.
- 4 Dependence: There is increase overpowering dependence and desire resulting in lack of satisfaction from initial satisfactory dose. This leads to tolerance (desire to increase the dose to maintain same effect. (1mk each of any correct point)

B. Management of Mr. Duke from admission to discharge

- Admission:** During the acute stage i.e withdrawal phase. Admit patient in the ward and perform all the admission procedures.
- Prevent alcohol withdrawal delirium through administration of drugs that duplicate the depressive action of alcohol on the CNS.
 - Prevent over-dehydration by giving correct fluid and electrolytes.
 - Establish good rapport with the client.
 - Approach client in a accepting and nonjudgmental manner.
 - Maintain minimal stimuli.
 - Adapt a firm, consistent, accepting and reasonable attitude.
 - Speak to him calmly.
 - Allow client to sleep off.
 - Observe the patient.
 - Remove tight clothing around the neck, chest and waist.
 - Monitor vital signs-temperature, pulse, respiration, and blood pressure.
 - Perform diagnostic measures such as evaluating the client for medical conditions associated with alcoholism such as liver damage, pancreatitis, altered blood glucose levels.
 - Evaluate client for nervous system disorders associated with chronic use, eg peripheral polyneuritis and temporary nerve palsies.
 - Encourage client to eat and finish his meals.
 - Give balanced diet.
 - Among the drugs give Vit B. Complex.
 - Encourage daily bathing, care of the mouth, hair and nails.

Discharge-process

- Work with an interdisciplinary team to develop a discharge plan.
- Invite the social worker to help organize a social and work settings to avoid joining others in drinking.
- Always invite and introduce him into AA or a similar self help group by introducing him to a therapist or counselor.

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- Help him to have a plan where to obtain help before leaving.

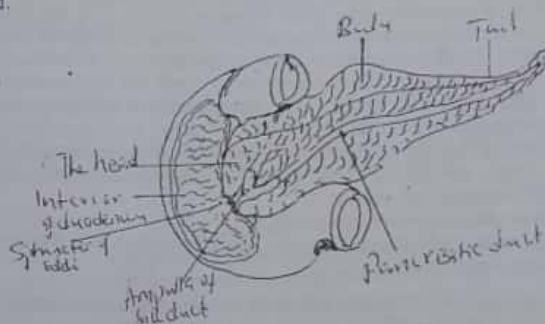
QUESTION

Mr. John, 44year old man, and known alcoholic reported to your hospital with complaints of m/ld upper abdominal pain, back ache, fever and vomiting. after critical observations, and the investigation physician on duty made a diagnosis of acute pancreatitis.

- A. Utilizing a well labeled diagram, described the human pancreas (1mk)
Diagram (1mk)
Labeling (4mks)
Description (2mks)
- B. List 6 causes/predisposing factors to Acute pancreatitis (2mks)
- C. Enumerate four likely investigations that assisted the physician in reaching at Mr. John's diagnosis. (3mks)
- D. Discuss the medical and Nursing managements of Mr. John (7mks)
- E. List 4 likely complications the patient may suffer if he is not properly managed.

ANSWER

Ai.



Bii

The pancreas

C. Description of the Pancreas

Definition: The pancreas is elongated racemose gland that lies behind the stomach with the head in the "C" shaped curve of the duodenum and it's tail in contact with the spleen.

It acts as an exocrine and endocrine gland.

Shape: It is about 12-15 cm long and 4cm wide. It is widest in the head and narrows out to the tail.

Weight: It weighs about 60g

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Position: It is located in the epigastric and left hypochondrial regions of the abdominal cavity. The head extends from the curve of the duodenum while the tail is in contact with the spleen.

Organs in association:

Anteriorly: Stomach and anterior abdominal wall.

Posteriorly: Abdominal organs, common bile duct, inferior vena cava, left kidney and the spleen.

Superiorly: Abdominal aorta, portal vein and the liver.

Inferiorly: Small intestine and transverse colon

Laterally: Spleen and duodenum

Structure:

The pancreas consists of a large number of lobules made up of small alveoli, with the walls consisting of secretory cells. Each lobule is drained by a tiny duct and these ducts unite to form the pancreatic duct which transverse the whole length of the gland and opens into the duodenum. The pancreatic duct joins the common bile duct just before it enters the duodenum to form the hepatopancreatic ampulla. A sphincter known as hepato-pancreatic sphincter (sphincter of oddi) controls the opening into the duodenum.

In its endocrine nature, the gland consists of cells that have close association with blood vessels. There are two distinct cells found in the pancreatic islets of Langerhans which produce hormones:

1. The alpha cells which secrete glucagon.
2. The beta cells which secrete insulin. These hormones help in the control of blood glucose level.

Blood supply: Splenic and mesenteric arteries which are branches of celiac artery and abdominal aorta respectively.

Venous drainage: Splenic and mesenteric veins which join the portal vein to the liver.

Nerve supply: Sympathetic nerve depresses the secretion of pancreatic juice while parasympathetic nerve increases the secretion.

Functions:

- Production of pancreatic juice under the influence of secretin and pancreatico-zymin.
- It houses the islets of Langerhans which secrete insulin and glucagon which are concerned with the control of blood glucose levels.

Predisposing factors to acute pancreatitis

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1. Alcoholism
2. Gallstones
3. Idiopathic
4. Endoscopic retrograde cholangio pancreatography.
5. Drugs- eg Azathioprine, thiazide diuretics.
6. Dysfunction of sphincter of oddi
7. Infection eg mumps, coxsacki virus infection.

C. Diagnostic Investigation for Acute Pancreatitis.

1. Serum Amylase level estimation to determine the elevation of amylase which is common within 24 hours of onset of pancreatitis.
2. Serum lipase level estimation- This is elevated in pancreatitis.
3. X-ray studies of the abdomen and the chest to differentiate pancreatitis from other disorders and to detect pleural effusion.
4. Ultra sound scanning- To note change in the size of the pancreas and detect pancreatic cysts, abscess or pseudocysts.
5. Haematocrit and haemoglobin levels estimation

B. Medical Management of pancreatitis

The goal of medical management is to relieve symptoms and prevent complications.

- **Drugs:** Likely drugs to be prescribed by the physician include
 - Pethidine 25-50mg prn to relieve pain
 - Propantheline bromide to inhibit vagal stimulation of pancreatic enzymes and enable relaxation of sphincter of Oddi.
 - Insulin- Give in case of hyperglycaemia.
 - Vitamin K- This is given parenterally to aid prothrombin production and prevent bleeding.
- Parenteral nutrition- To control severe catabolic state and need for nutritional support.
- NG tube inserted to suction gastric contents to relieve nausea and vomiting.
- Prophylactic broad spectrum antibiotics such as cefuroxime 250mg twice daily may be good for improvement in severe cases.
- Low dose subcutaneous heparin may be given to prevent thromboembolism.

C. Nursing management

- **Admission:** Admit patient in a medical ward or intensive care unit depending on the severity of clinical manifestations.

- **Position:** Allow patient to assume any comfortable position that gives him relief from the pain.

- **Psychological care:** create nurse-patient therapeutic

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relationship, reassure patient and involve him in his care.

- **Observation:** Observe patient's vital signs and record properly.
- Examining all stools passed by the patient for bulkiness, greasy and foul-smelling
- Assess the abdomen for distension and rigidity.
- Observe the sclerae and skin for yellowish colouration- jaundice.
- Assess the intensity of pain on the patient.

Rest and Exercise: Encourage patient to have adequate bed rest. Passive and active exercise will be instituted when patient's condition improves.

Personal hygiene: Oral care and bed bathing will be done for the patient until he is able to care for himself.

Diet: Patient is placed on nil per oral due to nausea and vomiting but maintained on parenteral nutrition until his condition improves. When oral intake is permitted, he is served adequate nutritive diet high in carbohydrate with low fat and protein.

Drugs: Serve doctor's prescribed drugs like pethidine, 25-200mg. Insulin, Antibiotics eg cefuroxime 250mg bd, etc. monitor the patient for signs of drug reaction or side effects.

Advice on Discharge

Advise patient to continue with adequate rest.

- Patient should be advised to abide by the dietary restrictions.
- Patient should take his drugs as prescribed.

QUESTION

A nurse was travelling to Calabar from Owerri in a bus and they suddenly came to an accident scene which involved about 20 passengers who were seriously affected. The nurse decided to implement her knowledge of triage to save as many lives as possible

- What is triage? (1mk)
- Discuss how the nurse will apply her knowledge of triage at the accident scene. (10mks)
- One of the victims had fracture of the humerus. Discuss the emergency management the nurse will render to the victim. (5mks)
- Distinguish between disaster and emergency situation. (4mks)

ANSWER

- Triage is a process of deciding how seriously ill/sick or injured person is so that the most serious cases can be treated first. It is a method applied in

selecting victims for care in emergency situation. In other words, it is the sorting of casualties to determine priority of health care needs and the proper site for treatment.

- The nurse with assistance of other people should perform the triage and other applicable resuscitation measures. Select the victims according to priority and use colour tags to identify them.

Priority I

Persons who require only minimal attention for their survival to be achieved. They may have minor injuries or airway obstruction. Attend to this group of victims first and if possible, integrate them into the work force for the rescue activities. If they are neglected, they may progress to expectant. Red colour tag/band is used to identify them.

Priority II

These are persons whose chances of survival can not be jeopardized by delay in this group are those with fracture without open wound and haemorrhage. The colour tag for this group is yellow.

Priority III

These are persons whose chances of survival can not be jeopardized by delay in attending to them. Victims in this group are those with fracture without open wound and haemorrhage. Delay in immobilizing these people will not affect their chances of survival green colour tag is used to identify them.

Priority IV

This group of people are critically ill and require extensive care with little or no chances of survival. They are unresponsive victims. Example of victims in this category are those with penetrating head injuries, high spinal cord injuries, 2nd and 3rd degree burns in excess of 60% of body surface area, people without pulse etc.

C. Emergency management of a victim with fracture.

- Keep by-stander/spectators away from the victim.
- Reassure the victim.
- Place the affected extremity in alignment with minimal movement.
- Arrest haemorrhage (if any)
- Immobilize the affected limb with any available binder eg head scarf.
- Move the victim to hospital without delay.

D. Difference between disaster and emergency

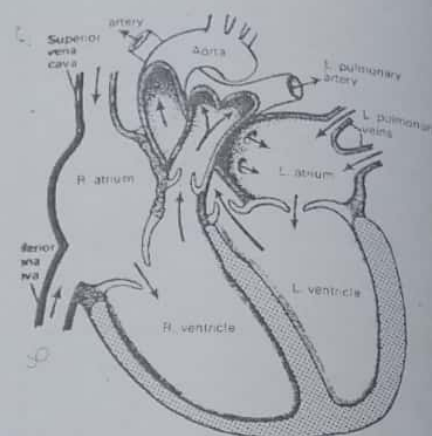
Disaster situation	Emergency situation
Definition: Disaster is an event that causes suffering or unfortunate consequences to greater many people and requires resources beyond the readily available one.	This is an event that causes suffering or unfortunate consequence to an individual or more.
Many people/victims are involved at a time.	An individual may be involved at a time.
Many hands/people must be required to render assistance.	An individual may be enough for intervention.
Example includes flood, earthquake, plane crash.	Example include epistaxis, choking.
Victims may not managed at the scene but transferred to hospital.	Victim can be managed at the scene.

QUESTION

Mr. Dunkan, a known hypertensive patient was brought to your hospital in an unconscious state suggesting a complication of the disease.

- A. With a well labeled diagram describe the flow of blood in the heart.
 Diagram (1mk)
 Labelling (1mk)
 Description (4mks)
- B. Define hypertension (mk)
- C. Briefly describe the pathophysiology of hypertension (4mks)
- D. Identify three(3) nursing problems of Mr. Dunkan and utilize the nursing care plan to solve the identified problem. (9mks)

ANSWER



(iii) Flow of blood in the heart

The right atrium of the heart receives deoxygenated blood from the superior and inferior vena cavae. The blood in the right atrium passes through the right atrio-ventricular valve from the right ventricle the blood enters the left and right lungs through pulmonary arteries by passing the pulmonary valves. In the lungs, the blood becomes oxygenated and returns through the pulmonary veins to the left atrium. From the left atrium, the oxygenated blood enters the left ventricle by passing through the left atrio-ventricular valves. From the L ventricle, the oxygenated blood is pumped into circulation through the aortic valves. Contraction of the atria leads to movement of blood into the ventricles and the contraction of the ventricles brings about the flow of blood into the pulmonary and general circulations.

- B. Hypertension is persistent ^{elevation} of arterial blood pressure above 140/90mmHg checked for not less than two times and possibly at different positions.
- C. **Pathophysiology of hypertension**
 Hypertension is a persistent elevation of arterial blood pressure above

arteriosclerosis - thickening & sclerosis of blood vessels
 atherosclerosis - Arteroma in blood vessels.

140/90 MMHg. In event of persistent elevation of arterial blood pressure due to narrowing of the arteries or atherosclerosis, the intima of the arteries becomes weak and increases its permeability to cholesterol which assists in the formation of arterioma. This affects mostly the blood vessels that supply the brain, the heart, the kidneys and the eyes. As these organs are affected, the patient may remain asymptomatic for a long period of time while complication continues in the affected organs. Those that experience symptoms may complain of occipital headache, dizziness, weakness, palpitation occurs as the heart keeps struggling to send enough blood into circulation through the narrowed arterial lumen. As the persistency of this elevation continues the heart muscles become hypertrophy with subsequent heart failure. The arteries in the brain may become weak due to excessive tension and rupture leading to Cerebro Vascular Accident (CVA). Arteroma formed as a result of collection of cholesterol may detach and cause coronary thrombosis which may occlude the arterial lumen. Thickening and sclerosis of the coronary arteries may cause ischaemic heart disease i.e. patient may experience angina pectoris or myocardial infarction.

The renal arteries may become thickened and sclerosed leading to loss of renal function and subsequent renal failure. The tiny blood vessels in the nose may rupture due to excessive tension leading to epistaxis. The blood vessels in the retina may rupture leading to retinal haemorrhage and oedema of the optic nerve with accompanying blindness which may be mistaken for glaucoma.

D. Nursing problems of Mr. Dunken, a known hypertensive patient brought to the hospital in unconscious state.

1. Impaired tissue perfusion related to reduced volume of blood reaching the organs secondary to narrowed arterial lumen as evidenced by shortness of breath.
2. Deficient self care (hygiene, feeding) related to unconscious state as evidenced by patient's inability to carry out self care activities.
3. Imbalanced nutrition less than body requirement related to unconscious state as evidenced by patient's loss of weight.

NURSING CARE PLAN OF A PATIENT WITH HYPERTENSION IN AN UNCONSCIOUS STATE

S/N	Diagnosis	Objective	Action	Rationale	Evaluation
1	Impaired tissue perfusion related to reduced volume of blood reaching the organs secondary to narrowed arterial lumen as evidenced by shortness of breath.	Patient will show signs of good tissue perfusion throughout the period of hospitalization.	- Lie patient supine with head turned to one side - Suction patient pm. - Administer artificial O₂ pm. - Serve prescribed antihypertensive e.g. propranolol 40mg tds.	- Lying patient supine with head turned to one side enables patient's airway and easy exchange of O₂. - Suctioning helps to remove any mucous obstruction which impairs gaseous exchange. - Artificial O₂ helps to increase O₂ level in the blood and increase tissue perfusion. - Antihypertensives dilate the blood vessels and enable easy flow of blood to the tissue.	Patient maintained good tissue perfusion throughout the period of hospitalization.
2	Deficient self care related to unconscious state as evidenced by patient's inability to carry out self care activities.	Patient will develop self care within 5 days of nursing action.	- Assist patient with self care activities when he is still unconscious e.g. bathing, feeding, etc. - Monitor patient's response using Glasgow coma scale. - Serve doctors prescribed drugs. - Encourage patient to carry out activities once he regains consciousness.	- Assisting patient in self care when he is unable helps him maintain personal hygiene, and nutrition. - Monitoring his response using Glasgow coma scale helps the nurse to know whether there is improvement from the unconscious state to enable the patient develop self care. - Prescribed drugs will assist in the patient's management, reviving him from unconscious state and making him develop self care. - Encouraging patient to carry out activities when he regains consciousness makes him to develop self care.	Patient developed self care after 4 days of nursing actions.
3	Imbalanced nutrition: less than body requirement related to unconscious state as evidenced by patient's loss of weight.	Patient will maintain nutritional throughout the period of hospitalization.	- Pass NG tube - Serve fluid diet to the patient which is adequate and nutritive. - Encourage patient to eat when he regains consciousness.	- NG tube will help for easy feeding of patient to maintain nutritional balance. - Balanced fluid diet will help to maintain fluid balanced. - Encouraging patient to eat will enable him to maintain nutritional balance.	Patient maintained nutritional balance throughout the period of hospitalization.

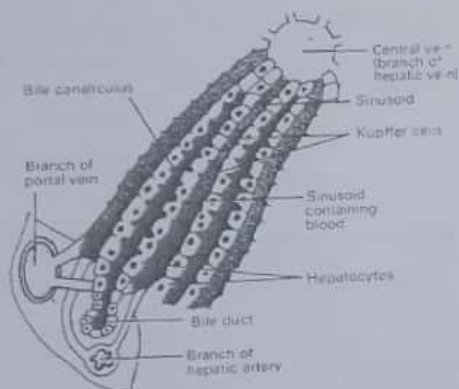
QUESTION

Ms. Joyce a 45-year-old commercial sex worker reported to your hospital with complaints of itching all over her body, tenderness at right upper quadrant, fever, weight loss, and yellowish colouration of the sclera and mucous membrane of the mouth.

On examination the doctor on duty noted hepatomegaly and made an impression of Hepatitis B.

- Describe the microscopic structure of the liver using a sketch (4mks)
- What is Hepatitis (1mk)
- Describe the pathophysiology of Hepatitis (4mks)
- Discuss the nursing management you will render to Mrs. Joyce (6mks)
- Enumerate the likely complications the patient may suffer following this disease.

ANSWER



Microscopic structure of the liver.

The lobes of the liver are made up of tiny lobules just visible to the naked eye. The lobules are hexagonal in shape and are formed by cubical-shaped cells, the hepatocytes, radiating from central vein, between two pairs of columns of cells there are sinusoids (blood vessels with incomplete walls) containing a mixture of blood from the tiny branches of the portal vein and hepatic artery. This arrangement allows the arterial blood and portal venous blood (with high

concentration of nutrients) to mix and come in close contact with the liver cells. Among the cells lining the sinusoids are hepatic macrophages (Kupfer cells) which function to digest and destroy any foreign particle through the liver. Blood drains from the sinusoids into the central or centrilobular veins which then join to form larger veins until they eventually become hepatic veins which leave the liver and empty into the inferior vena cava just below the diaphragm.

Blood supply: Hepatic artery and portal vein.

Venous drainage: Hepatic vein which drain into the inferior vena cava.

Nerve supply: Sympathetic and parasympathetic nerves

Functions

- Deamination of protein.
- Converts glucose to glycogen and convert glycogen to glucose.
- Desaturation of fats
- Storage of vit ADEK, Iron & copper.
- Secretes the bile

B. Definition of Hepatitis

Hepatitis is the inflammation of liver cells resulting from viral infection, drug toxicity or obstruction between the hepatocyte and biliary flow.

B. Pathophysiology of Hepatitis

Hepatitis is an inflammation of the liver cells due to viral infection or toxicity. In event of viral infection of the liver, there is inflammation of the liver cells and hyperplasia of the Kupfer cells. The inflammation causes the death of the hepatocytes and destruction of liver parenchyma with subsequent necrosis. The swelling and congestion affect the functioning of the liver. There is stretching of the peritoneum over the liver which results into abdominal pain. Disruption in portal circulation results into non-specific gastro-intestinal symptoms such as anorexia, nausea, vomiting, constipation or diarrhea. As a response to the inflammation, fever, chills, malaise and headache occurs. The flow of bile is affected leading to obstructive jaundice. As bile circulates in the blood, it brings about itching and there is discoloration of urine (dark coloured urine) due to excess bilirubin level and this also leads to change in the colour of faeces to clay.

C. Nursing Management of a patient with Hepatitis B

Admission: Admit patient in a cardiac bed with bed cradle in a quiet corner of the wards.

Position: Allow the patient to assume any comfortable position that gives him relief from the pain.

Psychological care: Create nurse/patient therapeutic relationship and involve in her care. Reassure her and provide diversional therapies.

Observation: Observe patient's vital signs and record observe the colour of stool and the consistency, observe the colour of urine and conjunctivae.

Intake and output recording: Keep strict intake and output recording monitor the I.V. fluid set by the doctor.

Rest and exercise: Encourage patient to have enough rest to conserve energy but manageable exercise must be undertaken to aid blood circulation.

Diet: Soft nutritive diet should be served to the patient. Serve patient diet supplement, glucose and fruit drinks, vit supplements eg vit A & C patient should avoid intake of alcohol.

Barrier Nursing: Patient should be nursed in a separate corner of the ward and screened. Do not use materials/instrument with patient and other patients. Wash hand before and after attending to the patient. Use gloves & gowns while attending to the patient. Do not have direct contact with the patient's urine, blood and faeces. Bed pan and urinals used by the patient must be disinfected.

Drugs: Administer prescribed drugs such as pentasacon 30mg, vit K supplement, anti pyretics criemetic etc.

D. Complications of Hepatitis.

- Aplastic anaemia
- Relapsing hepatitis
- Cholestatic viral hepatitis
- Post hepatitis syndrome
- Fulminant hepatitis failure.

QUESTION

Mrs. Jiddo, a 45year old teacher with secondary infertility reported the maternity unit of your hospital with bleeding per vagina. The gynaecologist on duty conducted series of observations and sent her for ultrasound scanning which confirmed the foetus dead inutero.

- A. What is abortion (2mks)
- B. Enumerate 6 causes of abortions (5mks)
- C. With the use of only sketch classify abortion (3mks)
- D. Discuss your psychological care of Mrs. Jiddo (10mks)

ANSWER

- A. Abortion is termination or loss of pregnancy before 24 weeks of gestation. It is expulsion from the uterus of the product of conception before the 24th week of pregnancy, the foetus not being alive.
- B. Fetal causes
 - Mal-development or chromosomal abnormalities.
 - Hypoxia of the foetus is viable.

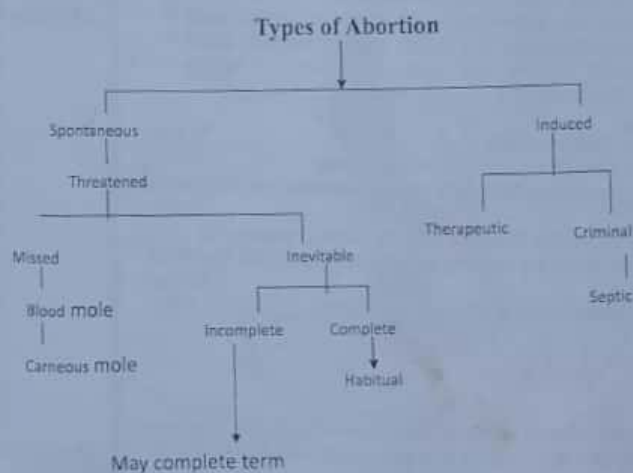
Maternal causes

- Maternal illness such as malaria, diarrhoeae, dysentery, hypertension, diabetes mellitus, pyelonephritic.
- Hormonal imbalance.
- Cervical incompetency
- Violent exercise
- Uterine infection eg endometriosis
- Uterine malformation
- Certain drugs take oxytocin

Social causes

- Unwanted pregnancies
- Excessive emotional stress
- Rape trauma syndrome

C. Classification of Abortion



A. Psychological Management of a woman with abortion

- Ensure that client's condition is stabilized through checking of the vital signs and close monitoring.
- Reassure patient.
- Counsel patient regarding her condition.
- Make her to understand that every hope is not lost as she can still become pregnant.
- If there are known women who have passed through related problems but are having their children now, introduce them to her.
- Encourage her to hold onto the child she has already as some women do not even have not all.
- Discuss option like abortion, artificial insemination and test-tube baby with her.
- Encourage her to visit a gynaecologist for proper investigation and treatment if she has not done so.
- Encourage her to keep praying as God can do all this that will gladden her heart.
- Follow her up to the home and encourage the husband to always give supportive care to her.

QUESTION

Corporal Jide, a police officer, was rushed to your hospital from a nearby village where he was stabbed at the chest by unknown person. At the time he was brought to the hospital, he was experiencing shortness of breath, pain at the chest, back and arm. On observation the doctor noted cyanosis, decrease of respiration, flapping sound in the left lung. Diagnosis of spontaneous pneumothorax was made by the doctors.

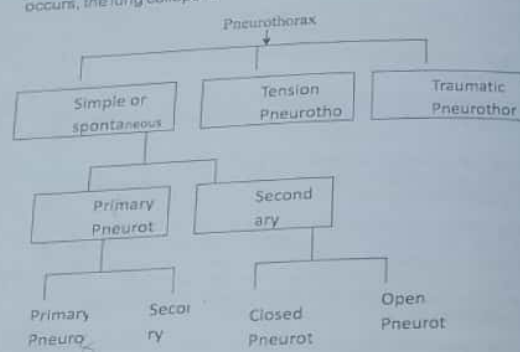
- What is spontaneous pneumothorax? (1mk)
- With the aid of a flow chart, classify pneumothorax (5mks)
- Briefly describe the pathophysiology of spontaneous pneumothorax (3mks)
- Identify in order of priority, three immediate nursing problem of corporal Jide and utilize the nursing care plan to solve the identified problem (9mks)
- Enumerate two complications of pneumothorax (1mk)

ANSWER

- Spontaneous pneumothorax is the accumulation of air in the interpleural space secondary to altered integrity of surrounding membrane. It is air in

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the pleural cavity caused by perforation of the chest wall or of the lungs pleura leading to entrance of air through the bronchi. When any of these occurs, the lung collapses.



Classification of pneumothorax using flow chart

- Pathophysiology of Pneumothorax**
Spontaneous Pneumothorax occurs when there is interruption in the pleural integrity which permits air from either outside (open) or inside (closed) to enter the pleural space. This can be as a result of trauma or occurs spontaneously.
In event of trauma to the lung, air is drawn into the pleural space which cannot leave again. This helps to build pressure in the pleural space, collapse of the whole lungs and displacement of the trachea away from the affected side. When this occurs the lung decompresses leading to shortness of breath and cyanosis due to inadequate supply of oxygenated blood to the peripheral tissues. With continue entrance of air into the lungs the individual develops severe difficulty in breathing. Major blood vessels become kinked and blood flow to the brain may be seriously impaired. As the condition progresses there is formation of piece of tissue which allows a one way movement of air into the lung with no exit (one way valve). Excessive pressure is built up in the lung which compresses the tissues of the lung and the thoracic cavity. If this is still not properly managed it leads to further collapse of the lungs and the client will not survive. (Walsh, 2002)

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- C. Nursing problem of a patient with spontaneous Pneumothorax.
1. Ineffective breathing pattern related to build up of pressure in the lung and compression of the thoracic contacts as evidenced by patient's decreased depth of respiration.
 2. Ineffective tissue perfusion (cardiopulmonary and peripheral) related to collapsing lungs as evidenced by cyanosis.

NURSING CARE PLAN OF A PATIENT WITH SPONTANEOUS PNEUMOTHORAX

N	Diagnosis	Objective	Action	Rationale	Evaluation
1	Ineffective breathing pattern related to build up of pressure in the lung and compression of the thoracic contacts as evidenced by patient's decreased depth of respiration.	Patient will breath with ease at the rate of 18 - 20 cycle per minute within 1 hour of nursing action.	- Monitor patient's respiratory rate and depth and keep record. - Place patient in a cardiac position (head of bed at 45°). - Encourage patient to lie on deep breathing. - Suction out mucus if any.	- Monitor to not base rate data and when there is improvement. - Cardiac position will enable the lungs to expand maximally and reduce the risk of aspiration. - Deep breathing allows for easy air flow through the airway and improves breathing pattern. - Suctioning of mucus helps to keep the airway patent and improve breathing pattern.	Patient breath with ease at the rate of 20 cycles per minute of nursing action.
2	Ineffective tissue perfusion (cardiopulmonary and peripheral) related to collapsing lungs as evidenced by cyanosis.	Patient will maintain adequate tissue perfusion within 30min of nursing action.	- Maintain patient in a cardiac position. - Institute measures to manage the collapsed lungs without delay. - Administer artificial oxygen at the rate of 4l per minute. - Observe patient's vital signs especially respiratory and the nail beds for cyanosis.	- Cardiac position will ease breathing and enable better gaseous exchange thereby permitting tissue perfusion. - Measures to manage collapsed lungs like closing the open wound into the lungs and enable gaseous exchange thereby permitting tissue perfusion. - Artificial oxygen helps to augment the volume of O₂ in the lungs thereby permitting tissue perfusion. - Observation of vital signs and nail beds enable the nurse to note improvement in the tissue perfusion.	Patient maintained adequate tissue perfusion after 25minutes of nursing actions as evidenced by disappearance of cyanosis.
3	Acute pain (in the chest cavity) related to compression of surrounding nerves as evidenced by patient's verbalization.	Patient will verbalize reduction in pain within 1 hour of nursing actions.	- Encourage patient to the cardiac position as per fowlers. - Give diversional therapy eg. Turn to T.V. station. - Administer doctor's prescribed analgesic eg. Tramadol 50mg bd.	- Fowler's position lifts abdominal weight of the patient thoracic cavity thereby relieving pain. - Diversional therapy helps to relieve pain psychologically. - Tramadol provides analgesic effect to the nerves thereby reducing pain.	Patient verbalized reduction in pain after 45 minutes of nursing actions.

3. Acute pain (in the chest cavity) related to the compression of surrounding nerves as evidenced by patient's verbalization.
- D. Complications of Spontaneous Pneumothorax
1. Infection may occur if aseptic technique is not observed during surgical management or insertion of chest tube.
 2. Cardiac arrest.
 3. Toxaemia following infection.
 4. Total lung collapse.

QUESTION

Mr. Kachi, a 56 year old petty trader was brought to your hospital with complaints of fatigue, intermittent mild fever, ankle oedema. On examination the doctor on duty noted, Hepatomegaly, clubbing of fingers and spider angioma and made a diagnosis of liver cirrhosis.

- A. With the aid of a well labeled diagram described the macroscopic structure of the human liver. (1mk)
- i. Diagram (1mk)
- ii. Labelling (1mk)
- iii. Description (4mks)
- B. What is liver cirrhosis (1mk)
- C. Enumerate 4 investigations that will assist the doctors in authenticating his diagnosis. (2mks)
- D. Utilize the nursing care plan and nursing problems of Mr. Kachi. (10mks)

ANSWER

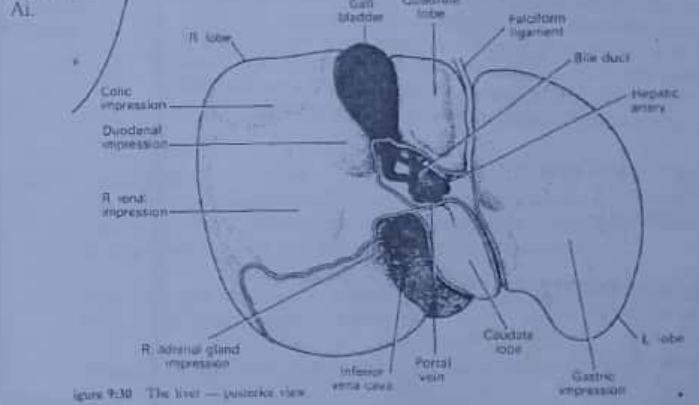


Figure 9:30 The liver — posterior view.

D. Description of macroscopic structure of the liver.

Definition: The liver is the largest gland in the body and can be considered a chemical station. **Position:** It is situated in the upper part of the abdominal cavity occupying the greater part of right hypochondriac regime, epigastric regime and extends to the left hypochondrium.

Size: It weights between 1-2.3kg.

Organs in association

Superiorly and anteriorly: Diaphragm, anterior abdominal wall.

Inferiorly: Stomach, bile duct, duodenum, right colic flexure of the colon, right kidney and adrenal gland.

Posteriorly: Oesophagus, inferior venacava, aorta, gall bladder, vertebral column.

Laterally: Lower ribs and diaphragm

Structure

The liver is enclosed in a thin inelastic capsule and covered completely by the peritoneum which also form supporting ligament. The folds of peritoneum attaches the liver to the inferior surface of the diaphragm. The liver is divided into lobes

- The caudate lobe
- The quadrate lobe.

The left and right lobes are visible at the anterior and posterior area but the caudate and quadrate lobes are visible only from the posterior aspect.

The portal tissue: This is a region at the posterior surface of the liver where structures end and leave the liver. Structures that enter through the tissue include, the portal vein, the hepatic artery, and nerve fibres. Structures that leave through the tissue are the right and left hepatic ducts, and lymph vessels.

Blood supply – hepatic artery and portal vein.

Venous drainage – hepatic vein

Functions of the liver

- The liver assists in the metabolism of carbohydrate, protein and fats.
- Production and release of bile
- Detoxification of drugs.
- Storage of vitamins A, D, E, K.
- Production of heat in the body
- Deamination of protein

- Storage of Vitamin A from carotene.
- Inactivates hormones.
- Forms the plasma protein

B. Liver cirrhosis is a fibrotic change which occurs in the liver as a result of chronic inflammation. It is a chronic disease of the liver characterized by replacement of normal liver tissue with diffuse fibrosis that disrupts the structure and function of the liver.

C. Investigations in liver cirrhosis

1. Enzyme tests to detect the release of serum glutamic oxalate transaminase (SGOT) and serum pyruvic transaminase (SPT).
2. Bilirubin test - to estimate bile excretion and retention.
3. Ultrasound scanning.
4. Computed Tomography scan
5. Magnetic Resonance Imaging (MRI).
6. Radioisotope liver scan
7. Liver biopsy.

D. Nursing Diagnoses of a patient with liver cirrhosis

1. Chronic pain (abdominal) related to enlarged abdominal organs and ascites as evidenced by patient's complaints/verbalization.
2. Imbalanced nutrition: Less than body requirement related anorexia as evidenced by patient's inability to finish a plate of meal served.
3. Deficient self care (bathing) related to fatigue as evidenced by patient's inability to carry out self care activities.
4. Excess fluid volume related to portal congestion and reduced colloid osmotic pressure as evidenced by ascitis and weight gain.

NURSING CARE PLAN OF PATIENT WITH LIVER CIRRHOSIS

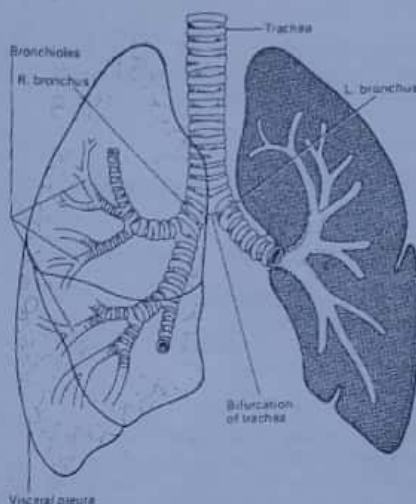
Diagnosis (Problem)	Objective	Action/Order	Rationale	Evaluation
Chronic pain (abdominal) related to enlarged abdominal organ and ascites as evidenced by patient's complaints	Patient will verbalize reduction in pain within 72 hours of nursing actions	- Assess patient's level of pain by touching the abdominal cavity. - Allow patient to assume any comfortable position. - Implement diversional therapy eg. turning to Radio or T.V. stations of interest to the patient. - Serve doctor's prescribed analgesics eg. Tramadol 25mg bd. - Serve prescribed diuretics.	- Understanding the level of pain will help to note improvement in the management. - Comfortable position to the patient will make him to experience reduction in pain. - Diversional therapy will help to remove his mind from the pain and bring about psychological relieve of pain. - Analgesics eg. Tramadol relieves pain by acting on the nervous system and anesthetizing the nerves thereby reducing pain. - Diuretics enhance fluid excretion via urine thereby reducing ascites and congestions via -4-vis reducing pain.	Patient verbalized reduction in pain after 48 hours of nursing actions.
Imbalanced nutrition: Less than body requirements related anorexia as evidenced by patient's inability to finish a plate of meal period.	Patient will maintain good nutritional status throughout the period of hospitalization.	- Place patient on adequate nutritive fluid or soft diet provide oral care. - Serve little quantity of food at a time but several times. - If patient cannot tolerate oral food due to his condition, advise doctor to set IV infusion. - Encourage patient to eat his food.	- Adequate nutritive fluid/soft diet make swallowing easier thereby enabling patient to eat and maintain nutritional status. - Oral care helps to stimulate appetite and enable the patient to eat and maintain nutritional status. - Serving little quantity of food at a time encourages the patient to eat and doing this severally will assist him to maintain nutritional status. - Encouraging the patient to eat will motivate him to eat and prevent nutritional imbalance. - IV fluid helps to maintain the patient's nutritional status if he cannot tolerate oral food.	Patient's maintained good nutritional status throughout the period of his hospitalization.
Deficient self care (bathing) related to fatigue as evidenced by patient's inability to carry out self care activities.	Patient will develop self care within 48 hours of nursing actions.	- Assist patient in self care activities like bathing. - Encourage him to carryout minimal self care activities unassisted. - Serve energy drinks to the patient eg. Lucozade. - do not allow patient to embark on stressful activities.	- Assisting patient helps him to conserve energy for future activities. - Carrying out minimal activities unassisted help him to develop self care gradually. - Energy drinks provide glucose which increases the energy to carry out self care activities. - Stressful activities gap energy and when avoided, energy to carry out self care is conserved.	Patient developed self care after 48 hours of nursing actions.

QUESTION

Mrs. Mary, a 60 year old woman was brought to the Accident and Emergency Unit of your hospital in obvious respiratory distress and weakness. After physical examination, the doctor on duty made a diagnosis of Bronchial Asthma.

- Draw, label described the bronchial tree and the lungs
 - Describe the pathophysiology of Bronchial asthma
- B. Identify three nursing problems of Mrs. Mary and utilize the nursing care plan to solve the identified problem in order of priority

ANSWER



A1. Bronchial tree

A1. Description

This is a hollow structure that branched out of the trachea through which air passes in and out of the lungs. of the bronchial in the lungs which convey air to and within lungs with the right or left principal divisions.

Size: The right bronchus is approximately 2.5cm long and the left is about 5cm long and narrower than the right.

Shape: The right bronchus is wider, shorter and more vertical than the left bronchus.

Structure:

The bronchi are composed of tissues as found in the trachea. They are lined with ciliated columnar epithelium. The bronchi subdivide to form the bronchioles, terminal bronchioles, respiratory bronchioles, alveolar ducts and the alveoli.

Towards the distal segment of the bronchi, the cartilages become irregular in size and shape and the walls become thinner as muscle and areola tissue disappear. The walls consist of a single layer of flattened epithelial cells at the terminal end. Interchange of gases take place across the walls of the alveolar duct and alveoli and those of the pulmonary artery capillaries.

Blood supply: Right and left bronchial arteries.

Venous drainage: Bronchial veins - The right empty's into the azygos vein while the left empty's into the superior intercostal vein.

Nerve supply: sympathetic and parasympathetic nerves.

Lymphatic drainage: Thoracic duct on the left side and right lymphatic duct on the right side.

Functions of the bronchi: Regulate the volume of air that enters the lungs.

- Warms and humidifies air the air before it gets to the alveoli.
- Maintains patency of the air way in and out of the lungs.
- Assists in cough reflex when foreign body enters the airway.
- It helps to remove particulate matter from the airway.

B. Pathophysiology of Asthma

Bronchial Asthma is an inflammatory disease of the airway in which the mucous membrane and muscle layers of the bronchi become thickened and the mucous glands enlarge reducing the airflow in the lower respiratory tract and is characterized by wheezing respiration. When an offending agent gets to the bronchi of a predisposed individual either by inhalation or ingestion, there is release of chemical mediators like histamine, prostaglandins in response to the antigen.

These chemical mediators act on the receptor sites on the membrane of the bronchial muscles leading to contraction of the bronchial trees and narrowing. The antigen within the bronchi and the bronchioles stimulate mucus-secreting cells in the epithelial lining to produce more mucus in an attempt to sweep away the antigen. The mucus further narrows down the bronchial tree, obstructing the expiration of carbon iv oxide through the air passage. This poses a problem as air leaves the bronchioles leading to

characteristic wheezy breathing with expiratory dyspnoea which leads to patient uses accessory muscles of respiration. Patient appears distraught and assumes upright position with accumulation of carbon iv oxide in the blood with obvious cyanosis. The broncho-constriction causes painful tightness in the chest with the patient displaying obvious anxiety and insomnia. The presence of the offending agent in the respiratory tract irritates the cough reflex. Sputum produced during coughing is usually scanty and thick.

Excess CO_2 in the blood stimulates the respiratory centre in the medulla oblongata leading to increase inspiration. Difficulty in breathing leads to increase heart beat in an attempt to increase the volume of blood pumped to the heart for oxygenation. This increases the pulse rate.

Nursing problem of a patient with Asthma.

1. Ineffective airway clearance related to bronchoconstriction as evidenced by patient's use of accessory muscles of respiration and wheezing respiration.
2. Impaired tissue perfusion related to reduced oxygenation at the alveoli as evidenced by cyanosis.
3. Anxiety related to tightness in the chest and difficulty in respiration as evidenced by patient's facial expressions.

NURSING CARE PLAN OF PATIENT WITH BRONCHIAL ASTHMA

S/N	Diagnosis	Objective	Action/Order	Rationale	Evaluation
1	Ineffective airway clearance related to bronchoconstriction as evidenced by patient's use of accessory muscles of respiration and wheezing respiration.	Patient will maintain a patent airway within 45 minutes of nursing actions.	- Loose tight clothing on the patient. - Place patient in an up right position. - Suction airway of mucus. Administer prescribed Broncho dilators eg. Aminophy 1hr 250 500mg in 19mls slowly.	- Loosening tight clothing will help to ease respiration. - Upright position helps to reduce abdominal pressure on the thoracic region and aids easy respiration. - Suctioning mucus from the airway helps to keep the airway patent. - Bronchodilators eg. Aminophy relaxes smooth muscles of the bronchioles thereby allowing patent airway.	Patient maintained a patent airway after 30 minutes of nursing actions.

patient's tissue perfusion related to oxygenation of the blood as evidenced by cyanosis.	Patient's tissue perfusion will improve within 2 hours of nursing actions.	Place patient in an up right position. - Suction the airway using suction machine. - Administer artificial oxygen 4-5 l/min per hour. - Serve prescribed bronchodilation eg. Ventolin 4mg tds.	Upright position reduces the weight of the abdominal content for the thoracic cavity thereby aiding easy respiration and oxygen exchange. Suctioning of mucus allows patency of the airway and improving oxygen exchange. Artificial oxygen helps to increase the oxygen content in the blood. Broncho dilators relax smooth muscles thereby allowing easy airflow to the lungs for oxygenation.	Patient's tissue perfusion improved after 1 hour of nursing actions as evidenced by disappearance of cyanosis.
Anxiety related to tightness in the chest and difficulty in respiration as evidenced by patient's facial expressions.	Patient will show signs of reduction in anxiety within 2 hours of nursing action.	- Admit patient close to another patient that is recovering from the same condition. - Create nurse therapeutic relations with the patient. - Implement all management protocols that will help to reduce tightness & difficulty in breathing. - Use diversional therapy on the patient eg. Switch on TV.	- Admitting him close to a recovery patient will instill hope and reduce anxiety. - Nurse patient therapeutic relationship enables the patient to feel relaxed and co-operate in his care. - Implementing management protocol will help to reduce bronchial constrictions and reduce tightness thereby reducing anxiety. - Diversional therapy removes the patient's mind from the condition thereby reducing anxiety momentarily.	Patient showed signs of reduction in anxiety as evidenced by his freely discussing with his visitors.

QUESTION

- A. Doctor's prescription for a 2 year old child with fever of unknown origin reads: Administer 20mg Gentacin 8 hourly for 5 days. If the drug comes 80mg per ampoule of 2ml, how many ampoules will be supplied at the pharmacy unit for the patient? (4mks)
- B. Prescription for a 25 year old by for infusion read: infuse 500ml D/W to run for 4 hours. If the drop factor for the infusion set is 15, at what rate will the nurse regulate the drop per minute? (4mks)

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C. Write briefly on the following:

- Anticoagulants
- Antiemetics
- Antihypertensives

ANSWER

A. Doctor's prescription

Gentacin 20mg 8 hourly = 3 times dly

No of days = 5 days

One ampoule = 80mg = 2ml

$20\text{mg} \times 3 \text{ times} = 60\text{mg dly}$

Fo. 5 days = $60\text{mg} \times 5 = 300\text{mg}$

Total mg for the child's treatment

No of ampoules required = $x = 300\text{mg}$.

1 ampoule = 80 mg

X ampoule = 300mg

No of ampoule = $80x = 300$

$X = \frac{300}{80} = 3.75 \approx 4$

No of ampoule required for the Childs treatment = 4 ampoule

B. Infusion prescribed = 500ml D/W

Duration = 4 hours

Drop factor = 15

Rate per minimide

$= \frac{500 \times 15}{4 \times 60} = 7500$

$= 240$

$= 31.25$

$= 31$

No of drops per minute = 31drops

Ci. Anti-coagulants:

These are substances that are gives to interfere with the clotting of blood. They are used in

Uses: Used in low dosage to prevent transient ischaemia. They are used to prevent deep venous thrombosis and thrombo in cardiac chamber.

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They can also be used during dialysis to prevent clotting but at low dosage.

Example of anti coagulants includes Heparine, Watarin.

Action: It prevents formation of thrombin from prothrombus.

Dosage: For Hebarine I.V injection 500-1000 units six hourly or by continuous infusion 40,000 units over 24 hours.

Side effects: May cause bleeding and haematuria.

Contraindication: Peptic ulcer, haemophilia and severe hypertension.

Nursing responsibilities

- Do not give I.M - it is very painful.
- Do not give to peptic ulcer patient or patients with haemophilia.
- Do not exceed prescribed dosage.

Cii. **Antiemetics:** These are drugs which ^{stop} nausea and vomiting by depressing the vomiting centre in the brain.

Uses: They are used in travel sickness, pregnancy uremia, radiation sickness and patients on cytotoxic drugs.

Side effects: Drowsiness, nervous depression.

Contra indication: Do not give to alcoholics.

Nursing responsibilities

- Advise patient not to operate machinery if on drug administration.
- They should be given at the same time with CNS depressants.
- Patient should not take antiemetics with alcohol.
- Administer drug 30 minutes before emetic stimulus if possible.

Example include

Hyocin (Buscopan)

Chlorpromazine (Largactil)

Promethazine (Phenergen)

Ciii. Antihypertensives

These are drugs used in the treatment of hypertension. They are of different types which includes

- Central acting antihypertensives
- Alpha adrenergic blocking antihypertensive
- Vasodilator antihypertensive.

- Angiotensin converting enzyme (ACE) inhibitors.
- Calcium channel blocking antihypertensive.
- Beta adrenergic blocking antihypertensives.

Uses: They are used to manage mild to severe hypertensive.

Side effect: Hypotension, tachycardia, fluid retention, dry mouth etc.

Nursing responsibilities

- Monitor patient's blood pressure before administration.
- Administer with diuretics to prevent fluid retention
- Administer with caution concurrently with other anti-hypertensives.

QUESTION

1. Mr. Yohanne, a 50 year old man, was admitted in the male medical ward of your hospital with -fever, headache, photophobia, irritability, nausea, vomiting and positive kerning's sign. The doctor in-charge made diagnosis of meningitis.
 - a. With the aid of a well labelled diagram, describe the production and flow of cerebro-spinal fluid.

Diagram - 1 mark
labelling - 2marks, description - 4marks)
 - b. Outline four (4) diagnostic procedures that should be carried out to confirm the diagnosis (2marks)
 - c. Identify and solve in order of priority, three (3) nursing problems of Mr. Yohanne. (9marks)
 - d. List four (4) possible complications of meningitis (2marks).

Answer

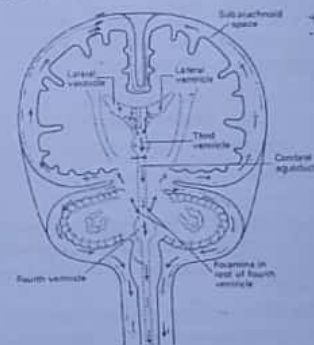


Figure 12.10 Arrows showing the flow of cerebrospinal fluid.

Production and flow of cerebrospinal fluid

- CSF is found in and around the brain and spinal cord.
- It forms a liquid cushion that protects the brain and spinal cord.
- It is produced by the choroid plexuses that hang from the roof of each ventricle.
- These plexuses are formed by, thin-walled capillaries, that have fairly permeable walls which allow tissue fluids to filter continuously from the blood stream.
- They have ion pumps that allow them to modify this filtrate by actively transporting only certain ions across their membranes into the CSF pool.
- Once produced, CSF moves from the foramen at the roof of the fourth ventricle into the subarachnoid space and completely surrounds the brain and spinal cord.
- This movement is aided by pulsating blood vessels, respiration and changes of posture.
- It is absorbed into the blood through arachnoid villi that project into the venous sinuses.
- This movement into the venous sinuses depends upon the difference in pressure on each side of the walls of the arachnoid villi, which act as one way valve.
- When CSF pressure is higher, CSF passes into the blood.
- When venous pressure, the arachnoid villi collapse preventing the passage of blood constituents into the CSF.
- Cells in the walls of the ventricle also reabsorb CSF.
- CSF is produced continuously at the rate of about 0.5ml per minute. That is 720 ml per day.
- The amount around the brain and spinal cord remains fairly constant at about 120ml, which means that absorption keeps pace with secretion.

Four diagnostic measures of meningitis

- Microscopic culture and sensitivity of CSF
- X-ray of skull
- CT Scan of the brain
- MSC of sputum and Nasopharyngeal secretion
- EEG
- MRI of brain
- Blood culture

B. Three Nursing Problems of a patient with meningitis

1. Hyperthermia related to presence of microorganism in the meninges as evidenced by increased body temperature.
1. Acute headache related to increased ICP evidenced by patient's verbalization.

Hyperthermia related to infection caused by body temp of 39.2°C

2. Risk for deficit fluid volume related to nausea and vomiting

39.2°C

See Care Plan after Number D

(D). Four possible complications of meningitis

- Neurological dysfunction
- Facial paresis
- Hearing loss, vertigo, tinnitus
- Compression of optic nerve-blindness
- Hemiparesis
- Dysphasia
- Hemianopsia
- Non communicating hydrocephalus
- Water house-friderichsen syndrome: petechiae, disseminated intravascular coagulation, and adrenal haemorrhage
- Cerebral abscess
- Subdural empyema
- Subdural effusion
- Cerebral oedema.

NURSING CARE PLAN OF A PATIENT WITH MENINGITIS

SN	Date	Nursing diagnosis	Nursing objectives	Nursing intervention	Rationale	Evaluation
1		Hyperthermia related to infection and abnormal temperature regulation by the hypothalamus as evidenced by thermometric reading	Patient's body temperature will reduce by 1°-2°C within 48 hours of nursing intervention	1. Expose him 2. Fan patient 3. Tepid sponge (Tillichy) until his body temperature reduces 4. Give prescribed antipyretic drug e.g. Iparacet P.C.M 1000mg tid 5. Give prescribed antibiotics eg. Ofloxacin 500mg tid	1. Exposing patient reduces body temperature by radiation. 2. Fanning reduces body temperature by convection. 3. Tepid sponging reduces body temperature by conduction and evaporation. 4. PCM acts as heat regulating centre in the hypothalamus to reduce fever. 5. Ofloxacin interferes with the protein synthesis of the bacteria thereby slowing the activities and fever.	Patient's body temperature reduced by 1.3°C after 48 hours of nursing interventions.

1	19/02/17	Patient (acute headache) related to increased intracranial pressure evidenced by patient's complaint	Patient will verbalize reduction in headache within 30 minutes of nursing action.	1. Assess the severity of the headache by asking patient to rate it. 2. Administer analgesic drug to achieve a comfortable position. 3. Ensure a quiet environment. 4. Encourage patient to lie down with head and back elevated. 5. Give prescribed analgesic like Tablet P.C.M. 1000mg 4x4.	1. Assess patient's level of consciousness. 2. Comfortable position patient is in after analgesic action. 3. Quiet environment is necessary for rest to relax and decrease pain which will ease the headache. 4. Intracranial pressure is usually lower when the client is lying down or in a quiet, dark environment. 5. P.C.M. 1000mg 4x4.	Patient verbalized reduction in headache after 30 minutes of nursing action.
1		Patient has deficit fluid volume related to nausea and vomiting	Patient will not show any signs of dehydration e.g. dry lips, sunken eye etc. throughout the period of hospitalization.	1. Ensure that the prescribed intravenous infusion is given. 2. Regulate the infusion to flow at the prescribed rate. 3. Encourage him to take oral fluids. 4. Monitor his input and output and record. 5. Assist and encourage him on oral care. 6. Give any prescribed antiemetic eg. metoclopramide 10mg stat or prn.	1. This replaces the body fluid and electrolytes lost through vomiting. 2. Regulating the infusion ensures that he gets the prescribed volume of fluid. 3. Oral fluid helps to replace lost fluid. 4. Input and output monitoring helps to determine whether input balances the output. 5. Oral care helps to diminish nausea. 6. Metoclopramide acts on the vomiting centre in the brain thereby stopping vomiting.	Patient did not show any signs of dehydration throughout hospitalization.

NB Nursing diagnosis 1/2 mark, Objective 1/2 mark, nursing orders 1 mark (1/2 mark each) scientific principle 1/2 mark, evaluation 1/2 mark. Each diagnosis = 3 marks. 3 diagnosis = 9 marks

QUESTION

(A) Write briefly on the following drugs using the guideline below.

- Haloperidol
- Syntocinon
- Lonart (Artemether-lumefantrine)
- Combantrin. (4 Marks each)

Guidelines

- Group 1/2 mark
 - Action 1/2 mark
 - Indication 1/2 mark
 - Adult dose 1/2 mark
 - Route of administration 1/2 mark
 - Side effects 1/2 mark
 - Contraindications 1/2 mark
 - Nursing responsibilities 1 mark
- (B) Baby Femi a one year child is to receive intramuscular injection E-mal 20mg daily for 3 days. The injection comes in 150mg in 2ml. how many ml will you give as a dose. (2 mark)
- (C) Patient's prescription reads: "Infuse 5% Dextrose saline one litre to run for 6 hours". How many drops per minute will the nurse regulate the infusion with infusion set with drop factor 15?

Answer

(A) HALOPERIDOL

- Group: -Antipsychotic

Action: Blocks postsynaptic dopamine receptors in the brain thereby decreasing psychotic behaviour.

- **Indications:** Schizophrenia, Mania, short-term adjunctive management of psychomotor agitation, violent behaviour, severe anxiety
- **Dose:** Adult 1.5-3mg 2-3 times daily or 3-5mg 2-3 times in severely or resistant patients.
- **Route of Administration:** Oral and Im injection
- **Side effects:** Extrapyramidal reaction, sedation, dizziness, insomnia, headache, confusion, blurred vision, transient leucopenia, leukocytosis, urial retention, menstrual irregularities, gynecomastia, pigmentation, photosensitivity.

Contraindications: Impaired consciousness, bone marrow depression, pheochromocytoma, porphyria, basal ganglia disease, breastfeeding, pregnancy, history of seizure or EEG abnormalities, glaucoma, urine retention, anticonvulsant, anticoagulant, antiparkinsonian or lithium medications.

Nursing responsibilities

- Given oral drug with food or milk.
- Give IM injection into large muscle, patient should lie down at least 30 minutes after injection.
- Supervise ambulation till patient is stabilized on medication.
- Avoid strenuous activities.
- Should not stand for long periods at a time.
- Increase fluid intake to prevent constipation, give sips of water for dry mouth.
- Do not withdraw the drug abruptly.
- Observed patient for side effects.
- Warn patient to avoid hazardous activities that require mental alertness.

(ii) SYNTOCINON

Group: Oxytocin, lactation stimulant.

Action: Causes potent and selective stimulation of uterine and mammary gland smooth muscle, induces labour and milk ejection and reduces post partum bleeding.

Indications: Prevention and treatment of post partum and post abortion haemorrhage, induction of labour.

- **Dose:** Induction of labour 1ml (10 units) in 1 litre of D/W or N/S infusion. Infuse at 1-2 milliliters/minute at 15-30 minute interval until normal contraction pattern is established.
- Reduction of postpartum bleeding: 10-40 units in 1 litre of D/W or N/S at the rate of 20 to 40 milliliters/minute.
- 10 units IM after delivery of placenta
- Incomplete abortion: 10 units in 500ml of N/S or D/W at the rate of 20-40 drops/minutes

Route of administration: IV and IM.

Side effects: Uterine spasm, uterine hyperstimulation, fetal distress, tetanic contractions, soft tissue damage or uterine rupture, water intoxication hyponatremia, nausea, vomiting, arrhythmias, rashes. Anaphylactic shock.

Contraindication: Hypertonic uterine contractions, mechanical obstruction, fetal distress, when vaginal delivery inadvisable, pre-eclampsic toxemia, severe cardiovascular disease.

Nursing Responsibilities

- Administer IV infusion in hospital where critical care facilities are available.
- Give only when pelvis is adequate and fetal maturity and position assured and favourable.
- Monitor and record uterine contractions, heart rate, fetal heart rate, character of vaginal blood loss ¼ hourly.
- Observe for fetal distress.
- Do not give by bolus IV, may cause short-lasting drop in Bp.
- Monitor fluid input and output
- IM injection is given after delivery of placenta

(iii) **Artemetherine (Artemether-lumefantrine).**
Group: Anti malarial (ACT)

Action: Inhibits nucleic acid and protein synthesis within the malaria parasite

Indication: Uncomplicated malaria, especially in area with significant drug resistance.

Dose: 4 tablets start, then 4 each day (2bd)

Route: Orally

Side effects: Abdominal pain, anorexia, diarrhea, nausea and vomiting, headache, dizziness, sleep disorder, palpitation, arthralgia, myalgia, cough, asthenia, fatigue, pruritus, rash.

Contraindication: Breastfeeding, history of arrhythmias, bradycardia, CCF, family history of sudden death, or of congenital prolongation of QTC, interval first trimester of pregnancy, electrolyte disturbances, severe renal impairment or hepatic impairment, severe malaria patients on drugs that is metabolized by cytochrome enzyme eg amitriptyline)

Nursing responsibilities

- Dizziness may affect ability to perform some task, advice to avoid hazardous task.
- Advise patient to swallow tablet with water.
- If vomiting occurs within one hour of taking the drug, the dose should be repeated.
- Take tablet with milk

(IV)

Combantrin

Group: Anthelmintics (Anti worm)

Action: Block neuromuscular action, paralyzing worm and causing its expulsion from normal peristalsis.

Indication: Ascariasis, hook worm, enterobiasis, trichostrongylasis, tissue nematode

Dose: 10mg/kg as single dose

Route of administration: Orally

Side effects: Mild GIT disturbances, headache, dizziness, drowsiness, insomnia, rash, elevated liver enzymes, fever, weakness, tenosmus

Contraindications: Severe malnutrition, or anaemia, hepatic dysfunction, pregnancy, breastfeeding, hypersensitivity

Nursing responsibilities

- Shake suspension form well before administration
- Take drug with food, milk or juice
- Strict hygiene to avoid reinfection
- No dietary restrictions, laxatives or enemas not necessary.

(B)

$$\text{Formular} = \frac{\text{Required dose}}{\text{Dose in stock}} \times \frac{\text{Volume}}{1}$$
$$\frac{20}{150} \times \frac{2}{1}$$
$$0.5333$$
$$0.53$$
$$= 0.5\text{ml}$$

(C)

$$\text{Formular} = \frac{\text{Volume}}{\text{Time in hours}} \times \frac{15}{1}$$

Or

$$\frac{1000}{6} \times \frac{15}{60}$$

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$$= \frac{1000}{6} \times \frac{1}{4} = 41.666 = 42 \text{ drops/minutes}$$

QUESTION

Delegation of duty is necessary for effective goal accomplishment in the ward.

(A) (i) Explain the elements involved in delegation of duty. (6marks)

(ii) Outline the importance of delegation of duty. (4marks)

(B) State two (2) principles of learning. (2marks)

(C) Write short note on the following

- Quality assurance
- Nursing audit (4marks each)

A. Elements of delegation

i. **Responsibility:** That is the task that is delegated e.g oral care, bed bath etc.

It should be those task that must or may be delegated

ii. **Authority:** The right to make and enforce decision by a person in position to do so.

- Responsibility must be backed with equal authority to commend and control resources for delegation to be effective.

- Higher authority leads to power intoxication while lower authority causes inability to command resources.

- Both leads to inefficiency and inability to achieve stated objectives.

iii. **Accountability:** Ability and willingness to assume responsibility for one's actions and to accept the consequences of one's behaviour.

- Irresponsible and non-trusted staff should not be delegated, as the manager assumes responsibility of any task he/she delegates to a subordinate. In view of this, delegation should be based on competence and knowledge.

- Adequate supervision should be done while the task is be done.

(ii)

Importance of Delegation

- Relieves the manager the burden of routine jobs and allows him to concentrate on jobs that cannot be delegated
- Permits effective decentralization of decision making authority
- Allows for autonomy and flexibility in organization
- Helps in training and skill development of subordinates
- Allows use of initiative by subordinates
- Aids goals accomplishment both personal and organizational goals.
- Reduces delay and promote efficiency in job accomplishment.

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(8)

1. **Law of readiness:** Learning is effective when the learners are physically, mentally and emotionally ready to learn. **Law of exercise:** Repetition enhances learning. The learners learn best when they have meaningful practice and repetition of skill.
2. **Law of effect:** Learning is strengthened when it is accompanied by a pleasant or satisfying feeling. Positive reinforcement enhances learning.
3. **Law of primary:** Learning occurs from simple to complex.
4. **Law of recency:** Most recently learned information/skills are best remembered.
5. **Law of intensity:** The more intense the material through, the more likely it will be retained. Learners learn more from real thing than a substitute.
6. **Law of freedom:** Things that are freely learned are best learned.
7. **Law of requirement:** Learners must have the required ability, skills, instrument etc, that may help them to learn.
8. Learning objectives should clearly set and communicated to the learners.
9. There must be a change in behaviour for learning to take place.
10. The change in behaviour must be enduring, not transient.

Quality Assurance

- A guarantee on client care that certain standard of excellence will be maintained in delivery of care.
- It involves the judicious use of available resources to confidently render effective health care to clients at all time.
- It entails the ability of the nurse to provide care that meets the client's needs based on professionally agreed standard which is measured outcomes or effects on clients.
- Quality problem exist when services rendered did not meet the desired needs.
- It is a quality control process with the following components:
- Defining quality/setting standards
 - Measuring quality- evaluation of care provided
 - Improving quality- taking actions to make correction
- Quality improvement approaches/components are
- Setting standards
 - Credentiaizing
 - Team approach
 - Continuing education
 - Performance appraisal
 - Nursing audit
- Quality improvement approaches demands team work, delegation, decentralization etc. it is a continuous process.

(ii)

Nursing Audit

- Part of the cycle of quality assurance
- Systematic and critical analysis by nurses of the planning, delivery and evaluation of nursing care, in terms of resources and outcomes for patients.
- Results in improvement in the quality of patient care.
- The sorting involves the following
 - Current and past practices
 - Comparison of actual practice with agreed standard.
 - Compares different types of practices for best use resources.
 - Types are organization and peer review
- Common approaches are:
 - Prospective review
 - Concurrent review
 - Retrospective review
- Audit process involves the following activities
 - Define the standard
 - Measure current practice
 - Identify gaps in services provided
 - Implement actions to solve the problem
 - Review standards

QUESTION

Mrs. Duke a divorce was admitted in a neuro psychiatric hospital diagnosis of psycho neurotic disorder.

- (A) What is psychoneurotic disorder (2marks)
- (ii) Outline the four (4) classification of psychoneurotic disorder. (2marks)
- (B) List six (6) clinical features of anxiety. (3marks)
- (C) Outline your nursing management of a patient with anxiety. (5marks)
- (D) Write briefly on the following
 - i Delusion
 - ii Hallucination (4marks each)

Answer

Psychoneurotic disorder is a group of mild or minor form of mental illness.

- Keeps patients mentally out of balance
- Patient maintains contact with reality
- Has insight

It is characterized by:

- abnormal emotional reaction to disturbing situations
- exaggerated sense of anxiety and tension.

(ii)

Types of psychoneurotic disorder

- Anxiety neurosis
- Phobic neurosis
- Hysterical neurosis
- Traumatic neurosis
- Depressive neurosis
- Depersonalization neurosis
- Obsessive compulsive neurosis
- Hypochondriacal neurosis

(B)

Six (6) Clinical features of anxiety

- Headache
- Palpitation
- Loss of appetite/dry mouth
- Tremor of various parts of the body
- Hypertension
- Chest pain
- Frequent micturition/stooling
- General weakness
- Insomnia
- Poor concentration
- Dizziness
- Tinnitus
- Tingling
- Restlessness
- Irritability
- Strained look

(C)

Nursing management of patient with anxiety

1. **Physical care:** Provide physical needs as usual, bathing, oral care, adequate nutrition, well ventilated room etc.
 - Plan a sleep schedule for the night; do not allow him to sleep in the day. Encourage activities during the day. Encourage activities during the day.
 - Provide comfortable bed, clam environment.
2. **Psycho social needs:** engage patient in a meaningful activities that are diversional in nature. Listening to radio.
 - Encourage friends and family members to visit him
 - Do not allow patient to be alone.

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3. **Psychotherapy:** Talk and listen efficiently to patient in order to detect cause and treat accordingly.
 - Be friendly to patient
 - Remain and stay near the patient
 - Allow visitor
 - Allow and encourage patient to participate in group and occupational therapy.

4. **Hydrotherapy:** Ensure that patient takes enough fluid to avoid dehydration, from micturition/stooling

5. **Chemotherapy:** Give drugs like phenobarbitone, Largactil etc according to prescription

(D)

Delusion

- A false belief held by a person despite appeals and logic from a person of the same socio-cultural background.
- Inconsistent with external reality
- Holds no reality to anyone except the patient
- A major sign of psychotic disorder
- Present in psychotic disorder, bipolar disorder, schizophrenic, dementia etc
- Loss of contact with reality and personality is disintegrated.
- Types are: Paranoid, persecuting, nihilistic, grandeur, somatic or hypochondriacal
- Management involves engaging patient in a conversation about the delusion/ideas.
- The nurse should express doubt over the reality of the delusion thought to patient.
- Redirect conversation away from delusion/ideas back to reality by introducing another topic.

(ii)

Hallucination

- A false perception without a corresponding stimuli
- A major psychotic symptom
- Loss of contact with reality, personality disintegration
- Found in schizophrenia, bipolar disorder, psychotic depression etc.
- Types includes: visual, auditory, tactile, gustatory, olfactory hallucinations
- Can be frightening to patients, relations and other patients in the unit
- Are real to patient and patient response to them and may be violent

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- Goal of nursing care is to establish safety and help the patient to re-define reality.
- Provide a quiet and non stimulating environment.
- Talk to the client and tell him that you do not see that he saw.
- Administer medications as prescribed.

QUESTION

Pressure sore (ulcer) are preventable complications that can occur due to failure of efficient nursing care.

- (A) Define pressure ulcer (1 mark)
- (ii) Enumerate four (4) characteristic of impending pressure sore. (2 marks)
- (iii) Briefly explain the pathophysiology of pressure sore. (4 marks)
- (iv) Outline eight (8) preventive nursing interventions you can carry out for patients at risk of the condition. (4 marks)
- (v) List eight (8) factors that contribute to the formation of pressure sore. (4 marks)
- (B) Outline the considerations you will bear in mind while caring for a patient on plaster of Paris cast. (5 marks)

Answer

(A) Pressure ulcer is any lesion caused by unrelieved pressure that results in damage to underlying tissue. It is a break in the continuity of skin due to pressure.

- (ii) Four clinical characteristics of impending pressure ulcer

- Redness of the skin on the area
- Stinging pain over the area
- Hot feeling over the area
- Tenderness

- (iii) Pathophysiology of pressure ulcer

- Pressure ulcers are due to localized ischaemia
- When the tissue is compressed between two hard surfaces, usually the surface of the bed and the bony skeleton.
- The blood cannot reach the tissue, the cells are deprived of oxygen and nutrients; and ischaemia occurs.
- The waste product of metabolism accumulates in the cells and the tissue consequently dies
- Prolonged, unrelieved pressure also damages the small blood vessels.

(iv)

Preventive measures against pressure ulcer.

- Adequate nutrition
- Maintain skin hygiene, minimize force and friction to the skin while bathing.
- Use mild cleansing agent that minimize irritation and dryness.
- Keep patient dry and free of irritation and maceration by urine, faeces, sweat, incomplete drying after bath.
- Change linen as soon as they are wet
- Routine attention to the pressure area and areas where two skin surfaces come in contact.
- Frequent changing of patient's position at least every 2 hours.
- Place patient in correct position
- Use special bed appliances eg bed cradle to lift the weight of the linen.
- Proper making of patient's bed. The bed should be made without wrinkles.
- Patched bed sheet should not be used.
- Chipped or cracked bed pans should not be used.
- Encourage early ambulation
- Provide supportive device eg head protectors, pillows to support patient's head.

V. Predisposing factors to pressure ulcer

- Friction and urinary incontinence
- Immobility
- Inadequate nutrition
- Decreased mental status eg unconsciousness
- Diminished sensation (paralysis)
- Excessive body weight
- Advanced age
- Chronic medical condition such as diabetes mellitus
- Poor lifting and turning of patient especially immobile patients
- Incorrect positioning of patient
- Lying on hard support surfaces.
- Lying on wrinkled bed linen

(B)

Nursing considerations while caring for a patient on plaster of Paris cast.

- Expose the cast to dry, leave it uncovered
- Avoid weight bearing until the cast is dry
- Assess for any skin lesion eg wounds, pre-existing skin conditions.
- Remove any jewellery close or in contact with the cast.
- Avoid getting the cast wet
- Observe for necrosis and swelling

- Assess for blood circulation in the affected limb by blanching text on the nails distal to the cast.
- Note the date the cast was applied
- Encourage patient to exercise the limb not affected by cast
- Encourage gradual weight-bearing and activities
- Encourage liberal fluid intake to prevent formation of renal calculi
- Pad cast edges to prevent skin abrasion
- When oedema subsides, look into the cast with flash light to check for skin changes to detect easily signs of infection or tissue breakdown and report appropriately.
- Note and report any sign of neurological deficit such as numbness, tingling and loss of sensation
- Discourage patient from use of sharp instrument to scratch the skin within the cast to avoid tissue breakdown/infection.

OBJECTIVE QUESTIONS (PART D)

Choose the most appropriate option from a-d

1. A client hit her head on the steering wheel during a road traffic accident. She complains of a headache and a nursing assessment reveals that she has difficulty comprehending language and diminished hearing. The nurse suspects injury to which lobe of the brain.
 - a. Frontal
 - b. Temporal
 - c. Occipital
 - d. Parietal
2. A client had just returned to the ward after an above-the-knee amputation of the right leg. A nurse places the client in which of the following most appropriate position?
 - a. Supine with flat on bed
 - b. Reverse trendelenburg
 - c. Prone
 - d. Supine with stump elevated
3. Calcitonin is secreted by
 - a. Parathyroid gland
 - b. Parafollicular cells
 - c. Paraventricular nucleus
 - d. Cuboidal epithelial cells
4. Every cell in the body has a nucleus except mature
 - a. White blood cells
 - b. Ovum
 - c. Red blood cells
 - d. Spermatozoa
5. All the following are active learning methods for the client except
 - a. Lecture
 - b. Group discussion
 - c. Demonstration
 - d. Role play
6. Nucleus contains loosely coiled fibres called
 - a. Chromosomes
 - b. Chromatin
 - c. Chromatids
 - d. Ribosomes
7. The following are ligaments of the knee joint except
 - a. Patella ligament
 - b. Oblique popliteal ligament
 - c. Pubofemoral ligament
 - d. Tibial collateral ligament
8. In research, which of the following described the behaviour, characteristics or outcome that the researcher wishes to explain or predict.
 - a. Independent variable
 - b. Dependent variable
 - c. Intervening variable
 - d. Extraneous variable
9. Which of the following client is at greatest risk of developing osteoporosis
 - a. A 25 years old woman who jogs
 - b. A 36 years old man who has asthma
 - c. A sedentary 65 years old woman who smokes cigarettes
 - d. A 70 years old man who consumes excess alcohol

OBJECTIVE QUESTIONS (PART I)

Choose the most appropriate option from a-d

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- b. A 36 years old man who has asthma
- c/ A sedentary 65 years old woman who smokes cigarettes
- d. A 70 years old men who consumes excess alcohol

10. The physician prescribed an infusion of 2,400ml of I.V. fluid over 24 hours, with half this amount to be infused over the first 10 hours, during the first 10 hours, the client should receive how many ml of I.V. fluid per hour.

- a. 120ml
- b. 100ml
- c. 50ml
- d. 240ml

11. Which of the following medical condition is commonly found in clients with bulimia nervosa.

- a. Allergies
- b. Cancer
- c. Hepatitis A
- d. Diabetes mellitus

12. A nurse is inserting an indwelling urinary catheter into the urethra of a male client. As the nurse inflates the balloon, the client complains of discomfort, the most appropriate nursing action is to:

- a. Remove the syringe from the balloon, discomfort is normal and temporary.
- b. Aspirate the fluid, withdraw the catheter slightly and re-inflate the balloon.
- c. Aspirate the fluid, advance the catheter farther and re-inflate the balloon.
- d. Aspirate the fluid, remove the catheter and insert a new catheter.

13. One of the following is not a function of the spleen.

- a. Erythropoiesis
- b. Synthesis of hormones
- c. Phagocytosis
- d. Immune response

14. A process that eliminates many or all micro organism with the exception of bacteria spores from inanimate object is

- a. Sterilization
- b. Sanitization
- c. Cleaning
- d. Disinfection

15. The oedema which occurs in nephritic syndrome is due to

- a. Increased nephritic pressure caused by sodium excretion.
- b. Decreased aldosterone secretion caused by adrenal insufficiency.
- c. Decreased osmotic pressure caused by decreased serum proteins
- d. Increased hydrostatic pressure caused by sodium retention.

16. Which of the following is a possible sign of pregnancy

- a. Amenorrhoea
- b. Morning sickness
- c. Fetal movement by palpation
- d. Uterine growth

17. All are stages of mitosis except

- a. Anaphase
- b. Interphase
- c. Telophase
- d. Metaphase

18. The mesoderm consists of

- a. Cardiac muscle, skeletal, red blood cells.

- b. Skin, brain, neuron.
- c. Skin, kidney, lung.
- d. Thyroid cell, lung cell, neurone.

19. A patient with gout should be observed for which of these possible complications

- a. Pulmonary emboli
- b. Cirrhosis
- c. Gastric ulcer
- d. Renal calculi

20. The major action of the mono-amine oxidase inhibitors (MOI) is that

- a. It causes amines normally present in the central nervous system to accumulate rather than be destroyed by enzymes.
- b. It causes amines normally present in the central nervous system to be destroyed by the enzymes rather than accumulate
- c. It destroys the mono-amine oxidase which break down acetylcholine.
- d. It allows amine to accumulate at the nerve endings and restore normal muscle tone.

21. Which of the following is not a vector borne disease

- a. Malaria
- b. Trepanosomiasis
- c. Tinea capitis
- d. Yellow fever

22. Which of these behaviours of a patient with primary dementia of the Alzheimer type would suggest that the patient is in the final state of this disorder.

- a. Loss of memory
- b. Indifference to food

- c. Wandering
- d. Impaired judgement

23. The maxillary sinuses lie

- a. Within the ethmoid above
- b. Below the orbit
- c. In the midline of the frontal bone
- d. In the body of the sphenoid bone

24. The patient who has a sensorineural hearing loss

- a. May have reversed of damaged caused by ototoxic drugs
- b. Has difficulty understanding speech
- c. Hears low-pitch sound better than high pitch-sound.
- d. Experiences clearer sounds with the use of a hearing aid.

25. The uterine tubes are lined with

- a. Mucous membrane
- b. Ciliated epithelium
- c. Fibrous tissue
- d. Pavement epithelium

26. The erythrocyte sedimentation rate indicates

- a. The level of white blood cells in the blood
- b. The level of sodium in the blood
- c. The level of antibodies in the blood
- d. The activity of infection in the body

27. In patent ductus arteriosus, there is an abnormal opening between the

- a. Right and left atria
 - b. Right ventricle and aorta
 - c. Aorta and pulmonary artery
 - d. Right and left ventricles
28. Which of the following is Not true of drug interactions
- a. Drugs that interact team up to provide fast relief since two drugs usually produce a combined effect that equals the single most active component of the mixture.
 - b. When two or more drugs are administered concurrently to a patient, they can produce additive effects that are usually equivalent to the sum of the effect of either drug administered alone in higher dose.
 - c. Interaction involving two drugs usually produces a combined effect equal to the single most active component of the mixture.
 - d. The combined response of two drugs can become less than response produced by either drug only.
29. Which of the following best described malpractice
- a. Sleeping on duty
 - b. Negligence of duty
 - c. Omission or commission of an act that an average prudent nurse would not do.
 - d. Threatening of people on the ward.
30. Xerophthalmis is caused by
- a. Lack of vitamin A
 - b. Lack of vitamin C
 - c. Excess vitamin B
 - d. Excess vitamin D
31. Which statement, if made by a client with cardiac disease regarding appropriate dietary measure, indicates an understanding of the information provided by the nurse.
- a. "I should increase my sodium intake during pregnancy"
 - b. "I should maintain a low-calorie diet to prevent any weight gain"
 - c. "I should lower my blood volume by limiting my fluids"
 - d. "I should drink adequate fluid and increase my intake of high fibre foods".
32. One of the following is a major cause of neonatal mortality.
- a. Diabetes
 - b. Erythroblastocia
 - c. Hypoxia
 - d. Trauma at birth
33. The following are examples of lower respiratory tract infections in children except
- a. Acute bronchitis
 - b. Bronchitis
 - c. Tuberculosis
 - d. Pneumonitis
34. Which of the following physical assessment findings is typical in a client with advanced COPD
- a. Increased antero-posterior chest diameter
 - b. Underdeveloped neck muscles
 - c. Collapsed neck veins

d. Increased chest excursion with respiration.

35. The client who experiences eighth cranial nerve damage will most likely report which of the following symptoms.

- a. Facial paralysis
- b. Impaired vision
- c. Difficulty swallowing
- d. Vertigo

36. Projectile vomiting is mostly seen in

- a. Oesophageal stenosis
- b. Volvulus
- c. Pyloric stenosis
- d. Intussusception

37. Best treatment approach for a tuberculosis patient is

- a. Single drug therapy
- b. Multi drug therapy
- c. Broad spectrum
- d. Haematinics

38. Kubler-Ross stages of grief is as follow

- a. Bargaining, depression, denial, acceptance and anger.
- b. Anger, acceptance, bargaining, denial and depression
- c. Denial, anger, bargaining, depression and acceptance
- d. Anger, denial, depression, bargaining and acceptance.

39. A transmissible parasitic infestation of the skin characterized by burrows, intense itching and excoriation is

- a. Urticaria
- b. Rediculosis
- c. Herpes zoster
- d. Scabies

40. A congenital or degenerative opacity of the lens that leads to gradual loss of vision is called

- a. Glaucoma
- b. Cataract
- c. Conjunctivitis
- d. Retinopathy

41. The following are Freud personality developmental stages except

- a. Concrete stage
- b. Oral stage
- c. Phallic stage
- d. Anal stage

42. Nursing is defined by as "the act of utilizing the environment of the client to assist him in his recovery

- a. Virginia Henderson
- b. Canadian Nurses Association
- c. American Nurses Association
- d. Florence Nightingale

43. The fetal circulation consist of

- a. Two umbilical veins and one umbilical artery
- b. Two umbilical arteries and one umbilical vein
- c. Arteries carrying oxygenated blood to the foetus
- d. Veins carrying deoxygenated blood to the foetus.

44. A nurse is performing an admission on a 6-month-old

infant with a diagnosis of hydrocephalus. The nurse assess for the major sign associated with hydrocephalus when the nurse.

- a. Test the urine for protein
 - b. Takes the apical pulse
 - c. Palpates the anterior fontanelle
 - d. Takes the blood pressure
45. Which of these is an early sign of congestive heart failure in an infant with congenital heart disease.
- a. Cough
 - b. Slow and shallow breathing
 - c. Pallor
 - d. tachycardia
46. a cancer of the bowel can be confirmed through
- a. magnetic Resonance Imaging (MRI)
 - b. Biopsy of the tumor
 - c. Computerized tomography (CT) Scan
 - d. Abdominal ultrasound
47. Which equipment should be used by the nurse to feed a 3months infant after surgical repair of cleft lip.
- a. Rubber dropper
 - b. Plastic spoon
 - c. Paper straw
 - d. Single-hole nipple
48. Which intervention should the nurse include in the care plan for a catatonic schizophrenic client
- a. Administering lithium carbonate as prescribed
 - b. Giving the client an opportunity to express concern

- c. Meeting all the client's physical needs
- d. Providing a quiet environment where the client can be alone.

49. Which is an early sign of acute renal failure in a paediatric client

- a. Hypertension
- b. Decreased urine output
- c. Anaemia
- d. Haematuria

50. A nurse carrying for a young child with tetralogy of fallot observes that the child is dyspenis and cyanotic. Which position would help relieve the child's dyspnoea and cyanosis

- a. Squatting
- b. Sitting in bed with the head of the bed at 45 degree angle
- c. Lying flat on bed
- d. Lying on his right side

51. Which of these clients is at risk for fluid volume deficit

- a. A client with congestive heart failure
- b. A client receiving frequent wound irrigations
- c. A client with decreased kidney function
- d. A client with colostomy

52. S client with atrial fibrillation that is not responding to medication was placed on warfarin sodium. The nurse during discharge dietary teaching advise the client to avoid which of the following food while taking this drug

- a. Cherries
- b. Potatoes
- c. Green vegetables
- d. Spaghetti

53. The most appropriate location to assess the pulse of an infant under one year of age is

- a. Carotid
- b. Brachial
- c. Popliteal
- d. Radial

54. Which of the following instructions would be least appropriate to include in the post operative discharge plan of home care for a client being discharged after a laparoscopic cholecystectomy

- a. Wound care
- b. Activity restrictions
- c. Follow-up care
- d. Deep breathing exercise

55. The nurse observes constant bubbling in the water s**** chamber of a closed chest drainage system. What should the nurse conclude.

- a. The system is functioning correctly
- b. The system has an air leak
- c. The client has a pneumothorax
- d. The chest tube is obstructed

56. A physician's order reads: ampicillin 125mg IV every 6hourss. The medication label reads 1g and reconstitute with 7.4ml of bacteriostatic water. A

nurse prepares to withdraw up how many ml to administer one dose.

- a. 0.92ml
- b. 0.5ml
- c. 1.1ml
- d. 7.4ml

57. Organic psychoses include all except

- a. Acute confusional state
- b. Epileptic psychosis
- c. Manic depressive psychoses
- d. Dementia

58. An expression of two opposing feelings or emotion at the same time refer to

- a. Mutism
- b. Ambivalence
- c. Elation
- d. Exstasy

59. Occupational rehabilitation of a footballer who had his lower limbs amputated after road traffic accident is an example of which level of prevention

- a. Primary
- b. Intermediate
- c. Secondary
- d. Tertiary

60. In the assessment of Glasgow coma scale (GCS) for an unconscious patient using best motor response, a score of 4 refers to

- a. Abnormal flexion to pain
- b. Eye opening
- c. Withdraws to pain stimuli
- d. Confused

61. Which of the following is a non-

modifiable risk of factor for hypertension

- a. Age
- b. Sedentary life
- c. Obesity
- d. Diabetes mellitus

62. Normal PH of blood is

- a. 6-7
- b. 7.35-7.45
- c. 6.35-6.45
- d. 7.05-8.05

63. Membranous gaps between the cranial bones of an infant are called

- a. Sutures
- b. Joints
- c. Fascia
- d. Fontanelles

64. Kussmaul respiration is typical of

- a. Diabetic ketoacidosis
- b. Diabetic neuropathy
- c. Diabetic nephropathy
- d. Diabetic retinopathy

65. The most common side effect of ECT is

- a. Confusion and memory loss
- b. Head ache and dizziness

- c. Nausea and vomiting
- d. Diarrhea

66. Communication is complete only when is elicited

- a. Response
- b. Reaction
- c. Repetition
- d. Residence

67. Which of the following pre-operative diet is most appropriate for a patient scheduled for a haemorrhoidectomy

- a. High fiber
- b. Plain liquid
- c. Bland
- d. Low residue

68. Mode of transmission of chicken pox includes all except

- a. Direct contact with shingles
- b. Contact with residue fluid of infected case
- c. Drinking of contaminated water
- d. Direct contact with respiratory droplets of infected cases.

Match the theorist in column A with their theories in column B.

Column A

- 69. Abraham Maslow E
- 70. Dorothy Orem G
- 71. Callista Roy I
- 72. Evelyn Dwallis C
- 73. Martha Rogers F
- 74. Erik Erickson D
- 75. Jean Piaget B

Column B

- A Environmental theory
- B Activity of daily living
- C Family developmental theory
- D Psychosocial theory
- E Hierarchy of need
- F Science of unitary of human being
- G Self care theory

76.	Virgnia Henderson	B	H	Cognitive developmental theory
77.	Florence Nightingale	A	I	Adaptation theory
78.	Hildegard Peplau	K	J	Psychosexual development
			K	Inter personal relation theory
			L	Functional health pattern.

Tetanus can not be prevented by

- a. Keeping wound clean
- b. Administering T.T to pregnant women
- c. Using sterile instruments delivery
- d. Administering DPT to infant

80. Naturally active and passive immunity can be gotten through the following means except

- a. From mother to child in utero
- b. Infection of tetanus toxoid
- c. Subclinical infection
- d. Clinical disease

81. All are ways of purifying water except

- a. Sedimentation
- b. Filtration
- c. Distillation
- d. Chlorination

82. Which of the following is not a vector borne disease

- a. Malaria
- b. Tinea capitis
- c. Trepanosomiasis
- d. Yellow fever

83. When the diaphragm, relaxes, it moves upwards because of:

- a. Contraction of the abdominal muscles
- b. Expansion of the chest
- c. Contraction of the diaphragm
- d. Elastic recoil of the lungs

84. When obtaining consent for surgery, initially, the nurse should

a. Explain the risk involved in the surgery

b. Explain that obtaining the signature is routine for any surgery

c. Evaluate if the client's knowledge level is sufficient to give consent.

d. Witness the signature because that is what the nurses' signature documents.

85. Which of the following is not an indication of an improving condition in a 10years old child being treated for acute attack of asthma

- a. Increased wheezing
- b. Decreased wheezing
- c. Warm, dry skin
- d. A pulse rate of 90beats per minute.

86. During physical examination of an infant, a nurse notes abnormally low set ears. This finding is associated with

- a. Renal anomalies
- b. Otogenous tetanus
- c. Congential heart defects
- d. Trncheoesophageal fistula

87. To evaluate a client's atrial depolarization, the nurse observes which part of the electrocardiogram waveform.

- a. Twave
- b. Pwave

- c. PR interval
- d. QRS complex
- 88. A business man loses an important contract and had a flat tire while driving home. That evening he began to find fault with everyone. What type of defense mechanism did he apply?
 - a. Sublimation
 - b. Regression
 - c. Projection
 - d. Displacement
- 89. All are characteristics of a community except
 - a. Person to person interaction
 - b. Common interest
 - c. Recognize rules and regulations
 - d. Creative
- 90. Which moral principle is the nurse applying by deciding what is right for a client and acting without consulting the client.
 - a. Beneficence
 - b. Paternalism
 - c. Fidelity
 - d. Autonomy

Indicate True (T) or False (F) for the following questions

- 91. Excoriation is avoided in colostomy patients by applying

barrier cream to the stoma (T/F)

- 92. Alcohol and coffee use decreases pancreatic stimulation (T/F)
- 93. During bladder washout, about 500ml of fluid is used to irrigate the bladder. (T/F)
- 94. Ethnic and cultural heritage can affect a person's reaction to pain (T/F)
- 95. The brain receives approximately 750ml of blood per minute. (T/F)
- 96. Vitamin D helps in calcium and sulphate absorption. (T/F)
- 97. In tracheostomy, non inflatable cuff cannot be used when patient is on radiotherapy. (T/F)
- 98. 5% iodine is used with 2 drops of urine in a test tube to test for bile pigment. (T/F)
- 99. In hepatic coma, a reduced intake of protein and electrolytes is recommended. (T/F)
- 100. Aspirin may act selectively to depress some part of the brain (T/F)

MARKING GUIDE TO OBJECTIVE QUESTIONS (PART I)

1. B	34. A	68. C
2. D	35. D	69. E
3. A ^B	36. C	70. G
4. C	37. B	71. I
5. A	38. C	72. C
6. B	39. D	73. F
7. C	40. B	74. D
8. B	41. A	75. H
9. C	42. D	76. B
10. A	43. B	77. A
11. D	44. C	78. K
12. C	45. D	79. D
13. B	46. B	80. B
14. D	47. A	81. A
15. C	48. C	82. B
16. D	49. B	83. D
17. B	50. A	84. C
18. A	51. D	85. B
19. D	52. C	86. A
20. B	53. B	87. B
21. C	54. D	88. D
22. B	55. B	89. A
23. B	56. A	90. B
24. A	57. C	91. F
25. B	58. B	92. T
26. D	59. D	93. T
27. C	60. C	94. T
28. A	61. A	95. F
29. C	62. B	96. T
30. A	63. D	97. T
31. D	64. A	98. F
32. C	65. B	99. T
33. B	66. A	100. F
	67. D	

OBJECTIVE QUESTIONS (PART II)

Choose the most appropriate option from A-D

1. Autoimmune disease is due to one of the following.
 - a. Predniselone
 - b. ☒ Antibodies produced by the body
 - c. Vaccine
 - d. Anti bodies produced by bacteria
2. Neutrophil polymorphs does all except
 - a. Ingest bacteria
 - b. Digest bacteria
 - c. ☒ Collect to form inflammation
 - d. Produce antibodies
3. One of these is not correct about antibodies
 - a. Antibodies are immunoglobulin
 - b. ☒ Produced by red blood cells
 - c. Produced by plasma
 - d. Produced by lymphocytes

4. The under listed provide immunity except
 - a. Antibodies and lymphocytes
 - b. Antibiotics
 - c. Phagocytes
 - d. Vaccination
5. Blood plasma consists one of these
 - a. Blood without its protein
 - b. Whole blood
 - c. The fluid portion of the blood
 - d. The fluid portion of the clotted blood
6. Production of red blood cells in the marrow does not require
 - a. Iron
 - b. Ascorbic acid
 - c. Folic acid
 - d. Hydrochloric acid
7. Blood for transfusion is best stored at temperature of
 - a. 2-2°C
 - b. 3-4°C
 - c. 4-6°C
 - d. 6-10
8. Deficiency of one of the under listed leads to hypochromic anemia
 - a. Iodine
 - b. Iron
 - c. Folic acid
 - d. B12
9. Blood pass the through the valves of the heart after is has entered the right atrium in the following order.
 - a. Mitral, pulmonary, tricuspid, aortic.
 - b. Pulmonary, tricuspid, mitral, aortic.
 - c. Tricuspid, pulmonary, mitral, aortic
 - d. Aortic, mitral, pulmonary, tricuspid.
10. Blood pressure depends the following
 - a. Peripheral resistance
 - b. Blood volume
 - c. None of the above
 - d. A and B
11. A common complication of rheumatic mitral stenosis is
 - a. Arterial fibrillation
 - b. Rheumatic fever
 - c. Hypertension
 - d. Rheumatoid arthritis
12. Distended jugular veins, cyanosis, dyspnoea, and oedema of the legs are all signs of
 - a. Left vein thrombosis
 - b. Severe hypertension
 - c. Congestive cardiac failure
 - d. Deep venous thrombosis
13. Arterial obstruction in the distal extremity may be due to the under listed to the listed except
 - a. Arteroma
 - b. Phlebothrombosis
 - c. Thrombosed aneurism
 - d. Embolism
14. The pleura composed of _____ layers
 - a. One
 - b. Two
 - c. Three
 - d. Four
15. Adrenalin does not carry out one of these functions
 - a. Increase the heart rate
 - b. Constrict bronchi
 - c. Dilate bronchi
 - d. Relax the bladder detrusor muscle

16. Which of the following is not used in bacteria pneumonia
- Oxygen
 - Semisolid diet
 - Diuretics
 - Tetracycline
17. Before the eruption of the wisdom teeth, an adults has _____ number of teeth
- 20
 - 28
 - 32
 - 36
18. One of the following is not a cause of dysphagia
- Glossitis
 - Multiple sclerosis
 - Stomatitis
 - Pharyngitis
19. The fundus of the stomach normally
- Secrets trypsin
 - Contains air
 - Is empty
 - Contains fluid
20. One of the under listed is indication for vagotomy
- Decreased gastric acid secretion
 - Decreased gastric mortality
 - Increased gastric acid secretion
 - Increased gastric mortality and emptying rate
21. Pale bulky stools which float on water may be due to
- Cholecystitis
 - Pancreatic
 - Gastritis
 - Colitis
22. The major accessory glands of the large intestine secrete
- Mucus
 - Water
 - Pancreatic juice
 - Salts
23. One of these is not part of blood vessels that supply blood to the digestive system
- Splenic artery
 - Oesophagal artery
 - Internal iliac artery
 - Mesenteric artery
24. Two muscles of the tongue are
- Glossal muscles (inner & outer)
 - Extrinsic and intrinsic muscles
 - Cricoid and hyoid muscles
 - Omo-hoid muscle
25. Salivary glands include all except
- Parotid glands
 - Sublingual glands
 - Submaxillary glands
 - Submandibular glands
26. Constituents of saliva include all except
- Water
 - Salt
 - Enzyme
 - Juice
27. The digestive tract is a tube that measure about
- 18-20 feet
 - 20-25 feet
 - 22-25 feet
 - 20-21 feet
28. Once the useful substances, water

and minerals in the food material are absorbed in the digestive tract, they are transported through

- a. The endocrine system
- b. The integumentary system
- c. The urinary system
- d. The circulatory system

29. The human teeth in each quadrant are mirror image of the other, and include all except,

- a. One central and one lateral incisors
- b. First and second premolars
- c. Two canine
- d. First, second and third molars

30. The amount of glomerular filtrate re-absorbed by the tubule is approximately

- a. 99%
- b. 50%
- c. 85%
- d. 1.5%

31. Corticosteroids and Aldosterone are secreted by the

- a. Renal cortex
- b. Adrenal cortex
- c. Renal medulla
- d. Adrenal medulla

32. The pituitary gland is attached to the hypothalamus by

- a. Infundibulum
- b. Thalamus
- c. Neurohypophysis
- d. Connector tissues

33. Acromegaly is an outcomes of

- a. Excess growth hormone in adults
- b. Excess growth hormone in children

- c. Excess thyroid hormone in adult
- d. Excess thyroid hormone in children

34. Insulin pump operates by the help of

- a. Chips that regular the quantity of insulin injected at a tip
- b. Syringe which is calibrated in units
- c. Insulin needle which allows little quantity of insulin to be given at a time
- d. Connector tubing which releases the required quantity of insulin at a time.

35. The bones that form the ocular orbit are all except

- a. Zygomatic
- b. Lacrimal
- c. Frontal
- d. Mandible

36. The length and width of the large intestine approximately measure _____ respectively

- a. 1.5m and 0.5m
- b. 1.5m and 0.75m
- c. 1.75m and 7.5cm
- d. 1.5m and 7.5cm

37. A hypertensive patient placed on tablet propranolol 80mg tds x 1/52. which of these comments by the patient demands that the nurse should re-educate the patients on the side effects of the drug

- a. " I will keep strickly to the doctor's prescription"
- b. " I do not need to re-check my blood pressure since I

have commenced treatment"

- c. "I will not take my drugs with other drugs except my doctor approves it"
- d. "Once my drugs finish, I will go back to consult my doctor".

38. Cerebrovascular fluid is secreted at about _____ mls per day

- a. 320
- b. 420
- c. 520
- d. 720

39. Which of the following food should not be included in the diet of a patient who has hepatic insufficiency?

- a. Fresh fruit
- b. Eggs
- c. Pounded yam
- d. pasta

40. Most peptic ulcers occurring in the stomach are in the

- a. Cardiac portion
- b. Oesophageal junction
- c. Pyloric portion
- d. Body of the stomach

41. In the management of a patient with rheumatic fever, the nurse lays emphasis on the following except

- a. Rest
- b. Nutrition
- c. Pain
- d. anxiety

42. When obtaining a health history of a client with medical diagnosis of acute rheumatic fever, the nurse must determine whether the client was recently ill with

- a. Measles
- b. Mumps
- c. A sore throat
- d. Influenza virus

43. During the management of a client with sickle cell crisis, the nurse's priority nursing diagnosis is likely going to be

- a. Ineffective coping
- b. Impaired cardiac output
- c. Pain
- d. Fluid volume deficit.

44. Which of the following clinical manifestation, would be most important when assessing a patient who is suspected of having breast cancer?

- a. Nipple retraction
- b. Breast enlargement
- c. Breast tenderness
- d. Nipple discoloration

Control odour in colostomy patient can be achieved through

45. _____

46. _____

47. _____

48. A nurse would assess a patient who experiences prolonged vomiting for signs of
- Hypovolemic shock
 - Water intoxication
 - Metabolic acidosis
 - Potassium excess
49. Over distension of alveoli is called
- Emphysema
 - Alveolitis
 - Pneumothorax
 - Empyema
50. One of the following is a reliable indicator of improvement in a patient who has a diagnosis of anorexia nervosa.
- Electrolyte balance
 - Increase energy level
 - Increased fluid intake
 - Desire to eat.

Marking guide Objective Questions (Part II)

- | | | |
|-------|-------|---|
| 1. B | 20. C | 39. B |
| 2. C | 21. A | 40. C |
| 3. B | 22. A | 41. A |
| 4. B | 23. C | 42. C |
| 5. C | 24. B | 43. C |
| 6. D | 25. C | 44. A |
| 7. C | 26. D | 45. Wearing clean and well fitted colostomy |
| 8. B | 27. B | 46. Emptying colostomy bag when necessary and clean thoroughly. |
| 9. C | 28. D | 47. Deodorant tablet may be used. |
| 10. D | 29. C | 48. A |
| 11. A | 30. A | 49. A |
| 12. C | 31. B | 50. D 51. |
| 13. C | 32. A | |
| 14. B | 33. A | |
| 15. B | 34. A | |
| 16. B | 35. A | |
| 17. B | 36. D | |
| 18. B | 37. B | |
| 19. B | 38. D | |

Last test in objective (Part A)

Choose the best option from a-d

1. One of these does not constitute the major ways of managing enlarged prostate gland (BPH)
 - a. Watchful waiting
 - b. Conservative therapy
 - c. Non-surgical invasive therapy
 - d. Surgical measure
2. The drug alpha adrenergic blockers such as Terazosin are used in the management of BPH because
 - a. They prevent the conversion of testosterone into a potent androgen (dihydrotestosterone)
 - b. They act like 5 α -reductase inhibitors
 - c. They relax the smooth muscles of the bladder neck and the prostate.
 - d. They prevent the conversion of angiotensin I to angiotensin II.
3. Using international prostate symptom scores (IPSS) to assess a patient with BPH, the following are correct except.
 - a. Total score of 0-7 is termed no symptom
 - b. Total score of 0-7 is mild symptom
 - c. Total score of 8-19 is termed moderate symptom
 - d. Total score of 20-35 is termed severe symptom
4. One of these is not assessed when IPSS is used in assessing a patient with enlarged prostate
 - a. Straining during micturition
 - b. Intermittency of micturition
 - c. Nocturia
 - d. Strangury
5. Investigation in BPH may include all except
 - a. Prostate specific antigen (PSA)
 - b. Streptolysin O+ titre test
 - c. Urodynamic flow rate
 - d. Urinalysis, with culture and sensitivity
6. In the management of BPH, sawpalmetto (a botanical remedy) functions like
 - a. Beta-adrenergic blockers
 - b. Alpha-adrenergic blockers
 - c. 5 α -reductase inhibitors
 - d. Angiotensin converting enzyme inhibitors
7. Pre-operative assessment of a patient with BPH involves the under-listed except
 - a. The last time patient voided
 - b. Volume of urine voided since last few hours
 - c. Desire to void but unable to initiate voiding
 - d. Colicky abdominal pain
8. The nurse sets catheterization tray and assists the urologist in catheterizing a patient with BPH to
 - a. Ensure that she does all that she suppose to do as a reasonable nurse
 - b. Ensure that she witnesses when the patient voids and to measure the volume of urine made by the patient.

- c. Prevent delay in the client care
- d. To reduce anxiety related to non-emptying of the urinary bladder

9. One of these statements made a patient discharged after prostatectomy shows that the nurse have to deepen her health education to the patient.

- a. "Thank you nurse, I thank God for everything but not happy that I will not perform my sexual functions"
- b. "I will always do my bladder training"
- c. "I will keep to my appointment date"
- d. "I will ensure that I complete my antibiotics"

10. One of these is not a possible complication in prostatectomy

- a. Infection
- b. Infertility
- c. Haemorrhage
- d. Disequilibrium syndrome

11. One of these is not true when managing a patient that is undergoing dialysis

- a. Procedure can be performed on the access hand
- b. Blood in the circuit time must return to the patient before disconnection is made after the procedure
- c. There is need for anticoagulant during the procedure
- d. The duration of the procedure is often determined by the level of toxic wastes in the blood.

12. One of these is not a type of urinary incontinent

- a. Urge incontinent

- b. Overflow incontinent
- c. Reflex incontinent
- d. Pressure incontinent

13. The incubation period for neisseria gonorrhea is

- a. 5-10days
- b. 2-10days
- c. 10-20days
- d. 2-20days

14. The incubation period for treponema pallidum is

- a. 10-30days
- b. 10-60days
- c. 10-90days
- d. 10-120days

(VDRL)
Veneral Disease Rother Laboratory Test

15. In the investigations to make diagnosis of syphilis, VDRL is a sure test to confirm the disease, this abbreviation stand for

16. In hypothyroidism, one of these hormones is found in excess in the blood

- ~~a. Thyroid stimulating hormone~~
- b. Thyroxin (T4)
- c. Growth hormone
- d. Androgen

17. One of these hormones play great part in metabolic activities in the body

- ~~a. Thyroxin~~
- b. Mineralocorticoids
- c. Glucocorticoids
- d. Glucagon

18. Adenococorticotrophic hormone (ACTH) resembles one of these hormones in chemical structure and effect

- a. Melanocyte stimulating hormone
- b. Antidiuretic hormone
- c. Growth hormone inhibiting hormone
- d. Parathormones

19. One of these does not form cardinal manifestations of Addison's disease

- a. Hypoglycaemia
- b. Weight loss
- c. Adrenocortical crisis
- d. ☒ Oedema

20. Hypertension is common in one of these endocrine diseases

- a. Addison's disease
- b. Diabetes mellitus
- c. ☒ Cushing's syndrome
- d. Simple goiter

21. Nursing management of a patient with Addison's disease aims at achieving the under-listed except

- a. Managing and monitoring Addisonian or Adrenal crisis
- b. Restoration of fluid and electrolyte balance
- c. Improving patient's activity tolerance
- d. ☒ Restriction of sodium from the diet.

22. The metabolism of the three major classes of food will definitely produce the following at the end except

- a. Energy in form of ATP
- b. ☒ Glucose
- c. CO_2 carbon IV oxide (CO_2)
- d. Water (H_2O)

23. When there is impaired glucose

metabolism, the possibility of dysfunction of one of these cannot be ruled out

- a. ☒ The pancreas
- b. The gall bladder
- c. The digestive enzymes
- d. The thyroid hormone

24. Exercise is essential in the management of diabetic mellitus because of the following reasons except

- a. It helps to metabolize carbohydrate and skeletal muscle receptors allow the muscles to take in glucose from the blood independent of insulin
- b. It lowers cholesterol and triglyceride levels and improves muscle tone
- c. It improves circulation and prevents macro and micro vascular complication
- d. ☒ It increases motor neuropathy

25. One of these is not a complication of diabetes mellitus

- a. Glaucoma
- b. Diabetic nephropathy
- c. ☒ Diabetes insipidus
- d. Hyperglycaemic hyperosmolar nonketotic syndrome

26. One of these may not be necessary investigation in diabetes mellitus

- a. Oral glucose tolerance test
- b. ☒ Occult blood test
- c. Postprandial glucose
- d. Random blood glucose

7. One of these is not a type of insulin

- a. Synthetic insulin
- b. Rapid acting insulin
- c. Insulin lence
- d. Intermediate acting insulin

28. The biguanides acts by

- a. Direct stimulation of glycolysis
- b. Rapid absorption from the git
- c. Decreasing hepatic gluconeogenesis
- d. Increase the quality of insulin secreted by the islets of langergharm

29. Pruritus vulvae is common in

- a. Newly diagnosed diabetics

b. Female diabetic

c. Patients with long standing diabetes

d. All diabetic irrespective of duration

30. In planning menu for a diabetic one of these should be considered

- a. The duration of the disease
- b. The management/treatment modality
- c. The age of the patient
- d. The culture of the patient

Last test in objective questions (Part B)

1. Consumers of Health care are faced with uncertainty which makes hen to relinquish their sovereignty and create specialized agency relationship because of the under-listed reasons except

- a. They may not fully understand the health care information
- b. Their state of health may not permit them
- c. They may not be able to make rational choices
- d. Their demand is derived demand

2. Expected beahviour of healthcare providers differ from that of a typical business man in the under-listed ways except that

- a. Advertizing and over

competition by healthcare providers are not allowed

b. Treatment is dictated by the needs of the case and should be limited by financial consideration.

c. Healthcare providers utilize the information disparity as an opportunity to overcharge the consumers of their products

d. The healthcare providers are relied upon as experts and are expected to do the best for the consumers of their services.

3. Externality arises when

a. Health prevention is undermined by an individual.

b. When economic activity of an individual's part generates positive effect on another.

- c. Individuals embark on health promotional activities.
 - d. One positive effect is achieved by one extra cost.
4. One of the following is not true about demand in healthcare arena.
 - a. Demand in healthcare is not a demand that must be met
 - b. Demand in healthcare be elastic
 - c. Demand in healthcare is unitary
 - d. Demand in healthcare can be inelastic
 5. Healthcare services are done at accost which occur mainly for the following reasons except
 - a. Service equipment
 - b. Purchase materials
 - c. Pay human resources
 - d. Advertise products and services
 6. Healthcare market is difficult by one of these in such a way that only those who can afford the cost are willing to trade there in
 - a. Increasing cost of services
 - b. Increase demand of services
 - c. Increasing cost of training healthcare providers
 - d. Increasing competition in healthcare market place
 7. Determination of ill-health by an individual is personal, emotional and often uncertain but healthcare provides one of these to the consumers
 - a. Psychic and monetary benefits
 - b. Emotional benefits
 - c. Feeling of wellbeing
 - d. Bridge the gap between

perceived current health status and desired health status

8. An individual may be said to demand for healthcare to achieve one of the following
 - a. Bridge the gap between his/her perceived current health status and desired better health status
 - b. Compete favourably in economic market place
 - c. Achieve the self desired goal in life
 - d. Attend maximum understanding of his/her health status
9. The cost of meeting the need and the demand for healthcare is so high that achieving a better health by the citizens may not become a reality. This therefore calls for the healthcare providers to:
 - a. Understand the various cost containment strategies
 - b. Allow issues that pertain to cost to remain sole activities of the hospital account unit
 - c. Think less about the cost of treatment on their clients but focus on their management of cases
 - d. Understand the federal government budgetary allocation to health
10. In order to involve individuals and communities in the consideration of healthcare cost. _____ was established as an avenue to reach the 70% of the

- population
- Primary healthcare
 - Cost consideration plans
 - Budget variance
 - Cost consideration strategies
11. Cost of healthcare will be reduced to both the individual and the government if one of these will be taken serious
- Monitoring the individual hospital bills to the consumers of healthcare.
 - Monitoring the activities of the private healthcare institution
 - Taking primary healthcare services serious
 - Establishing of good referral system
12. One of these is not among the cost content strategies
- Cost monitoring
 - Cost awareness
 - Cost avoidance
 - Cost budgeting
13. As an avenue to reduce cost, there is need for
- Healthcare providers to collaboration in patients care.
 - Highly killed staff should not be used when lesser skilled staff should not be used when lesser skilled staff can effectively carry out the procedures.
 - Alternative medical treatment should be encouraged
 - Used of technologies should be encouraged irrespective of standard of acceptability
14. Cost benefit analysis allows for
- Monitoring of healthcare status
 - and effectiveness
 - Identification, measurement and comparison of benefits and cost of a program or treatment alternative
 - Identification of alternative measure or approaches to particular
 - Exchange of information on the most available treatment in terms of cost.
15. Production of goods and services in a country is measured by
- The health status of the country
 - The country, technological emancipation
 - The number of scientists in the country
 - The country's Gross domestic product
16. Health indicators of a country is measuring data on the following except
- Infant mortality rate
 - Prevalent of endemic diseases
 - Maternal mortality rate
 - Child survival strategies
17. To ensure maximum utilization of the scare resources _____ come into play in the real economic planning
- Variance
 - Budget and budgeting
 - Monitoring
 - Policy making
18. Budget can be defined to mean
- Management technique to function optimally

- b. The financial contents in a balance sheet
 - c. Quantitative statement prepared and approved prior to a defined period of time for a purpose of attaining a given objective.
 - d. Financial statement defended by the executive officers in a fiscal year.
19. Budget serves the following purposes except
- a. States the organizations expectations in order to avoid confusion and ambiguity
 - b. Communicates expectation to all cadres of management
 - c. Motivating employees
 - d. To give partial plan of action and raise uncertainty
20. One of these is not a type of budget
- a. Basic budget
 - b. Cash budgets
 - c. Flexible budget
 - d. Organizational budget
21. Requirement for effective budgeting include all except
- a. Clear organizational chart i.e organogram of the organization
 - b. Good application of SWORT analysis
 - c. Well defined objectives and policies of the organization
 - d. Outline of various sectional or departmental budgets.
22. Once budget is approved, by the legislators, it passed into law as
- a. Bill of budget
 - b. Act of budget
 - c. Appropriation bill
 - d. Appropriation act
23. Negative effects of poor healthcare financing in Nigeria has led to deterioration of healthcare system as evidenced in all except.
- a. Number of dilapidated infrastructure
 - b. Efficient public private partnership
 - c. Lack of equipment and materials
 - d. Poor supply of drugs and vaccines
24. Evaluation of sources of healthcare financing is based on the underlisted except
- a. Efficiency
 - b. Judgment
 - c. Effectiveness
 - d. Quality of care
25. Looking at efficiency in healthcare services involves evaluating the following
- a. Allocative efficiency
 - b. Performance efficiency
 - c. Administrative efficiency
 - d. Technical efficiency
26. One of these is not part of the problems identified in Nigeria healthcare financing and provision
- a. Private partners in healthcare provision tend to be interested in profit maximization
 - b. Poor monitoring and evaluation of funds and funding of healthcare programmes
 - c. Most foreign aids come in form of importation of equipment and technical assistance
 - d. Government does not consider

the remuneration of healthcare providers

27. One of these is not part of the sources of healthcare financing
- a. Special taxes
 - b. Charitable appeals and donations
 - c. Budgeting
 - d. External assistance

28. National Health Insurance Scheme (NHIS) was established in Nigeria under
- a. Decree 25 of 1999
 - b. Decree 35 of 1999
 - c. Decree 25 of 1979
 - d. Decree 35 of 1979

29. Health Economics as a course tends to:

- a. Blend physical, emotional, social intellectual and spiritual issues revolving in economic aspect of human.
- b. Define the impact of economics on health-decision-making and economic well-being of individuals.
- c. Identify the possible ways of making healthcare services affordable to the citizens.
- d. Look into the mental and social wellbeing of individuals with consideration of their economic status.

30. Health happens due to one of the following

- a. Constant change and adaptation to stress which results in homeostasis.
- b. Constant medical review or checkup.

- c. Constant examination of extent of stress and proper stress management strategies.
- d. Mastering of developmental changes in life and strategies to adapt to such changes

31. Economic deals with the one of the following

- a. Means of raising the purchasing ability of individuals.
- b. Way in which money influences or is organized within an area of business or society.
- c. Way by which individuals are financial empowered financially.
- d. Means of expanding individuals ability to be independent of government in monetary matters.

32. Need can be best defined as stated below with exception of one.

- a. Urgent requirement
- b. Necessary commodity which one cannot do without.
- c. The things an individual requires in order to live in a comfortable way.
- d. The desire or wish in an individual for something.

33. Demand can be said to simply mean

- a. Firm request for commodity/services that somebody needs.
- b. Firm request for commodity/services which is backed up by willingness and

- ability to have the commodity at prevailing price.
- c. Firm request for commodity/services with or without willingness and ability to purchase the same at prevailing prices.
- d. Firm request for goods/services with reasonable ability to explain the reasons for such request.
34. For demand to remain the same, such demand is said to be
- "Ceteris paribus"
 - "Ceteris paribus"
 - "Ceteris paribus"
 - "Ceteris paribus"
35. One of the underlisted is a type of demand
- Personnel demand
 - Individual demand
 - Group demand
 - Public demand
36. Elasticity of demand reflects the ability of to increase or reduce and may be represented by the following phenomena EXCEPT
- Elastic demand
 - Imperfect demand
 - Unity demand
 - Inelastic demand
37. Demand for goods and services cannot be affected by one of the under-listed.
- Income
 - Market demand
 - Other prices
 - Taster
38. Supply of goods and services can be affected by the under-listed EXCEPT
- Weather
 - Technological changes
 - Demand for products
 - Cost of production
39. The under-listed affect demand for healthcare services EXCEPT
- State of one's health
 - Knowledge of healthcare provider
 - Level of education
 - Cost of available healthcare services.
40. Healthcare has some outstanding characteristics listed below with EXCEPTION of one
- Unpredictability
 - Derived demand
 - Understanding of information
 - Consumer sovereignty
41. Consumer sovereignty in healthcare is affect by the following EXCEPT
- Consumer of healthcare may not fully understand the information on healthcare.
 - Patient/consumer's educational level
 - Patient's/consumer's state of health
 - Decision making may be difficult for the patient/consumer.
42. Behvaiours of doctors differ from that expected of a typical business man in the following way EXCEPT.

- a. Advertising and over competition by physicians are not allowed
- b. Treatment of patients is claimed to be dictated by needs of the patient and not be limited by financial consideration.
- c. Advice given by physicians to consumers of healthcare remains outside self interest.
- d. Physicians' charge on patients is sometimes determined by the economic status of the patients.

43. In normal economic relationship _____ is arrived at through bargaining

- a. Treatment modality
- b. Contractual arrangement/agreement
- c. Relationship
- d. Consumer satisfaction

44. In economic goods prices of different commodities are not _____ according to the customer's income

- a. Stated
- b. Subsidized
- c. Discriminated
- d. Distributed

45. Conventionally, neoclassical economists assume that

responsibility over consumption decision is rested on

- a. The consumer
- b. The nurse
- c. The doctor
- d. The patient

46. The autonomy of the patient in decision making is likely to be affected by one of these

- a. Consumer sovereignty
- b. Agency relationship
- c. Asymmetry of information between the consumer and the care provider
- d. Utility theory

47. The under-listed make decision about healthcare difficult for consumers EXCEPT

- a. Uncertainty
- b. Ignorance about health status
- c. Effectiveness of the treatment
- d. Lack of knowledge of the ability of the healthcare providers

48. Three variables implicit in commodity include all EXCEPT

- a. Value
- b. Supply
- c. Demand
- d. Monetary

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6

Fill the spaces below with the most appropriate word or phrase

49. Budget can be broadly divided into _____ and _____
50. A budget that includes the procurement of human resources, purchase of utility vehicle and furniture's is classified as _____ budget.
51. When budget is approved by the legislators, it goes back to the President or Governor for _____ before it becomes a financial income/expenditure for the fiscal year
52. Healthcare financing of choice aims at improving the _____ of the health system
53. Equity in healthcare financing considers the financier and the _____
54. Most of the assistance from the foreign bodies (countries) are directed towards health prevention neglecting _____ and _____ aspects of healthcare.
55. Through NHIS employers are guaranteed good healthcare services for their _____ at a cheaper rate thereby increasing their _____
56. At registration with NHIS, the contributor and the dependents will be issued with _____
57. _____ is an avenue established to control budget
58. In Nigeria, budget proposal is often based on revenue from _____
59. _____ set in each zone and FCT considers complaints by aggrieved person under NHIS
60. Healthcare to a large extent depends on _____
61. Minimal budget allocation to the health sector affects the health sector and the _____ of the _____ of the country

Choose the best option from a-d

62. One of the under-listed is not a prepayment model in health system

- a. community financing
- b. NHIS
- c. Complementary
- d. Supplementary

63. Example of indirect tax from which fund is pooled to fund the health system may include all except

- a. Value Added Tax
- b. National Health Insurance Scheme

- c. Pay As You Earn
- d. Local taxes

64. Out-of pocket payment in health system has the following characteristics except one

- a. Individuals pay directly to the healthcare provider
- b. Common where payment system is available
- c. Payment is out of their own pocket
- d. It is mostly applied in expensive care services

65. Fee for service becomes a mode of payment when a contributor in NHIS is managed by all except
- Laboratory scientist
 - Nurse consultant
 - Specialist doctor
 - Private hospital
66. Arbitration Board for NHIS is set to take care of
- complaints by the HMOs only
 - complaints by parties under the scheme only
 - complaints by the agrived Primary Health care providers only
 - complaints by the secondary health care providers only
67. One of these is not an objective of NHIS
- ensure appropriate patronage of all levels of healthcare
 - limit the rise in the cost of health care costs
 - ensure equitable distribution of health care costs among different income group
 - to enable participation of stakeholders in the health care delivery system
68. In large organizations, the preparation of the budget is usually the responsibility of
- the Chief executives
 - the budget committee
 - the administrator
 - the government
69. One of these is not a decision making technique
- Cost Benefit Analysis
 - Cost Effectiveness Analysis
 - Marginal Analysis
 - Budgetting
70. Private Health Insurance has three main focus which include all Except
- Substitutive
 - NHIS
 - Complementary
 - Supplementary

MARKING GUIDE FOR LAST OBJECTIVE QUESTIONS (Part A)

1=A
2=C
3=A
4=D
5=B
6=C
7=D
8=C
9=A
10=D
11=A
12=D
13=B

14=C
15 =
Veneral
Disease
Research
Laborator
y test
16=A
17=A
18=A
19=D
20=C
21=D

22=B
23=A
24=D
25=C
26=B
27=A
28=A
29=B
30=D

(Part B)

1. D
2. B
3. B
4. C

5. D
6. A
7. A
8. B
9. A
10. A
11. C
12. D
13. B
14. A
15. D
16. D
17. B
18. C
19. D
20. D
21. B
22. C
23. B
24. B
25. B
26. D
27. C
28. B
29. B
30. A
31. B
32. D

33. B
34. D
35. B
36. B
37. B
38. C
39. B
40. C
41. B
42. D
43. B
44. C
45. A
46. C
47. D
48. D
49. CAPITAL AND
RECURRENT/C
URRENT
50. CAPITAL
51. ASSENT OR
SIGNATURE
52. FUNDING
53. BENEFACTOR
54. CURATIVE AND
REHABILITATIV
E
55. EMPLOYEE,

PRODUCTIVITY
56. PERSONAL I.D. CARD
57. VARIANCE
58. CRUDE OIL
59. ARBIBRATION BOARD
60. INFORMATION
61. HEALTH STATUS AND
CITIZEN

62.A

63.A

64.B

65.D

66.B

67. A

68.A

69.D

70.B